



GE VERNOVA

ADMS

16.25.1

OMS Client Configuration Editor User Guide

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Change History

Date	Change Description
May 31, 2023	ADMS 3.16: • Updated references to "Storm Assist" to new product name "Outage Assist". • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Dec 8, 2023	ADMS 16.3.0: • Updated Incident Settings. • Updated Incident PDF Export. • Updated Customizing the Order PDF Export.
Apr 1, 2024	ADMS 16.7.0: • Updated document format and styling.
Sep 25, 2024	ADMS 16.12.0: • Updated section Update Work Order.

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1. Introduction

The OMS Client Configuration Editor is a support tool provided with ADMS.

The main purpose of this tool is to configure Outage Management System (OMS) clients. It also allows an administrator to configure the appearance and effects of geographic network displays to meet the needs of individual utilities.

Some of the aspects that can be configured using the OMS Client Configuration Editor include:

- Appearance of polygon types
- Appearance of tabs and display names in the Incident Detail display in the OMS
- Configuration of the customer list
- Creation of several event types, and association of a default customer notifications message with each event

The OMS Client Configuration Editor provides a great deal of flexibility in designing the way the Network view appears.

The output of the OMS Client Configuration Editor is an XML file (`OmsClientConfig.xml`). This is a global file that is managed locally and then distributed to all clients, to ensure that all clients see the same view of the network.

2. General Usage

2.1. Starting the OMS Client Configuration Editor

To start the OMS Client Configuration Editor, from the Microsoft Windows Start menu, go to All Programs> Eterra> Distribution> Tools> Configuration Tools> OMS Client Configuration Editor.

The shortcut is configured to pass the default `OmsClientConfig.xml` file to the application, which is opened automatically.

When it starts, the OMS Client Configuration Editor opens the default configuration file. The default file name is defined in the shortcut that starts the OMS Client Configuration Editor. If there is a need to open another configuration file during startup, the file name defined in the Target field of the shortcut must be modified accordingly.

2.2. File Menu Options

In the OMS Client Configuration Editor, the File menu includes the following commands:

- **New:** Creates a new, empty OMS configuration file.
- **Open:** Opens an alternate or existing configuration file (usually found in the %IDMS_ROOT%/Config/Client folder).
- **Export to JSON Data:** Exports OMS client configurations as JSON files to the %IDMS_ROOT%\config\output folder. Used for customizing the Outage Assist system. You can also export configurations using the command line. For more information, refer to the *Outage Assist Software Installation and Maintenance Guide*.
- **Save:** Saves changes to the currently open configuration file.
- **Save As:** Saves all changes to another configuration file.

The OMS Client Configuration Editor allows saving the configuration file under different names. However, when it starts, it automatically opens the default configuration file (`OmsClientConfig.xml`). The default file name is defined in the OMS Client Configuration Editor shortcut.

- **Exit:** Closes the application.

2.3. Applying Changes to the ADMS Client

After changes have been made to the configuration using the OMS Client Configuration Editor, those changes must be saved as an XML file.

To apply the changes to the local ADMS client, the XML file must be saved to the correct location, and the file name must be specified in the client dynamics file. The default path is

%IDMS_ROOT%\Config\Client\OmsClientConfig.xml.

Since the client only reads the OmsClientConfig.xml file on startup, the client must be re-started for the changes to take effect.

Refer to the *ADMS Software Installation and Maintenance Guide* for more information about the following:

- **File Updater:** The tool used to broadcast the updated OmsClientConfig.xml file to all clients.
- **Net Dynamics Configuration Editor:** The tool used to define the name and location of the OmsClientConfig.xml file to be used in a particular client environment.

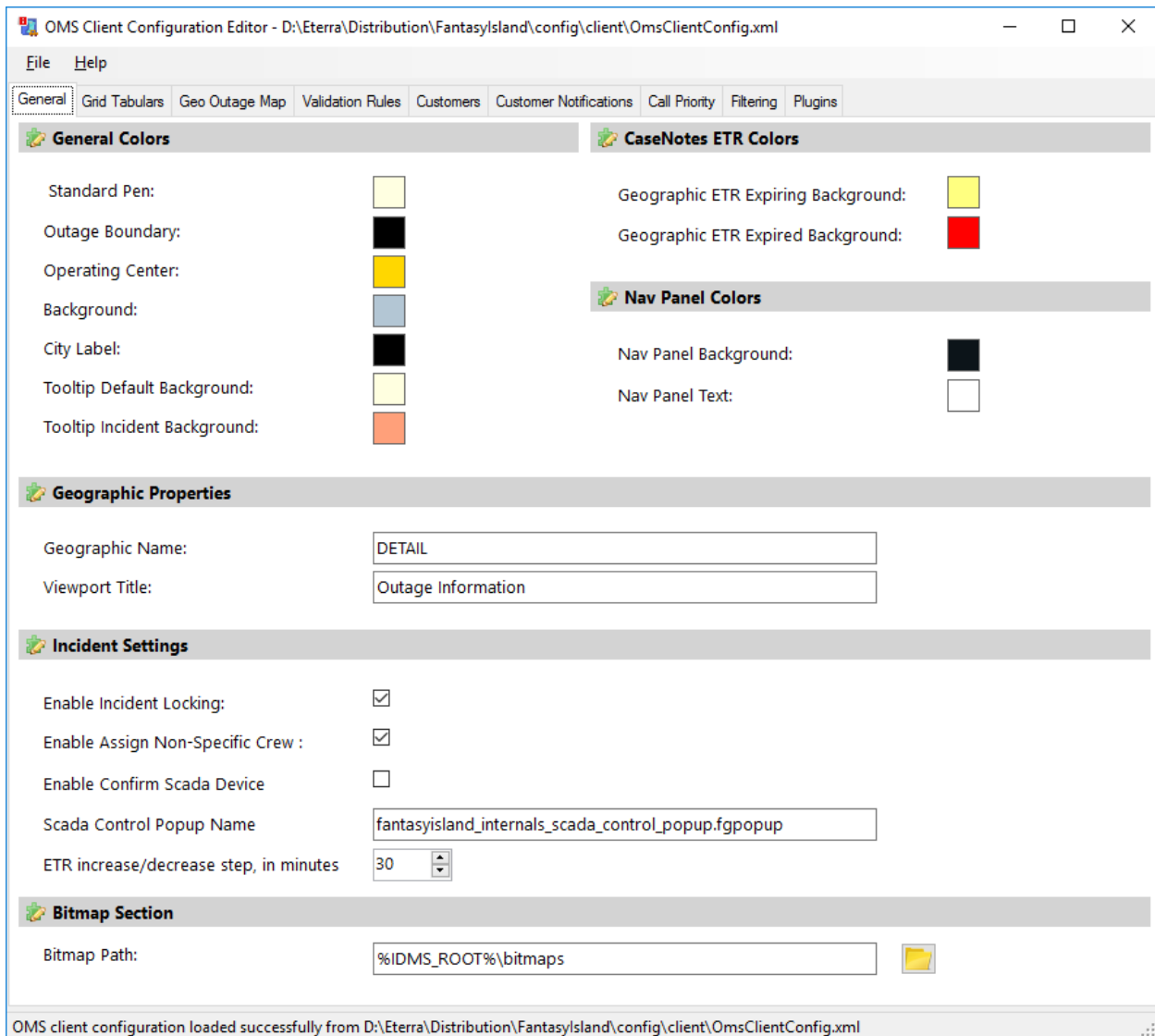
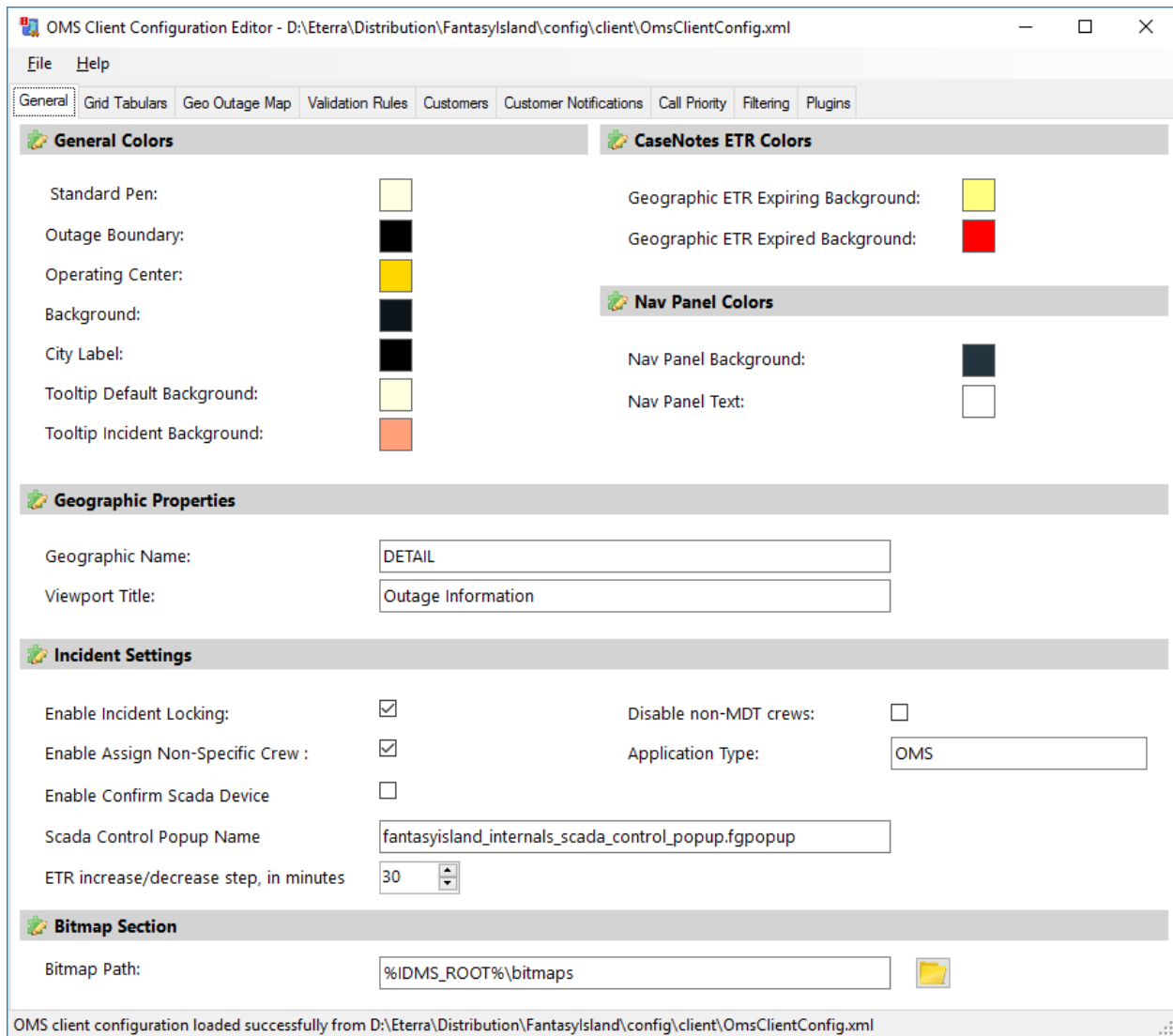


Figure 1. ADMS OMS Client Configuration Editor

3. General Tab

On the General tab, you can configure general properties of the OMS display plugin.



The screenshot shows the 'General' tab of the 'OMS Client Configuration Editor'. The window title is 'OMS Client Configuration Editor - D:\Eterra\Distribution\FantasyIsland\config\client\OmsClientConfig.xml'. The 'General' tab is selected, with other tabs like 'Grid Tabulars', 'Geo Outage Map', 'Validation Rules', 'Customers', 'Customer Notifications', 'Call Priority', 'Filtering', and 'Plugins' visible. The 'General Colors' section includes color pickers for 'Standard Pen' (yellow), 'Outage Boundary' (black), 'Operating Center' (yellow), 'Background' (black), 'City Label' (black), 'Tooltip Default Background' (yellow), and 'Tooltip Incident Background' (orange). The 'CaseNotes ETR Colors' section includes 'Geographic ETR Expiring Background' (yellow) and 'Geographic ETR Expired Background' (red). The 'Nav Panel Colors' section includes 'Nav Panel Background' (dark blue) and 'Nav Panel Text' (white). The 'Geographic Properties' section has 'Geographic Name' set to 'DETAIL' and 'Viewport Title' set to 'Outage Information'. The 'Incident Settings' section includes checkboxes for 'Enable Incident Locking' (checked), 'Enable Assign Non-Specific Crew' (checked), and 'Enable Confirm Scada Device' (unchecked), along with 'Disable non-MDT crews' (unchecked), 'Application Type' set to 'OMS', 'Scada Control Popup Name' set to 'fantasyisland_internals_scada_control_popup.fgpopup', and 'ETR increase/decrease step, in minutes' set to '30'. The 'Bitmap Section' has 'Bitmap Path' set to '%IDMS_ROOT%\bitmaps'. A status bar at the bottom indicates 'OMS client configuration loaded successfully from D:\Eterra\Distribution\FantasyIsland\config\client\OmsClientConfig.xml'.

Figure 2. General Tab

3.1. General Colors

Use the General Colors settings (shown in below image) to change the colors of different types of lines and tooltips in the Geographic view. You can choose the type of color to configure by selecting the appropriate square in the General Colors group.

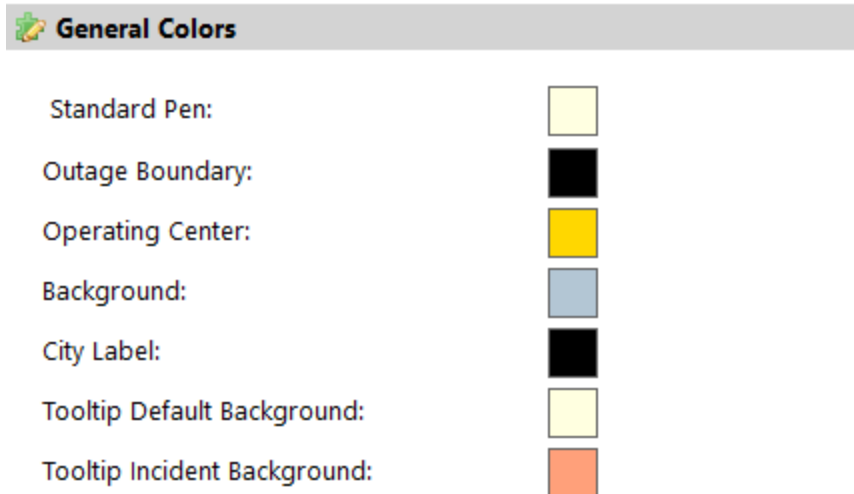


Figure 3. General Colors

When you click a color square, the Color window appears, enabling you to select the desired color.

3.2. CaseNotes ETR Colors

CaseNotes ETR assists in selecting a desired color to easily differentiate among Geographic Expected Time to Restore (ETR), Expiring Background Color, and Geographic ETR Expired Background Color.

Note

Expired ETR remains highlighted even after customers are restored.

3.3. Nav Panel Colors

In the Nav Panel Colors group, you can set the colors of the Navigation Panel background and text.

3.4. Geographic Properties

The Geographic Properties group includes the following fields:

- **Geographic Name:** Use this field to set the name for the Geographic Network view (for example, "DETAIL").
- **Viewport Title:** Use this field to set the title of the OMS viewport (for example, Outage Information – Viewport Station – ADMS).

3.5. Incident Settings

The Incident Settings group includes the following fields:

- **Enable Incident Locking:** Select this check box to prevent simultaneous access and editing of the same incident.
- **Disable Non-MDT Crews:** If you select this check box, the "Dispatch" and "Dispatch Evaluator" buttons on the Dispatch Crew To Incident display are disabled when crews that are not equipped with a Mobile Data Terminal (MDT) are selected.
- **Enable Assign Non-Specific Crew:** If you select this check box, the None item appears in the crew type menu in the Auto Dispatch section on the Crews tab of the Update Incident display. This enables WFM to dispatch a non-specific crew type to an incident. If this option is not selected, the crew type menu contains only items specified by the CrewType coded translation item.

Name	Crew Id	Crew Type	Work Status	Crew Area	Default Crew Area	Is Evaluator	CreatedBy	Idle	Work	Shift Status	Availability
Amoros...	FI2559	Distribution Sp...	Dispatched	West Cr...	West Cr...	☒	Yes	0	8	Unknown	Unf
Beaudet...	FI1522	Local Lineman	SignOn	Pebble ...	Pebble ...	☐	No	250	0	Unknown	Unf
Bertilso...	FI1571	Local Lineman	SignOn	3Lions D...	3Lions D...	☐	No	250	0	Unknown	Unf
Bertsch ...	FI2035	Crew Foreman	SignOn	West Cr...	West Cr...	☐	No	250	0	Unknown	Unf
Bradsha...	FI1975	Lead Lineman	SignOn	West Cr...	West Cr...	☐	No	250	0	Unknown	Unf
Brewer (...)	FI1604	Local Lineman	SignOn	Raven D...	Raven D...	☐	No	250	0	Unknown	Unf
Britten (...)	FI2036	Crew Foreman	SignOn	West Cr...	West Cr...	☐	No	250	0	Unknown	Unf

- **Application Type:** The application type. Used for custom fields configuration in the OMS Server Configuration Editor. The default value for the ADMS client is "OMS".
- **Enable Confirm SCADA Device:** Select this check box to show the Confirm Device State button on the Update Incident pop-up for SCADA devices.

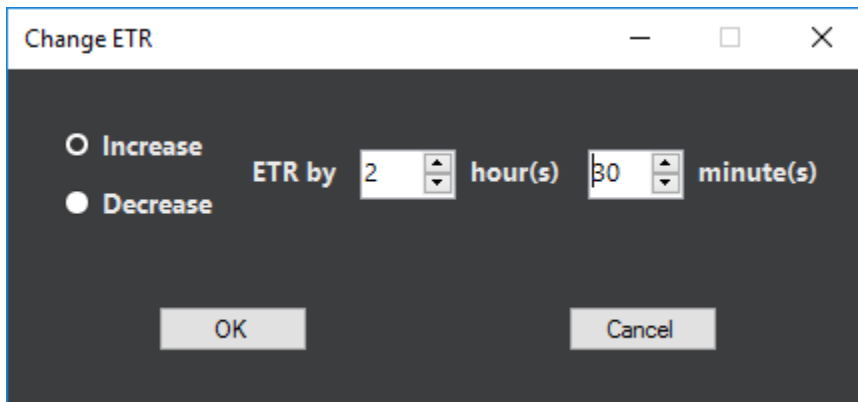
Closed Confirm Device Out

Note

SCADA devices cannot be controlled from the Update Incident pop-up. Clicking the Confirm

Device In and Out buttons opens the SCADA Control pop-up for the relevant device.

- **Enable Device Of Record On Split Incident:** Select this check box to show the Device Of Record popup for the Split Incident on the Restoration tab in the History Correction Update Incident pop-up.
- **SCADA Control Pop-Up Name:** The name of the SCADA Control pop-up. The default location of Scada displays is %IDMS_ROOT%\config\WebFGRoot\displays\scada.
- **ETR Increase/Decrease Step:** Specifies the magnitude of the ETR increase/decrease step on the Change ETR dialog box (in minutes).



3.6. Bitmap Section

The Bitmap Section group includes the following field:

- **Bitmap Path:** Path to the folder with bitmaps.

You can browse to the folder's location by clicking the Folder button.

4. Grid Tabular Tab

The Grid Tabulars tab shows the currently applied font sizes and styles of headers and fields.

Note

The data on the Grid Tabular tab is read-only. You can configure the grid tabular styles using the Net Dynamics Configuration Editor and Grid Tabular Builder. For more information, refer to the *Net Dynamics Configuration Editor User Guide* and *Grid Tabular Builder User Guide*.

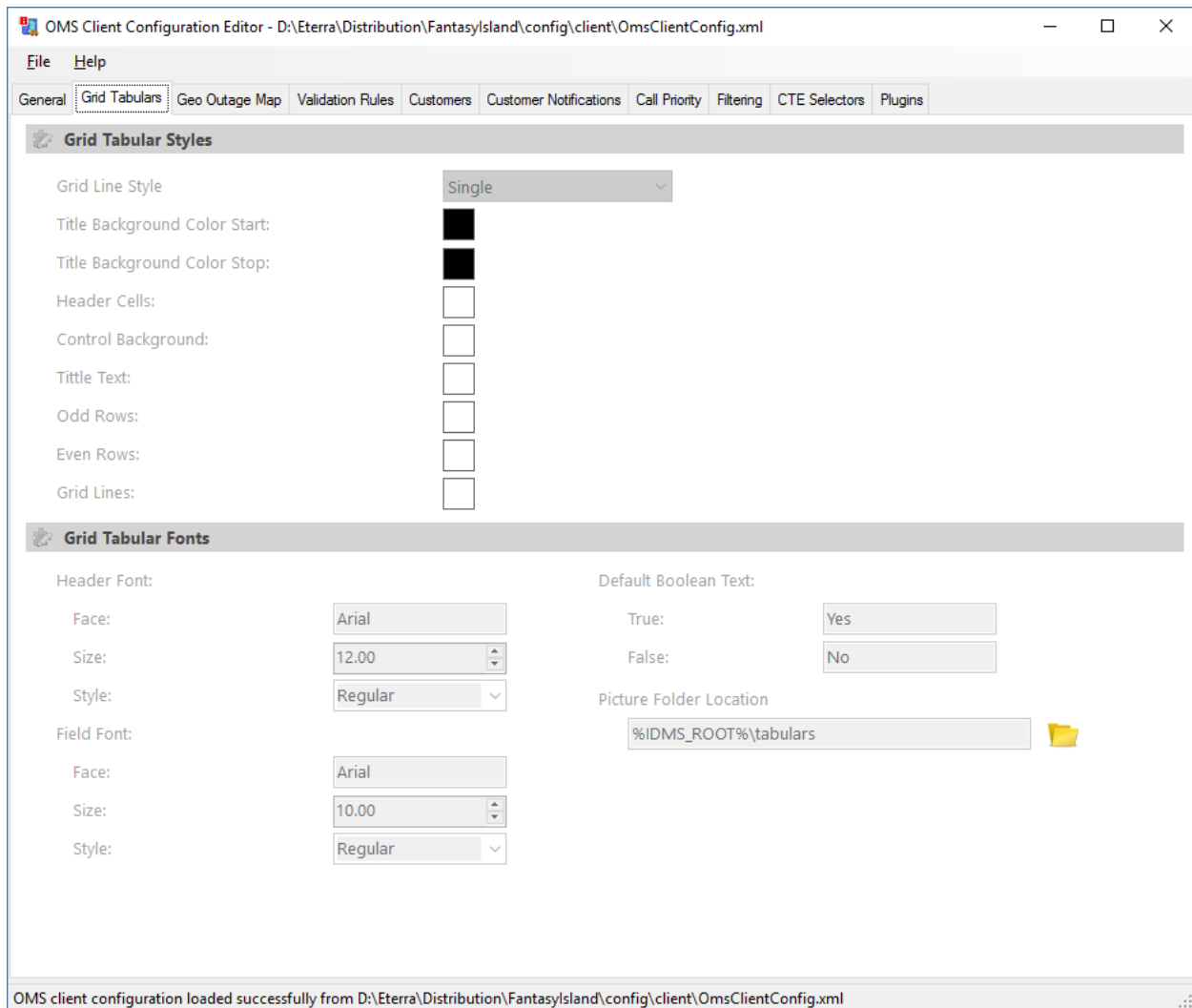


Figure 4. Grid Tabular Tab

4.1. Grid Tabular Styles

The Grid Tabular Styles group includes the following fields:

- **Grid Line Style:** The OMS grid tabular border style. Can be one of the following:

- Custom: For future use.
- Single: A single line border.
- Raised: A three-dimensional raised border.
- Sunken: A three-dimensional sunken border.
- None: No borders.
- SingleVertical: A vertical single-line border.
- RaisedVertical: A vertical three-dimensional raised border.
- SunkenVertical: A vertical three-dimensional sunken border.
- SingleHorizontal: A horizontal single-line border.
- RaisedHorizontal: A horizontal three-dimensional raised border.
- SunkenHorizontal: A horizontal three-dimensional sunken border.
- **Title Background Color Start/Stop:** The start and end color for the display's title background gradient. The title gradient is painted left-to-right using the specified colors.
- **Header Cells Color:** The color of the header cells for the OMS grid tabulars.
- **Control Background Color:** The basic background color for the OMS grid tabular.
- **Title Text Color:** The display's title color (black on default).
- **Odd Rows Color:** The color for odd rows.
- **Even Rows Color:** The color for even rows.
- **Grid Lines Color:** The color of the OMS grid lines.

4.2. Grid Tabular Fonts

The Grid Tabular Fonts group includes the following fields:

- **Header Font Face:** The font used for grid tabular headers.
- **Header Font Size:** The font size ranging from 0 to 100.
- **Header Font Style:** The font style (Bold, Italic, Regular, or Underline).
- **Field Font Face:** The font used for grid tabular fields.
- **Field Font Size:** The font size ranging from 0 to 100.
- **Field Font Style:** The font style (Bold, Italic, Regular, or Underline).
- **Default Boolean Text:** Default values for True and False flags.
- **Picture Folder Location:** The path to the tabulars folder.

5. Geo Outage Map Tab

The Geo Outage Map represents the footprint of each substation (i.e., the geographic extent of each feeder that is normally supplied from the substation).

The Geo Outage Map tab contains settings for configuring the appearance of polygons in different views on the OMS Geographic Overview (such as Feeder, Station, Zip Code, and Service Center).

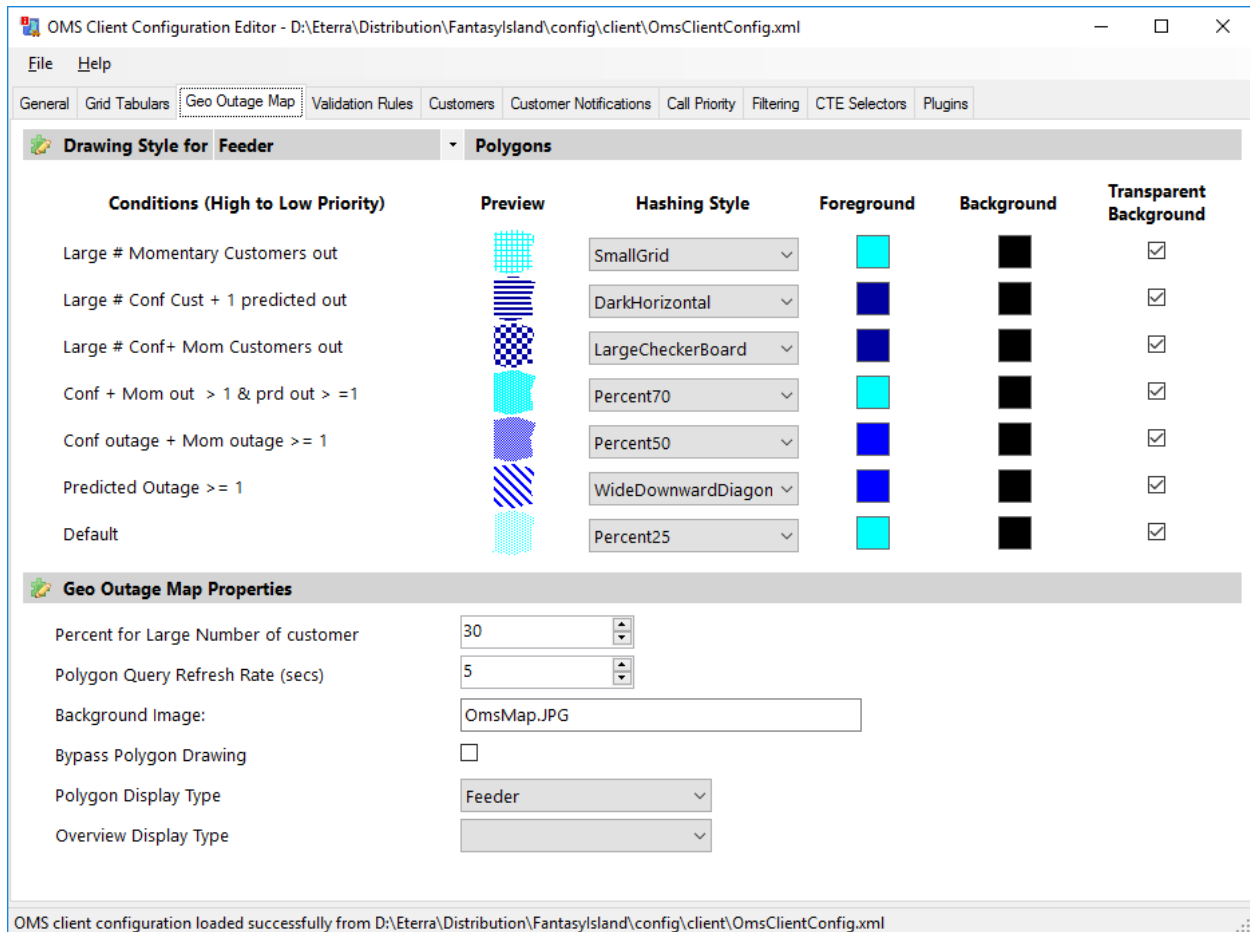


Figure 5. Geo Outage Map Tab

5.1. Drawing Style

Select polygons types for review or modification from the Drawing Style drop-down list. There are five types available:

- Feeder
- Station
- ZipCode
- OpCenter (Operating Center)

- Division

5.2. Conditions Higher to Lower Priority

Large # Momentary Customers Out: Sets the appearance of a polygon where a large number of momentary customer outages exists.

Large # Conf Cust + 1 Predicted Out: Sets the appearance of a polygon where a large number of confirmed customer outages and one predicted outage exist.

Large # Conf + Mom Customers Out: Sets the appearance of a polygon where a large number of confirmed outages and momentary customer outages exist.

Conf + Mom Out > 1 & Prd Out > = 1: Sets the appearance of a polygon where a number of confirmed outages and momentary outages is greater than one, and where a number of predicted outages is greater than or equal to one.

Conf Outage + Mom Outage > = 1: Sets the appearance of a polygon where a number of confirmed outages and momentary outages is greater than or equal to one.

Predicted Outage > = 1: Sets the appearance of a polygon where a number of predicted outages is greater than or equal to one.

Default: Sets the default appearance of a polygon where none of the conditions listed above exists.

5.3. Preview

Depending on the type of style, and the foreground and background colors selected, you can see a sample preview of the image to be displayed.

When you select a style in the Hashing Style list box for each condition, you can see the image in the square box of the Preview column change to the selected hashing style.

5.4. Hashing Style

You can select the desired style from the list box in the HashingStyle column.

5.5. Foreground

You can choose the desired color to be displayed in the foreground of the OMS by selecting the corresponding square image in the Foreground column.

5.6. Background

You can choose the desired color to be displayed in the background of the OMS by selecting the corresponding square image in the Background column.

5.7. Transparent Background

If you want the background color to be transparent, select the corresponding check box in the Transparent Background column.

5.8. Geo Outage Map Properties

Percent for Large Number of Customer: Use this field to set a value in percent for what the system considers a large number of customers. This value is used to calculate if the given condition exists and to change the polygons' appearance accordingly.

Polygon Query Refresh Rate: Set a time interval in seconds for refreshing the data on the Polygon panel of the OMS Summary display.

Background Image: Set the path of the background geographic map file. By default, the map image is stored in the Bitmaps folder. You can specify either the full path or just the file name.

Bypass Polygon Drawing: If selected, the Area World View polygons and toolbar do not appear on the OMS Summary display.

Polygon/Overview Display Type: To see a sample preview of the images to be displayed when certain conditions exist for this display type, select a display type for polygons or the overview.

6. Validation Rules Tab

Use the Validation Rules tab to specify the callback number format.

Note

As of ADMS 3.13, the incident clearance validation rules are configured in the OMS Server Configuration Editor.

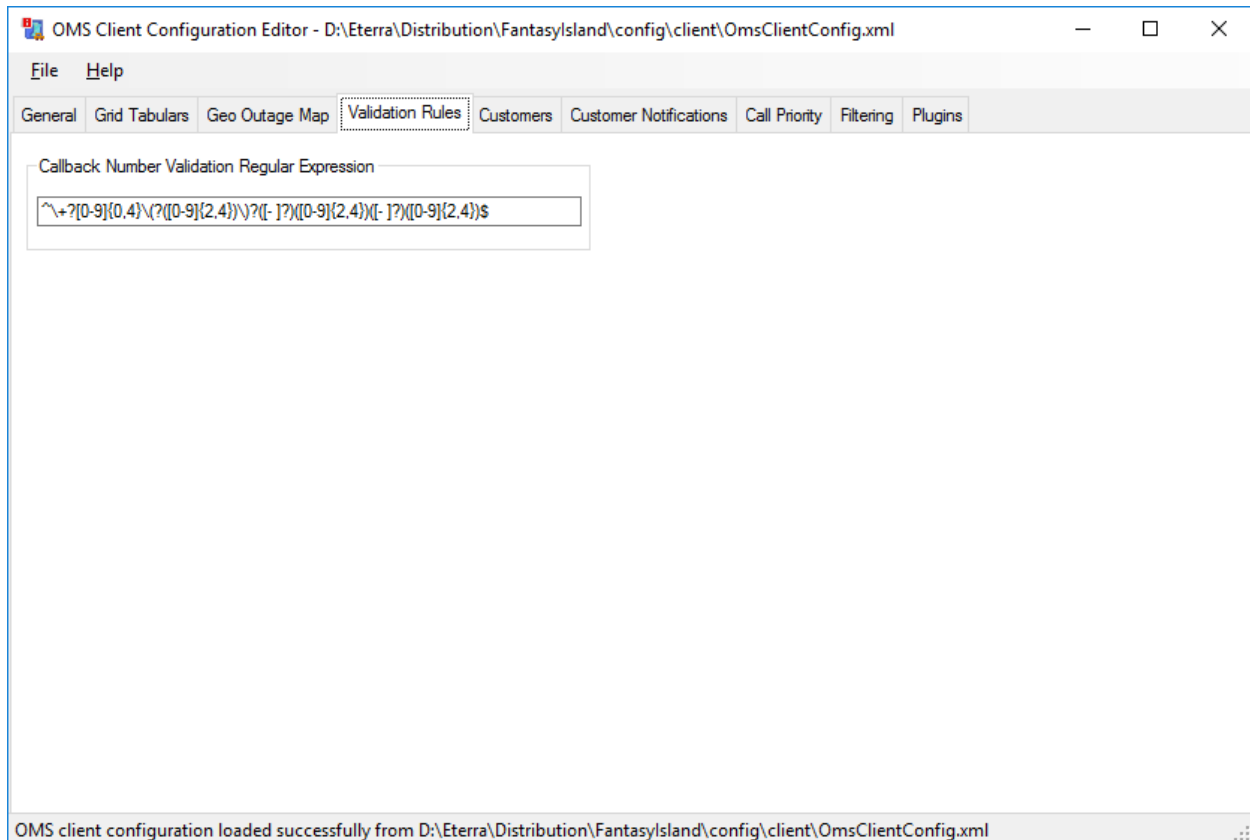


Figure 6. Validation Rules Tab

The Callback Number Validation Regular Expression field defines the callback number format on the Add/Update Customer Call display.

- **Callback Number Validation Regular Expression:** Specifies the validation pattern (in the form of a regular expression) that is used to validate the Callback Number field on the Add/Update Customer Call display.

The default regular expression (RegExp) pattern is:

```
"^\\+?[0-9]{0,4}\\(?([0-9]{2,4})\\)?([- ]?)([0-9]{2,4})([- ]?)([0-9]{2,4})$".
```

It supports the following phone numbers formats:

-

- + (123) 456 7899
- +2(123) 456 7899
- +23(123) 456 7899
- (123)-456-7899
- (555)5553444
- (555)5553-444
- 123-456-7899
- 123 456 7899
- 1234567899
- +123(456)7890

 Caution

Modifying the validation pattern requires experience with regular expressions.

7. Customers Tab

Previously, the Customers tab was used to add, remove, or change the order of columns in the Customer List tab on the Planned Outage display. These configurations are now deprecated. Use the TracedCLoutage_GridDisplay.xml file to modify the Traced Customer List grid tabular display.

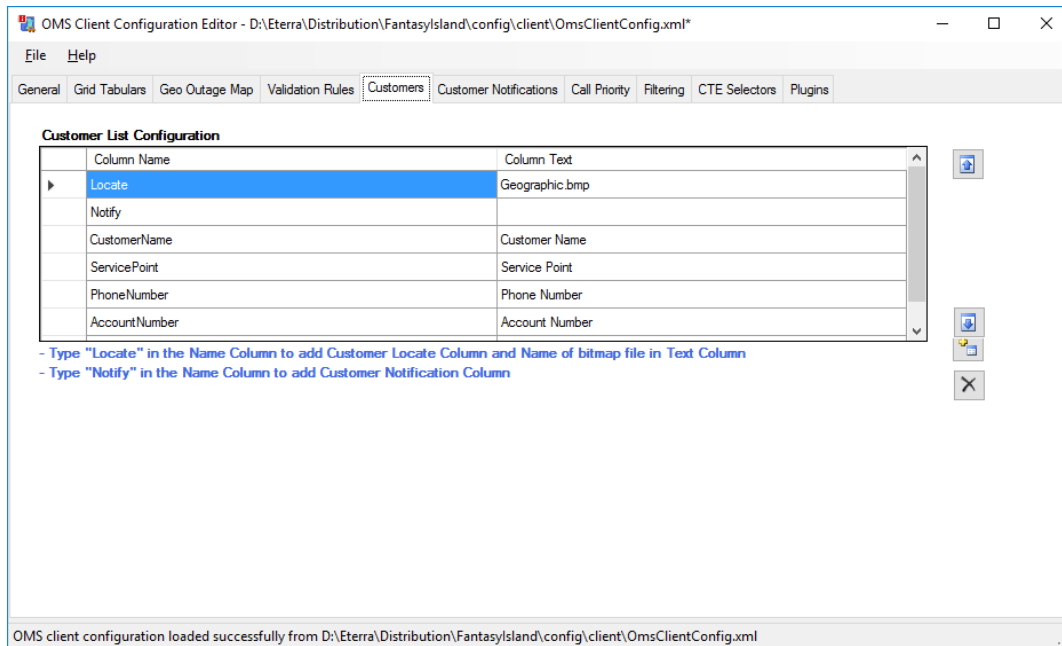


Figure 7. Customers Tab

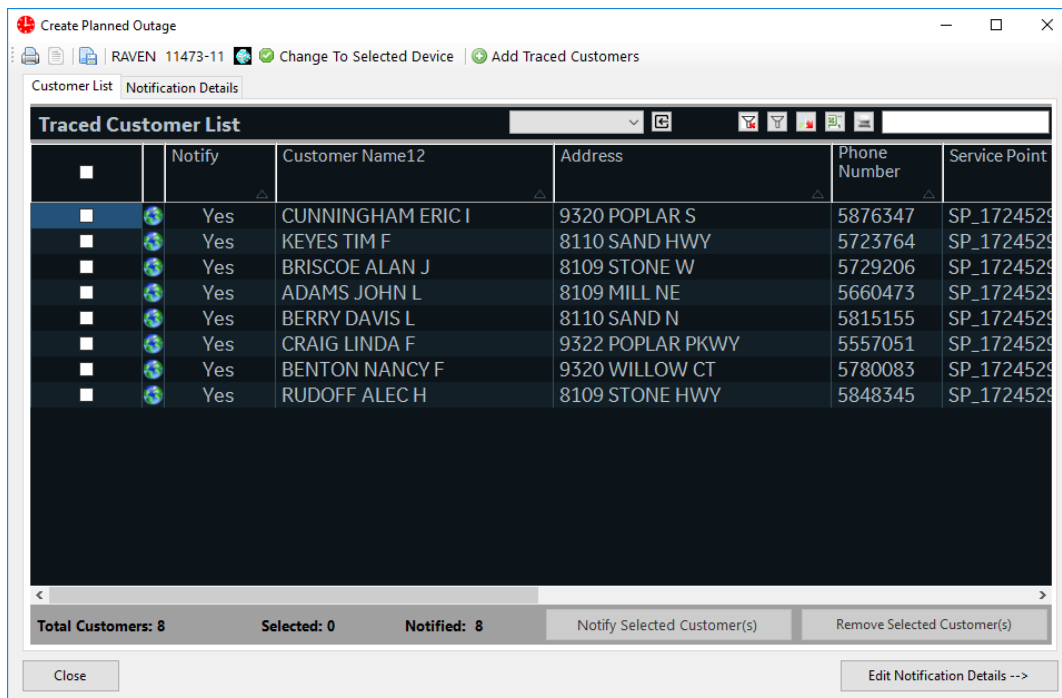


Figure 8. Planned Outage Display - Customer List Tab

8. Customer Notifications Tab

Use the Customer Notifications tab to configure planned outage settings and define a default notification message for each planned outage event type.

Events List

- replacing meters
- replacing transformers
- replacing poles

☒ Event Causes Outage

Event Type

replacing meters

Add Remove

Notification Message

Dear Customer, Our company will be %EventType% in your area on %StartDate%, from %StartTime% to %EndTime%. We estimate the duration of this planned outage to be %DurationHours% hours. We regret any inconvenience this may cause and thank you for your cooperation. If you need to discuss this with us, contact %ContactName% at %ContactPhone%.

Supported Templates - Substituted at Client Run-time

Template Name	Template Value
%EventType%	Event Type
%StartTime%	Event Start Time
%StartTimeAlt%	Event Start Time Alternate
%StartDate%	Event Start Date
%StartDateAlt%	Event Start Date Alternate
%EndTime%	Event End Time

Minimum Number of Days between Notification Creation Date and Planned Outage Date 0 **Minimum Days Start Time** 1:00 PM

Maximum Number of Days between Notification Creation Date and Planned Outage Date 28 ☐ Unlimited

Enable sending customer callbacks in case of ETR changing ☒

If ETR changes by X minutes, prompt operator to send callbacks to called customers 30 **From** 8:30 AM **To** 8:30 PM

Minimum outage duration in minutes 5

Notification Output Folder %IDMS_ROOT%\misc\output\USPS.xml

Notification Application Path %windir%\system32\notepad.exe

Notification Application Command Line Parameters %IDMS_ROOT%\misc\output\USPS.xml

Allow Full Edit On Update ☒

Enable Contact Info Validation ☒

Show Work Order Id ☒

Show Planned Outage XML ☐

OMS client configuration loaded successfully from D:\Eterra\Distribution\FantasyIsland\config\client\OmsClientConfig.xml

Figure 9. Customer Notifications Tab

The Customer Notifications tab includes the following fields:

- **Events List:** The list of available event types. You can add a new event or remove an existing event using the Event Type text box and the Add/Remove buttons.

- **Event Causes Outage:** Select this check box if the selected event type causes an outage.
- **Event Type:** The name of the new event type. To add a new event, enter the event name in this text box and click Add.
- **Notification Message:** Use this text box to create a default notification message for the selected event type. You can use available templates from the Supported Templates section. The templates are substituted with the data at client run time.
- **Supported Templates:** Shows the available templates. A template included in the notification message is replaced with corresponding value. For example, %StartDate% is substituted with the specified event's start date, such as Wednesday, October 29, 2014.
- **Minimum Number of Days Between Notification Creation Date and Planned Outage Date:** Sets the minimum time in days between the notification creation date and the planned outage start date.

For example, to notify customers at least two days before the planned outage date, enter the "2" value.

- **Minimum Days Start Time:** Sets the time of the day from which the minimum number of days between the notification creation date and the planned outage date is counted.
- **Maximum Number of Days Between Notification Creation Date and Planned Outage Date:** Sets the maximum time in days between the notification creation date and the planned outage start date.

For example, to prevent notifying customers more than a month before the planned outage date, enter the value as "30".

- **Unlimited:** Select this option to disable the Maximum Number of Days Between Notification Creation Date and Planned Outage Date option. In this case, operators can select any future date above the Minimum start date as the planned outage start date.
- **Enable Sending Customer Callbacks in Case of ETR Changing:** Select this option to prompt the operator to send callbacks to the called customers if ETR changes by the specified value.
- **If ETR Changes by X Minutes, Prompt Operator to Send Callbacks to Called Customers:** Sets the value in minutes for the Estimated Time to Restore parameter on the Update Incident pop-up. If the ETR changes by this value (for example, it takes X more minutes to restore the power), the system prompts the operator to send callbacks to the called customers.
- **From/To:** Sets the time frame to make callbacks to customers about the ETR changes.
- **Minimum Outage Duration in Minutes:** Sets the minimum outage duration (in minutes) for the planned outage.
- **Allow Full Edit on Update:** Enables the operator to edit all fields of the planned outage during the update. If the full edit is disabled, the operator can only modify the recipients list and select between the scheduled and alternative date and time.
- **Enable Contact Info Validation:** Enables validation of the company contact information on the Create Planned Outage display. If validation is enabled, the Name, Phone, Email, and Mailing Address fields cannot be blank.

- **Show Work Order ID:** Shows or hides the Work Order ID field on the Create Planned Outage display.
- **Show Planned Outage XML:** Select this option to show the USPS.xml file after clicking the Send Notification button on the Create Planned Outage display. The USPS.xml file contains data and outage message details for the selected customers.
- **Show SWO Link:** Select this option to show the Switching Orders section on the Notification Details tab of the Create Planned Outage display.

To specify which application should process the output notification file once it has been created, three paths need to be provided:

- **Notification Output File:** This is the location (folder path and file name) of the notification XML message.
- **Notification Application Path:** This is the name of the application (folder path and file executable name) that should process the notification XML message.
- **Notification Application Command-Line Parameters:** These are optional parameters that can be used by the application that processes the notification XML message.

Create Planned Outage

RAVEN 22056203 | Change To Selected Device | Add Traced Customers

Customer List | Notification Details | Custom Fields

General Info

ID: [] Status: Active

Company Contact Info

Name: Test User
 Phone: 425-555-0112
 Email: TestUser@ge.com
 Service Center: 3LIONSSC
 Mailing Address: 111 Main St. City NL 12345

Event Type

Event Type: replacing meters
 Planned Outage Type: Regular Planned Outage
☒ Causes Outage

Creation Date

10/26/2021
☐ Override Mailing Date
 Work Order ID: FI1947P002

Scheduled Event Date and Time

Start Time: 10/27/2021 12:00 AM
 End Time: 10/27/2021 12:05 AM
 Duration (h:m): 00:05

Alternate Event Date and Time

Start Time: 10/28/2021 12:00 AM
 End Time: 10/28/2021 12:05 AM
 Duration (h:m): 00:05 ☐ Use Alternate Date and Time

Switching Order

↻

Notification Status

Due Date: 10/26/2021 2:48 PM

Creator Comments

[]

Notification Message Preview | Edit Message

Dear Customer, Our company will be replacing meters in your area on Wednesday, October 27, 2021, from 12:00 AM to 12:05 AM. We estimate the duration of this planned outage to be 00:05 hours. We regret any inconvenience this may cause and thank you for your cooperation. If you need to discuss this with us, contact Test User at 425-555-0112.

Close | Send Notification

Figure 10. Create Planned Outage Display - Notification Details Tab

8.1. Creating Events and Adding Default Messages

To create a default notification for a planned outage event:

1. Enter the event type (for example, replacing wires) in the Event Type field.
2. Click the Add button.

The event type appears in the event list.

3. Set the Event Causes Outage flag, if needed.
4. Enter text in the Notification Message field.

5. Copy templates from the Supported Templates list, and then paste where needed.

For example, use %StartTime% to indicate the start time of the planned outage. This template is substituted with values at client run time.

6. Set other values for the selected event type (for example, minimum number of days between notification creation date and scheduled outage date).
7. Save the changes via the File/Save menu.

The changes are saved in the `OMSClientConfig.xml` file.

9. Filtering Tab

Use the Filtering tab to configure filtering options for fields on the Update Incident display.

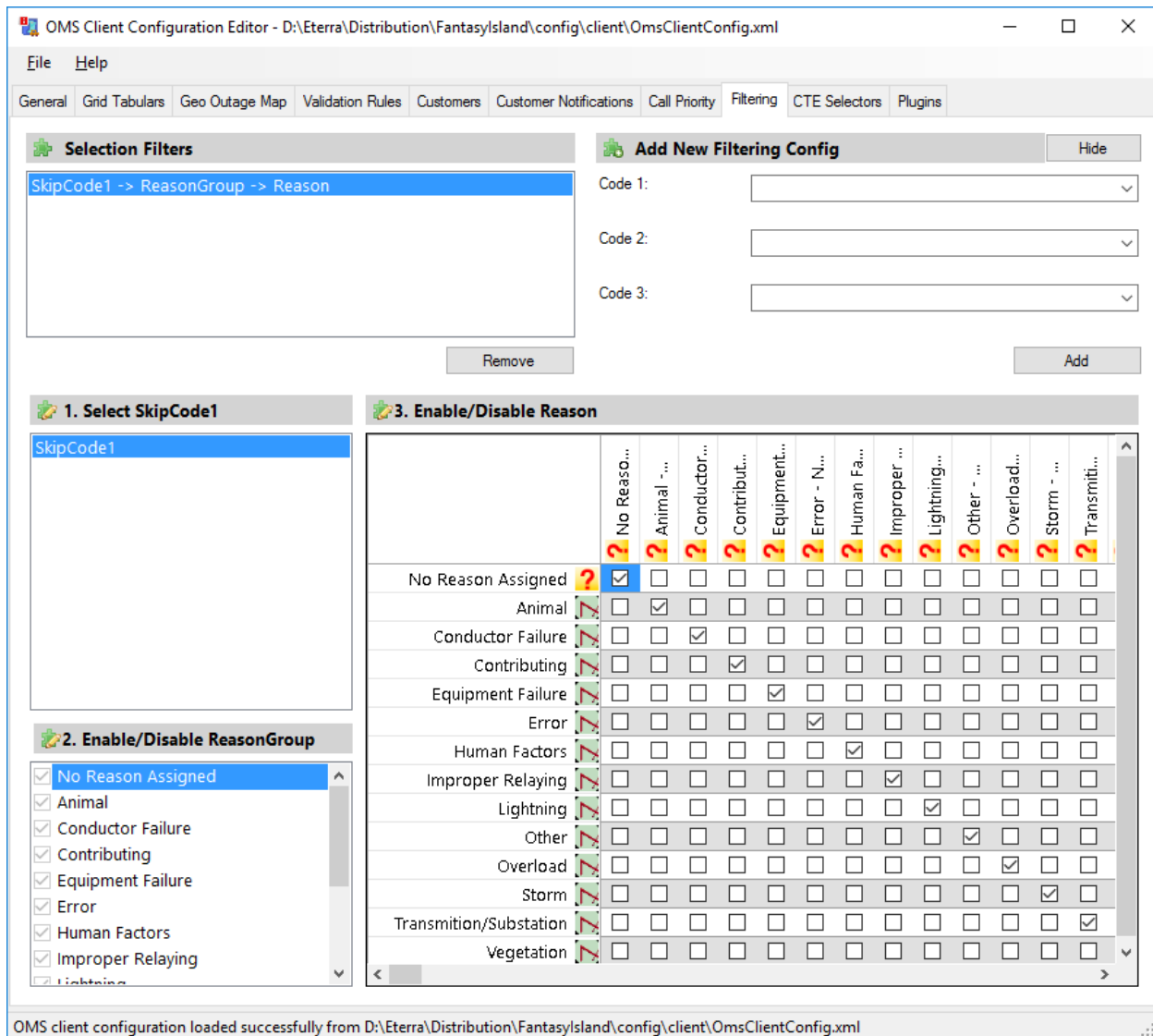


Figure 11. Filtering Tab

The Filtering tab includes the following fields:

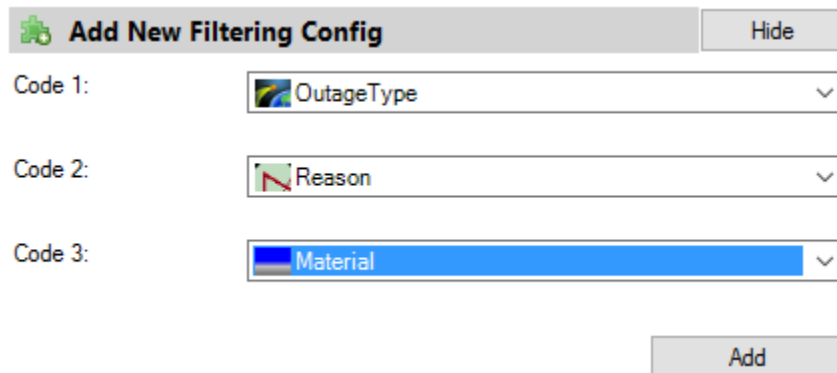
- **Selection Filters:** Lists new filtering configurations.
- **Add Filtering Config:** Contains three drop-down lists with a choice of coded translation items available for Code 1, Code 2, and Code 3.
- **1. Select Code:** Contains a list of options available for the selected Code 1 item.
- **2. Enable/Disable Codes:** Contains Code 2 filtering options for the selected Code 1 item.
- **3. Enable/Disable Codes:** Contains Code 3 filtering options available for the selected Code 1 and Code 2 items.

- **Show/Hide:** Click to show or hide the filtering code configuration options.
- **Add:** Click to add new filtering configuration.
- **Remove:** Click to remove the selected filtering configuration.

9.1. Adding a Filtering Configuration with Three Codes

To add a new filtering configuration for three types of codes:

1. In the Add New Filtering Config field, choose the coded translation items to be filtered via the Code 1, Code 2, and Code 3 drop-down lists.

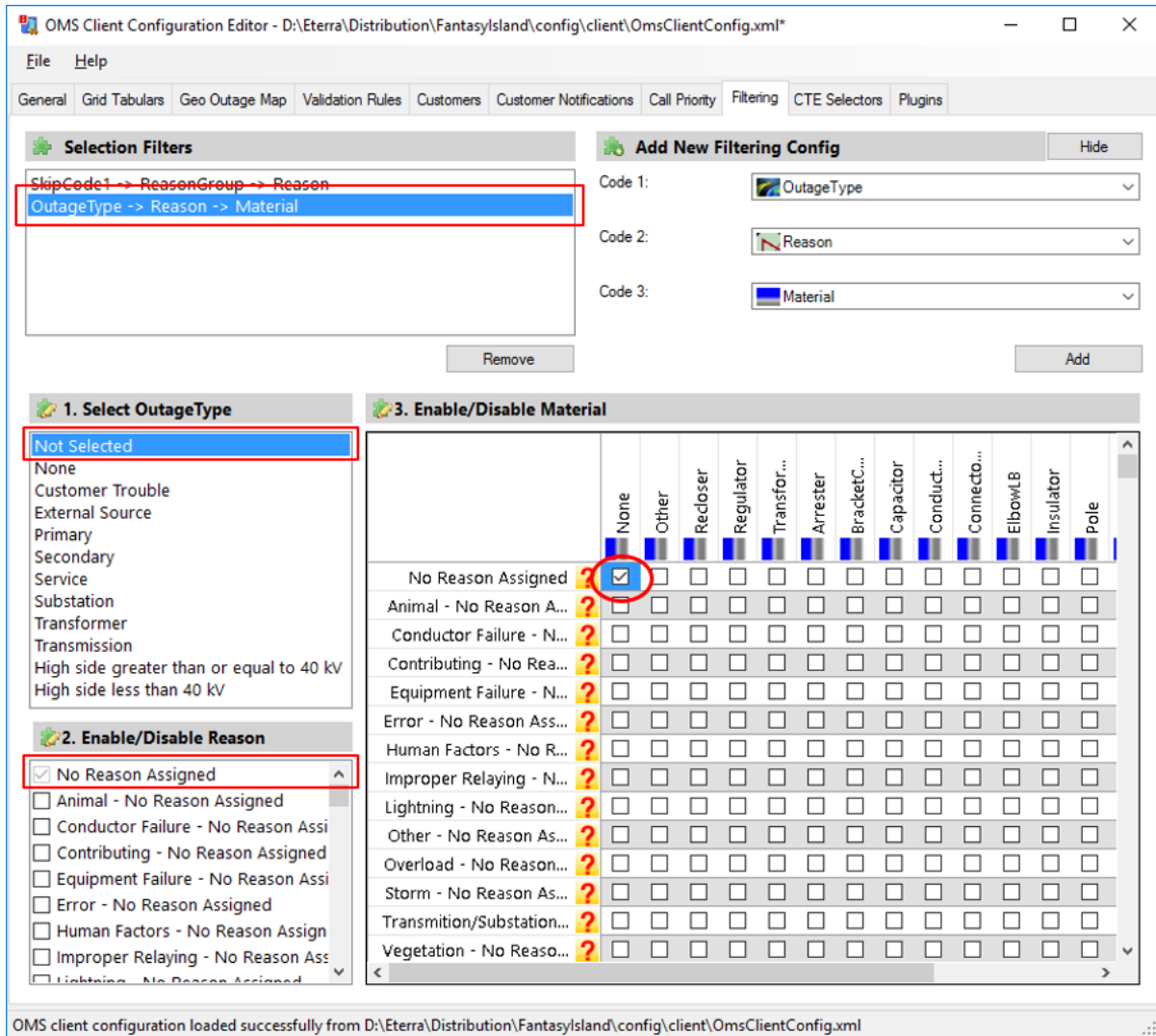


The screenshot shows a dialog box titled "Add New Filtering Config" with a "Hide" button in the top right corner. Below the title bar, there are three drop-down menus labeled "Code 1:", "Code 2:", and "Code 3:". The "Code 1:" menu is set to "OutageType", the "Code 2:" menu is set to "Reason", and the "Code 3:" menu is set to "Material". At the bottom right of the dialog is an "Add" button.

2. Click Add. The Selection Filters group is populated with the new filtering configuration. For this example, the following filtering options were used:
 - Code 1: Outage Type
 - Code 2: Reason
 - Code 3: Material
3. In the Selection Filters group, select the added filtering configuration. Fields 1, 2, and 3 are populated with options relevant to the selected coded translation items.
4. For each outage type in Field 1, select the materials that will be available in the OMS client and the reasons that will be available for each material.

For this example:

- Field 1: Select the Not Selected outage type.
 - Field 2: Disable all reasons except No Reason Assigned. Only this reason will be available when the Not Selected outage type is selected.
 - Field 3: Disable all materials except None. Only this material will be available when the Not Selected outage type is selected.
5. Save the configuration.



For this configuration, the Cause (Reason) drop-down list is filtered to only show the No Reason Assigned reason type, and the Material drop-down list is filtered to only show None for the Not Selected outage type, as shown in below image.

Update Incident: 1550000001 - Currently locking the incident

Properties Crews Customers Switching Restoration Chronology

Incident [1550000001](#)

Restoration

Outage Type: Not Selected

Action Taken: None

Material: None

Work Order Id:

Power Quality Type: Momentary Outages

Non-Outage Type: None

Weather: Calm

Failure Mode: Tree falls, dead

Manufacturer: 3-M Company

RestorationDetails

Step Number	Station Name	DeviceName	Device Id	Service Center	Phase	De-energized Time
1	RAVEN	52721	22233938	3LIONSSC	ABC	6/4/2018 5:00:17 AM

Figure 12. Filtering Configuration with Three Codes

9.2. Adding a Filtering Configuration with Two Codes

To add a new filtering configuration for only two types of codes:

1. In the Add New Filtering Config group, for the Code 1 field, select the SkipCode1 coded translation item.

Add New Filtering Config Hide

Code 1: SkipCode1

Code 2: ReasonGroup

Code 3: Reason

Add

2. For the Code 2 and Code 3 fields, select the coded translation items to filter.
3. Click Add.

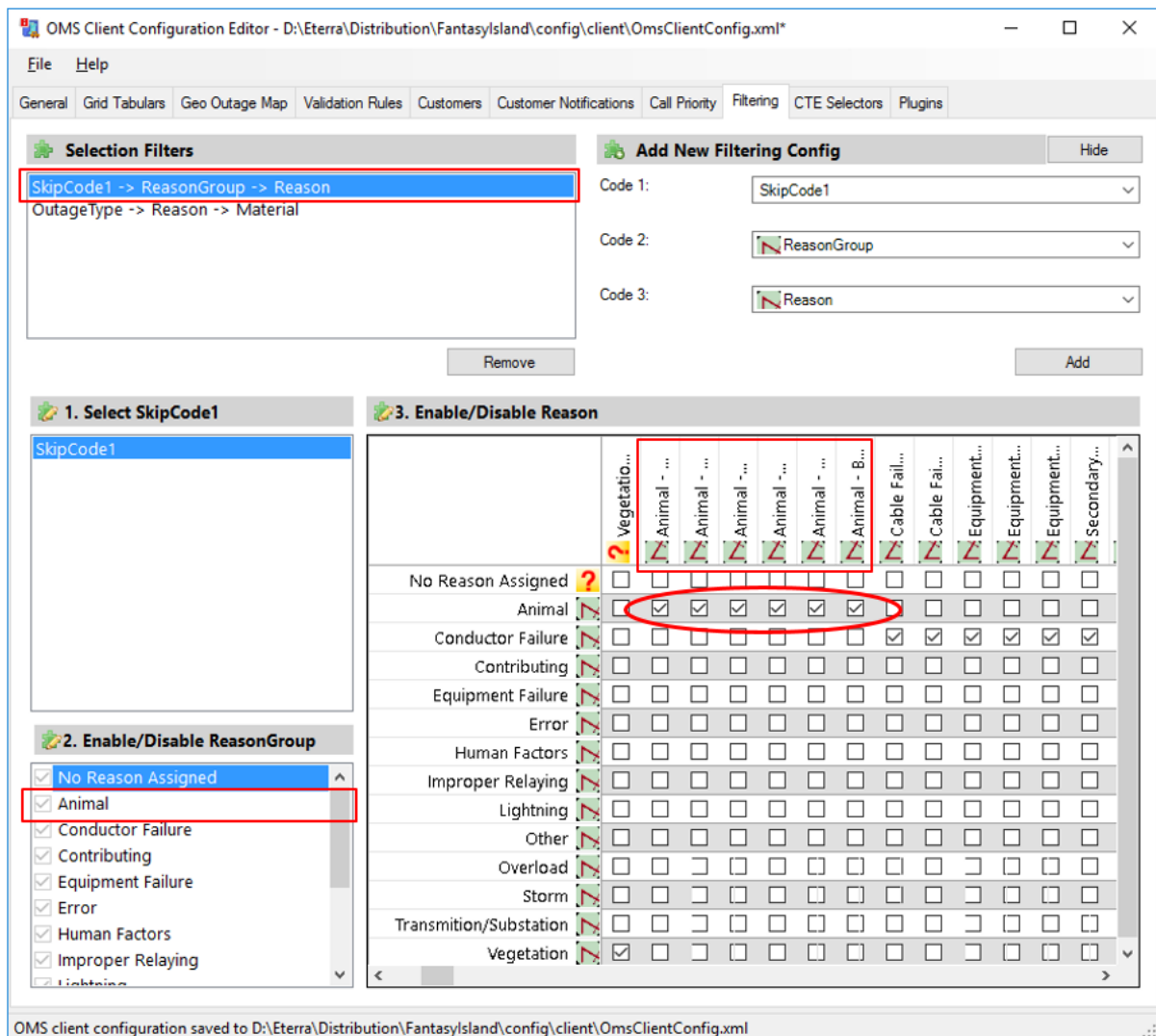
The Selection Filters group is populated with the new filtering configuration. For this example, the filtering options are:

- Code 1: SkipCode1
- Code 2: ReasonGroup

- Code 3: Reason
4. In the Selection Filters group, select the added filtering configuration.

Field 1 is populated with the SkipCode1 value, which is not applied for filtering. Fields 2 and 3 are populated with options relevant to the selected ReasonGroup and Reason coded translation items.
 5. In Field 3, for each outage reason in Field 2, select the action items that will be available in the Update Incident display.

For example, in Field 3, select only animal reasons for the "Animal" reason group.



6. Save the configuration.

For this configuration, the Sub-Cause drop-down list is filtered to show only animal sub-causes (reasons) for the "Animal" cause (reason group), as shown in below image.

Update Incident: 1550000001 - Currently locking the incident

Properties Crews Customers Switching Restoration Chronology

Incident 1550000001

CMI	0	Start Time	06/04/2018	05:46 AM
Calls	0	End Time	mm/dd/yyyy	--:-- --
Priority Calls	0	First Call	mm/dd/yyyy	--:-- --
Affected	3012	ITR	06/04/2018	08:00 AM
Current	3012	ETR	mm/dd/yyyy	--:-- --
Incident Type	Confirmed	ETR Sequence Number	0	
Priority	None	Callback	Automatic	
Cause	Animal	Working Status	Incident is still unassigned	
Sub-Cause	No Reason Assigned No Reason Assigned Bird Non-Reportable Incident Bird Reportable Incident (Mortality or Injury) Other (Describe in Remarks Field) Raccoon Snake Squirrel	Case Note	Outage Recorded	
LineSwitch 51		City	ROARKE	
Feeder		Service Center	3LIONSSC	
Phase		Substation	RAVEN	
Responsible System	Overhead Distribution			

Clear Incident Cancel Incident Ok Cancel Apply

Figure 13. Filtering Configuration with Two Codes

10. Plugins Tab

The Plugins tab allows the analyst to modify the OMS Display Plugin settings.

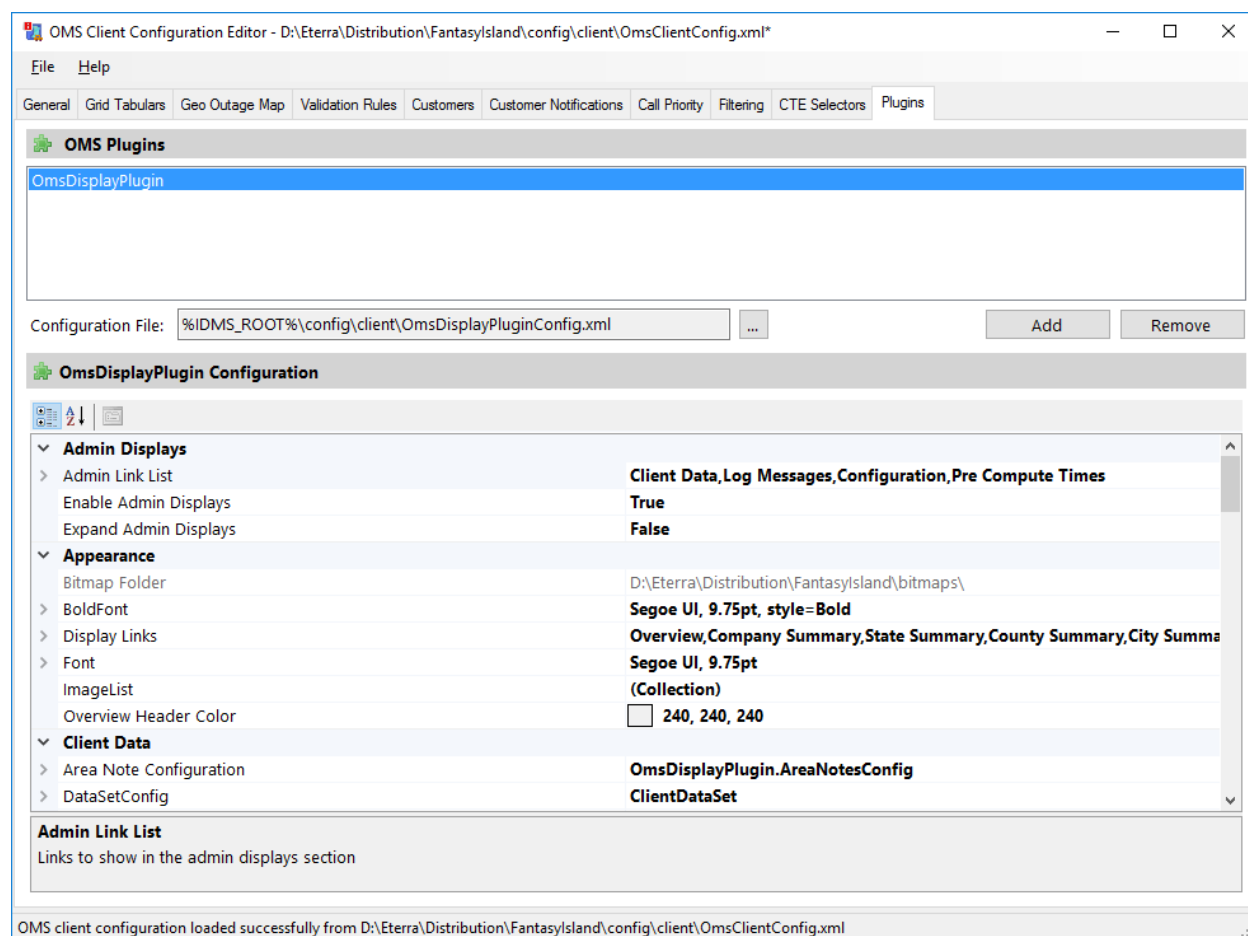


Figure 14. Plugins Tab

Click OmsDisplayPlugin in the OMS Plugins field to view the configuration controls in the Plugin Configuration panel.

If OmsDisplayPlugin is not present in the OMS Plugins field:

1. Click Add.

The Select the Plugin DLL dialog box opens.

2. Select OmsDisplayPlugin.dll and click Open.

A prompt to open the OMSDisplayPlugin XML file appears.

3. Select OmsDisplayPluginConfig.xml and click Open.

The OmsDisplayPlugin appears in the OMS Plugins field.

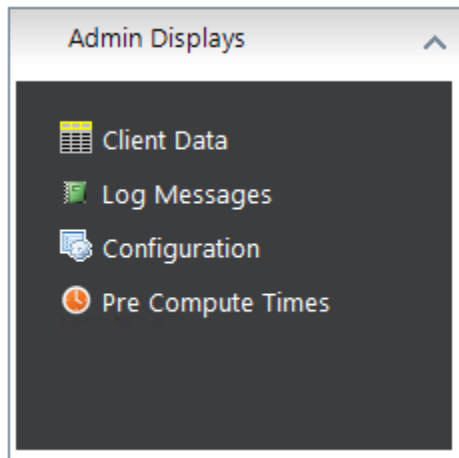
4. Click OmsDisplayPlugin.

The OMS Plugin Configuration settings are available in the Plugin Configuration panel.

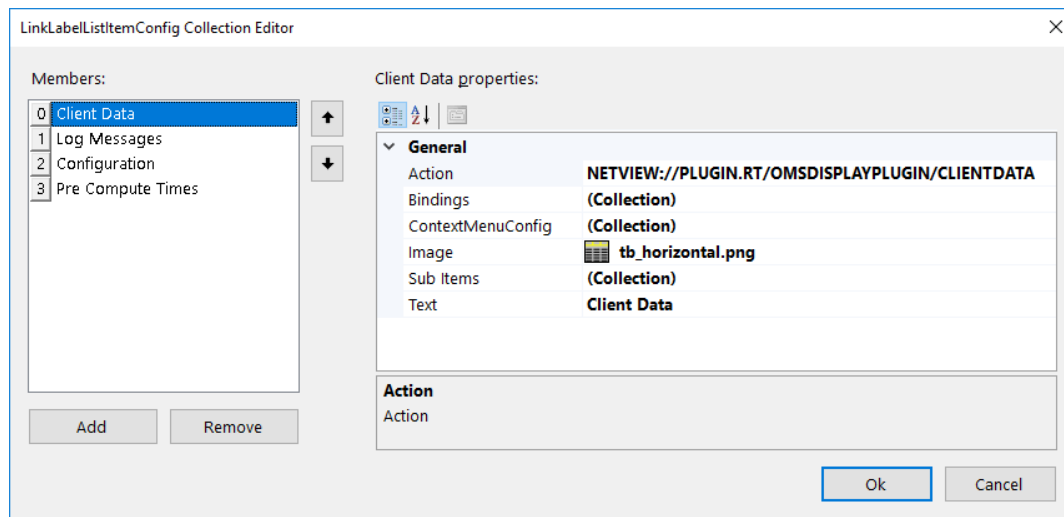
10.1. Admin Displays

▼ Admin Displays	
▼ Admin Link List	Client Data, Log Messages, Configuration, Pre Compute Times
Items	(Collection)
Enable Admin Displays	True
Expand Admin Displays	False

Admin Link List: Specifies links to show in the Admin Displays section in the navigation pane of the OMS Display Plugin.



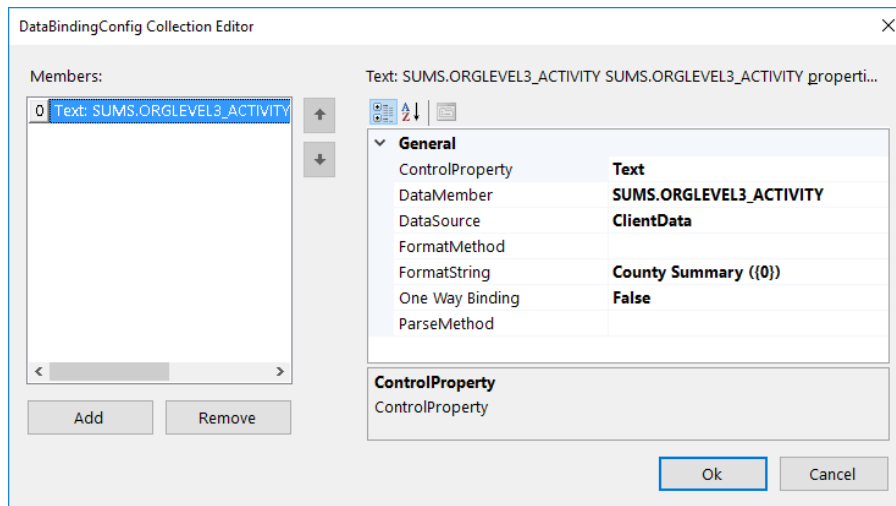
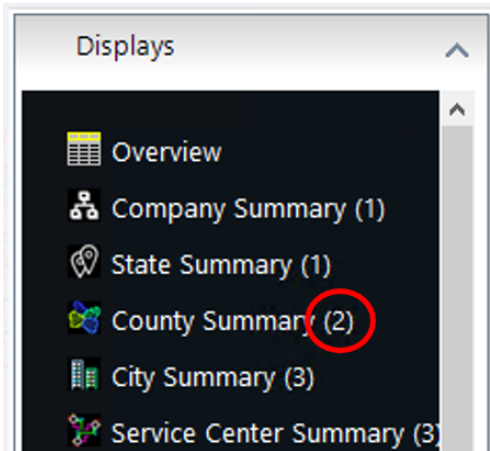
To review and configure the links, select the Items sub-element, and click the ellipsis  button at the right. This opens the LinkLabelListItemConfig Collection Editor dialog box.



- **Members:** Includes the list of links to be visible in the Admin Displays section. The links can be added, removed, and moved up and down in the list using the corresponding buttons.
- **Link Properties:** For each link in the list, the options include:

- **Action:** URL to call when the link is clicked. The URL format is `NETVIEW://PLUGIN.RT/OMSDISPLAYPLUGIN/<TabID>`, where TabID is the tab's ID, which is specified in the ID parameter of the General> TabDisplays group.
- **Bindings:** Specifies data bindings that are assigned to the link. To open the DataBindingsConfig Collection Editor, click the ellipsis  button next to Bindings.

For example, you can add a counter next to a navigation link. In the example below, the binding formats "Text" to include the count from the Incidents_AOR table.



Each binding item has the following properties:

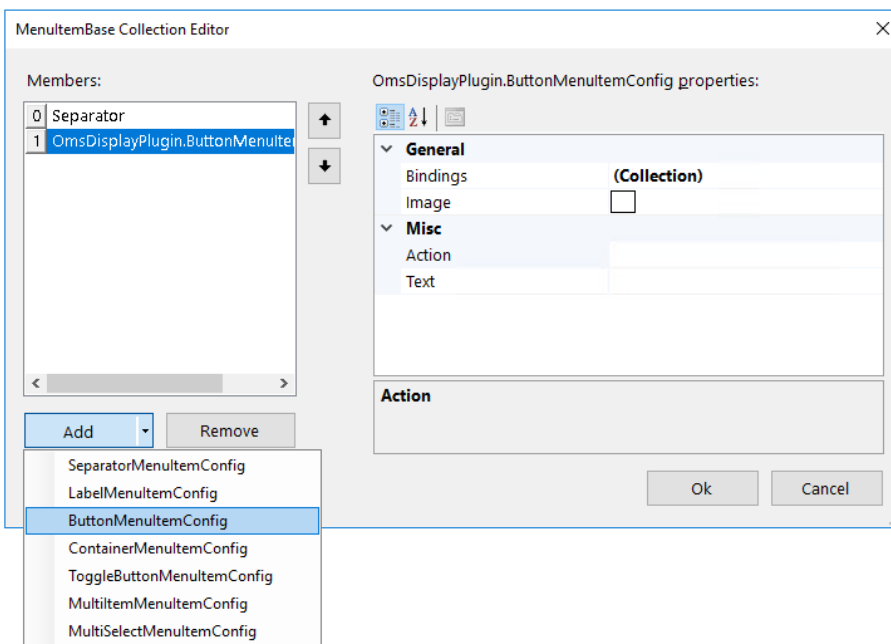
- **Control Property:** Control property type, such as Text or Checked.
- **Data Member:** Name of the item that is used by the binding.

To show the count of items in a table, use the SUMS<TableName> value.

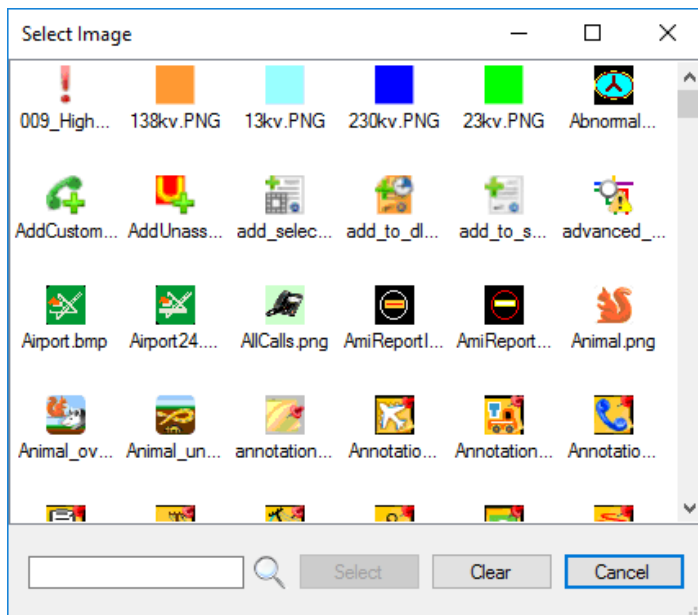
To show a filtered count of items in a table (for example, show the count of divisions with incidents only), define the filtering condition in the Client Data> Data Set Config> Table Counts field, and then use the SUMS<ColumnName> value.

To show elements on the geographic overview display when a button is clicked (for example, show incidents on the map), use the <ItemName>On value.

- **Data Source:** Name of the data source.
- **Format Method:** Format method.
- **Format String:** Format of the string. To include a counter, use the "{0}" item next to the string.
- **One Way Binding:** Specified whether this is a one-way binding.
- **Parse Method:** Parse method.
- **ContextMenuConfig:** Context menu configurations. Clicking the ellipsis  button opens the MenuItemBase Collection Editor dialog box, where you can configure the context menu.



- **Image:** The image file to display with the link and the image preview. Clicking the ellipsis  button opens the Select Image dialog box, which allows you to select an image from the bitmaps folder. The bitmaps folder is specified by the Bitmap Folder parameter in the Appearance group.



- **Sub Items:** Related links that will be shown directly below the main link. Clicking the ellipsis button opens the LinkLabelListItemConfig Collection Editor.
- **Text:** Text to display for the link.

Enable Admin Displays: If True, the Admin Displays section is visible on the OMS Display Plugin.

Expand Admin Displays: If True, the Admin Displays section is shown expanded on the OMS Display Plugin.

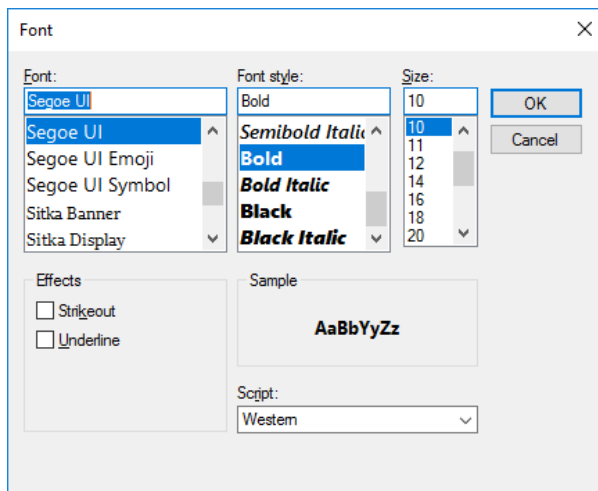
10.2. Appearance

▼ Appearance	
Bitmap Folder	D:\Eterra\Distribution\FantasyIsland\bitmaps\
▼ BoldFont	Segoe UI, 9.75pt, style=Bold
Name	Segoe UI
Size	9.75
Unit	Point
Bold	True
GdiCharSet	1
GdiVerticalFont	False
Italic	False
Strikeout	False
Underline	False
> Display Links	Overview, Company Summary, State Summary, County Summary, City S
> Font	Segoe UI, 9.75pt
ImageList	(Collection)
Overview Header Color	240, 240, 240

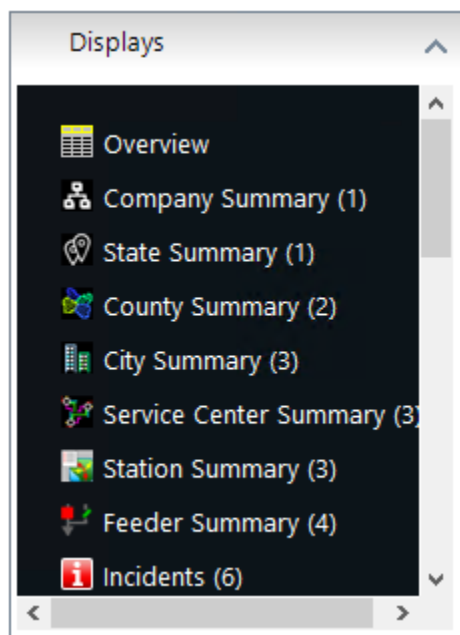
Bitmap Folder: Path to the folder that stores images used by the OMS Display Plugin.

Bold Font: Font to use for bold text on the OMS Display Plugin interface. To set the font, click the

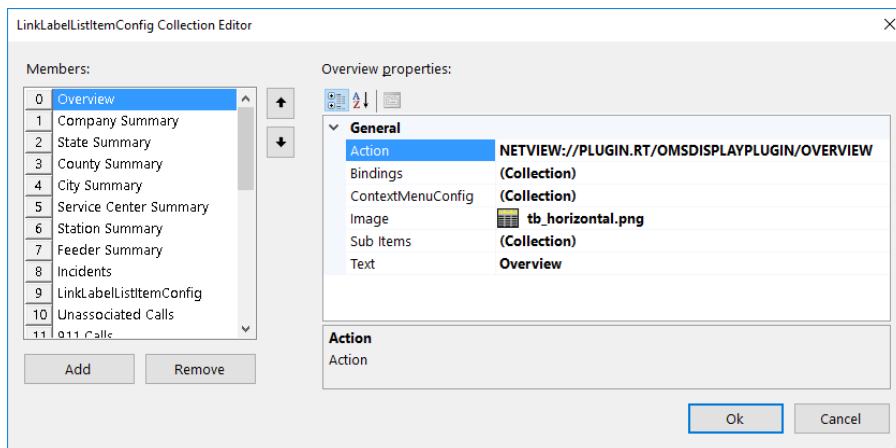
ellipsis  button at the right. This opens the Font dialog box, where you can customize the font.



Display Links: Specifies links to be shown in the Displays section in the navigation pane of the OMS Display Plugin.



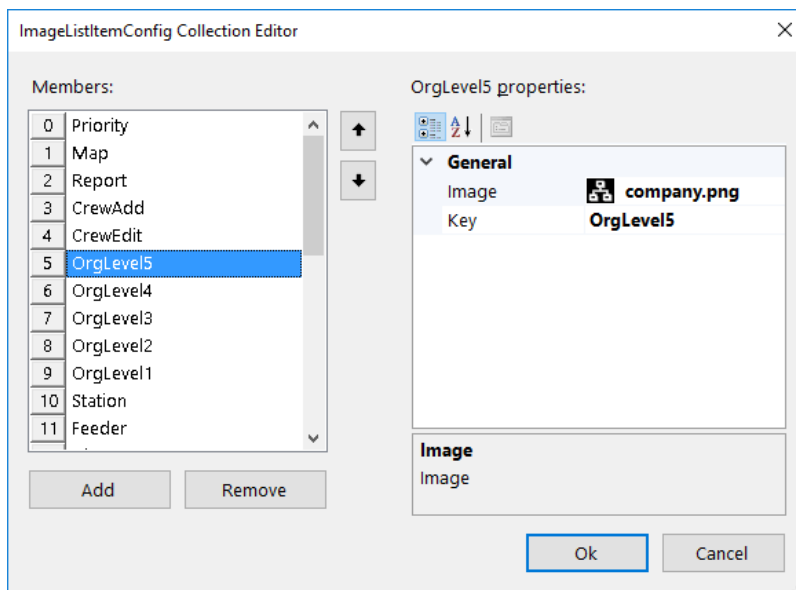
To review and configure the links, select the Items sub-element, and click the ellipsis  button at the right. This opens the LinkLabelListItemConfig Collection Editor dialog box.



For a description of the LinkLabelListItemConfig Collection Editor dialog box, see [Admin Displays](#).

Font: Font to use for bold, formatted, and normal text on the OMS Display Plugin interface.

ImageList: The list of images that are used for the plugin's tabs, filter columns, etc. To review and configure images, click the ellipsis  button at the right. This opens the ImageListItemConfig Collection Editor dialog box.



Each image item has the following properties:

- **Image:** The image file to display with the link and the image preview.
- **Key:** The ID to refer to this image.

Overview Header Color: Color used for the Overview tab column header row. To set colors, select a color item, and click the arrow  button at the right. This opens the color selection palette.

10.3. Client Data

This section describes Client Data parameters.

▼ Client Data	
> Area Note Configuration	OmsDisplayPlugin.AreaNotesConfig
> DataSetConfig	ClientDataSet
> Overview Configuration	OmsDisplayPlugin.OverviewConfig

10.3.1. Area Settings Configurations

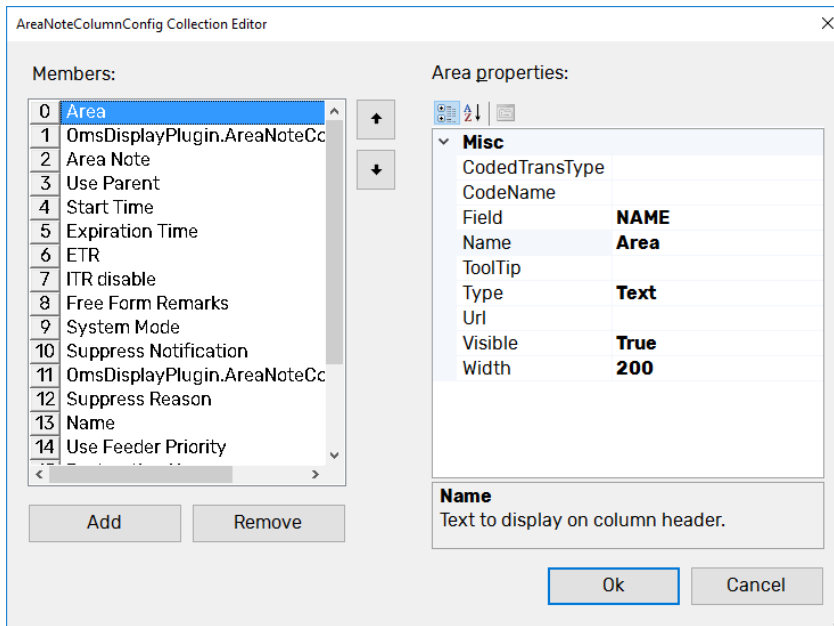
▼ Client Data	
▼ Area Settings Configuration	OmsDisplayPlugin.AreaNotesConfig
Column Mapping	(Collection)
Context Menu	(Collection)
Level Definitions	(Collection)
> ToolBar	OmsDisplayPlugin.ToolStripConfig
> DataSet Config	ClientDataSet
> Overview Configuration	OmsDisplayPlugin.OverviewConfig

Area Settings Configuration: Configuration options for the Active Area Settings tab in the OMS Display Plugin.

Area	Area Note	Use Parent	Start Time	Expiration Time	ETR	ITR disab	Free Form Remarks
General Electric							
Oregon							
Locke	CrewNotAvaila...	☒	5/26/2021 5:59...	5/27/2021 5:59...	5/28/2021 5:59...	Yes	
McDowell		☒					
Widgeon		☒					
STHC		☒					
STHD		☒					
DURBAN		☒					
EQSI		☒					
MARS		☒					
SATURN		☒					
STEVENS		☒					
Setzer							
Washington							
Gordon							

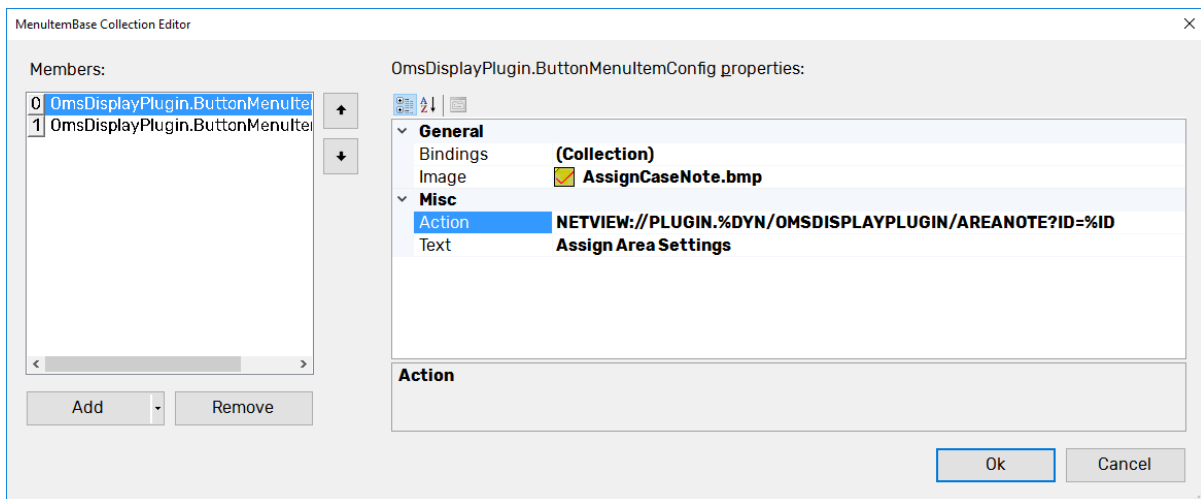
- **Column Mapping:** Column mapping settings for the Active Area Settings table.

Clicking the ellipsis  button at the right opens the AreaNoteColumnConfig Collection Editor dialog box:

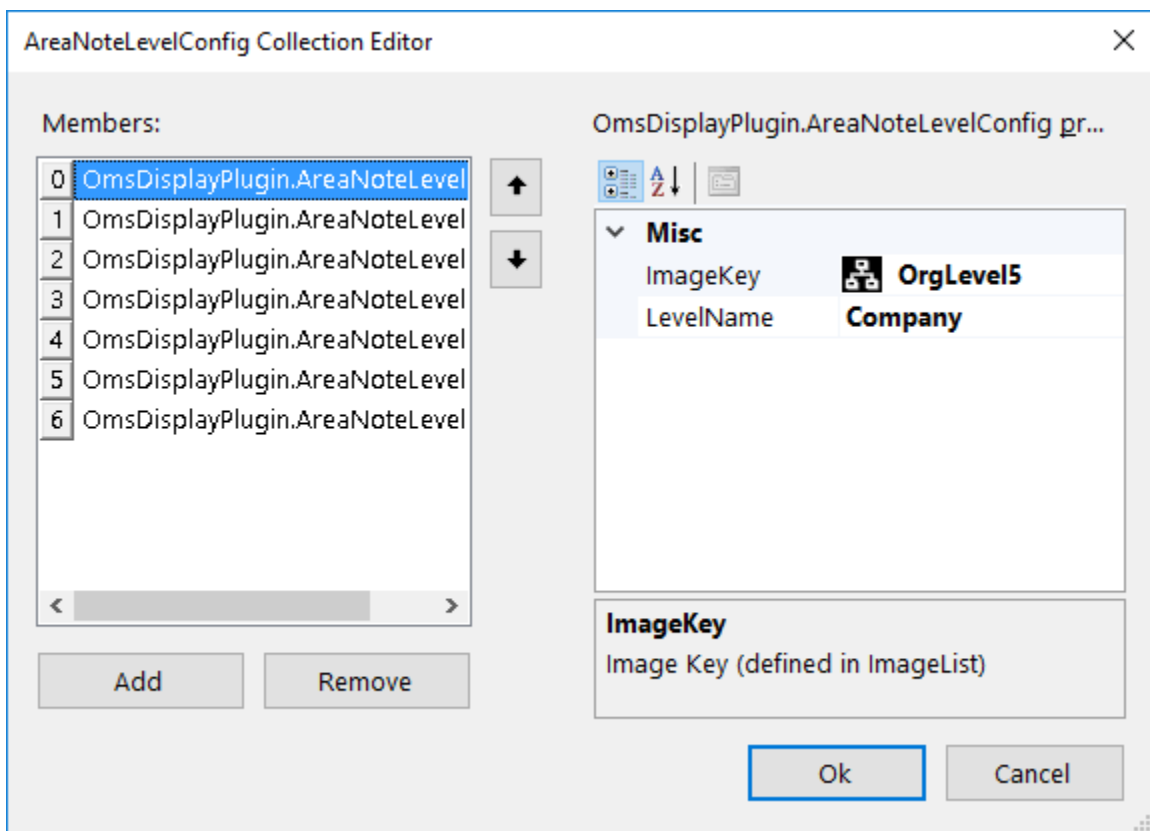


Each item has the following options:

- **CodedTransType:** Specifies the coded translation type (image or text), as appropriate.
- **CodeName:** Specifies the coded translation item that translates the field value.
- **Field:** Name of the OMS query field of the AREANOTE object to get the value from.
- **Name:** Column name to show in the Active Area Settings table.
- **ToolTip:** Field tooltip.
- **Type:** Field type (Text, Image, or CodeTranslation).
- **URL:** URL to navigate to from this column.
- **Visible:** If true, the column is visible.
- **Width:** Column width.
- **Context Menu:** Context menu configurations. Clicking the ellipsis  button opens the MenuItemBase Collection Editor dialog box, where you can configure the context menu.



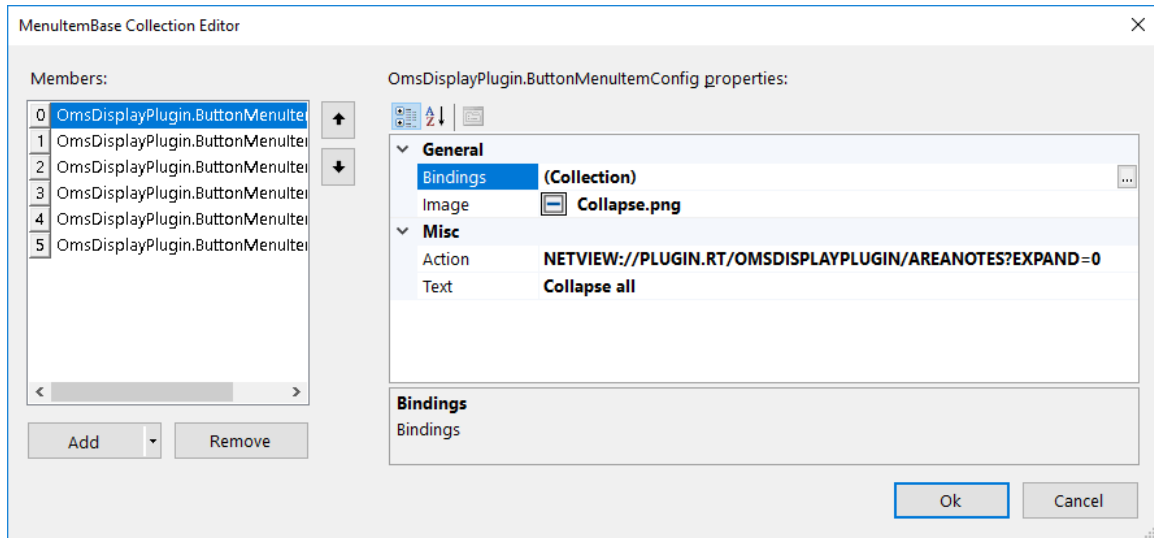
- **Level Definitions:** Hierarchy level configuration on the Active Area Settings tab, in display order. Each level is defined by a data relation or by specifying table, child, and column names. Clicking the ellipsis  button at the right opens the AreaNoteLevelConfig Collection Editor dialog box:



Each item has the following options:

- **ImageKey:** Image key, defined in the ImageList.
- **LevelName:** Name of the management areas hierarchy level.

- **Toolbar:** Configuration for the Active Area Settings tab toolbar's items.
 - **Enabled:** If True, the toolbar is visible on the Active Area Settings tab of the OMS Display Plugin.
 - **ImageSize:** Size for all toolbar buttons.
 - **Items:** Click the ellipsis  button to open the MenuItemBase Collection Editor dialog box.

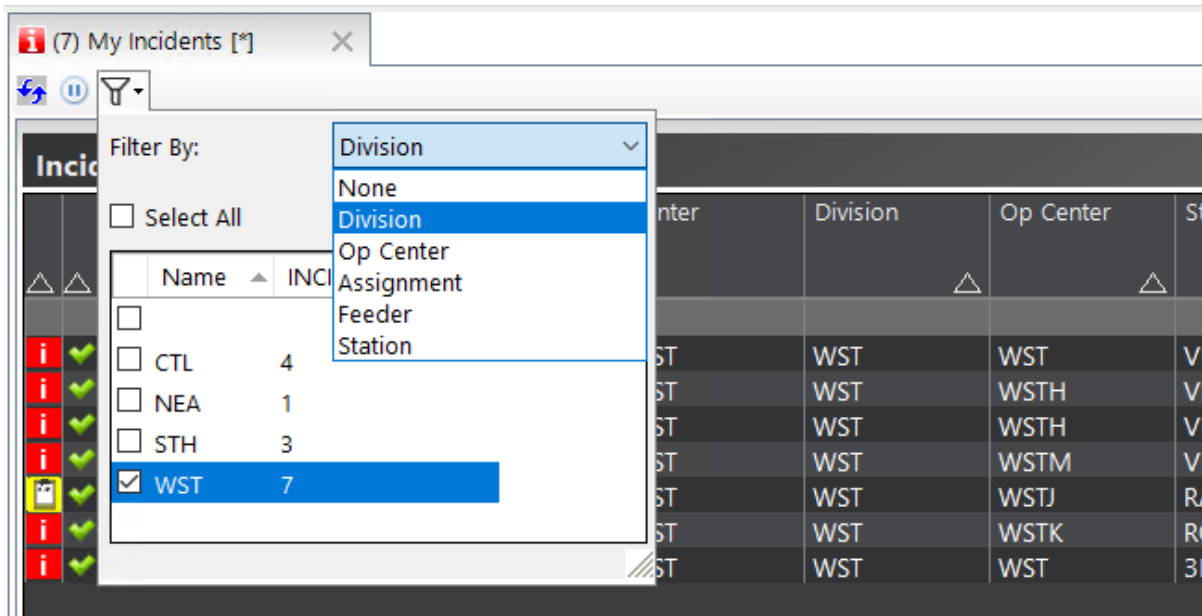


10.3.2. Data Set Config

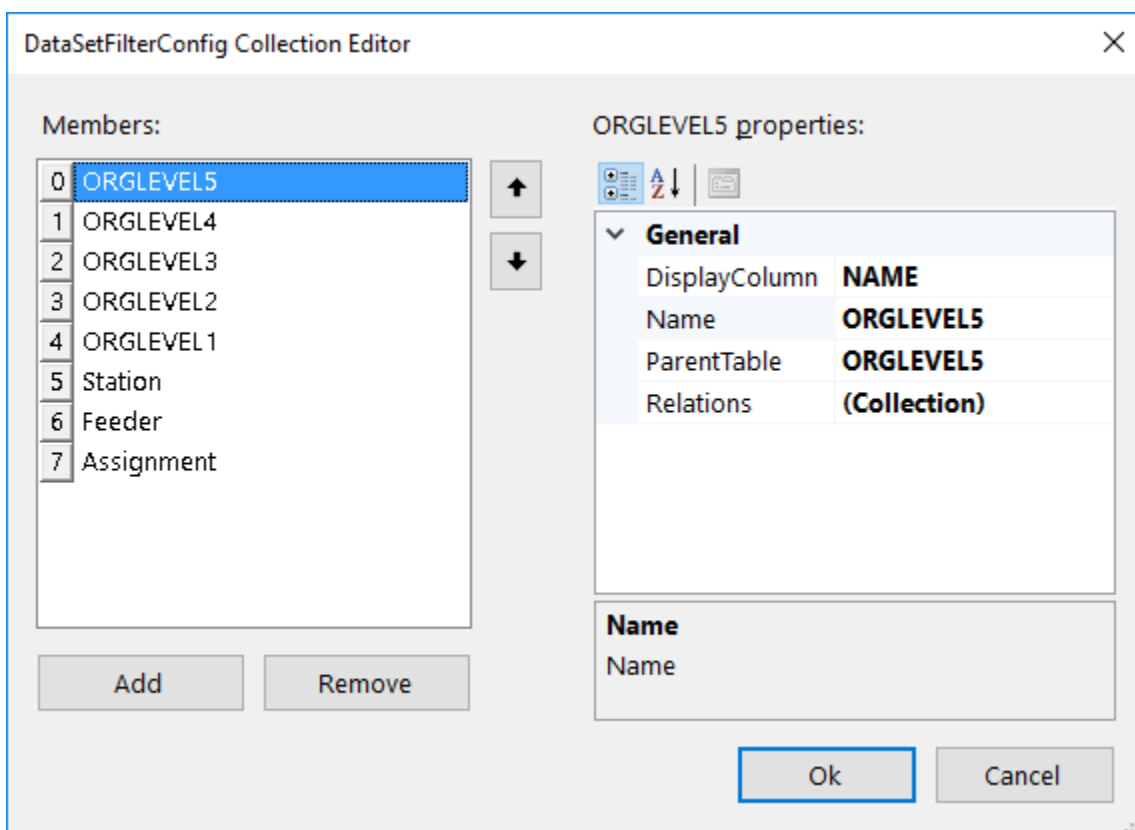
DataSetConfig	ClientDataSet
EnableDataRelations	True
EnableExpressionColumns	False
EnableHistoryCorrectionRefresh	False
Filters	(Collection)
Name	ClientDataSet
RefreshRate	5
Relations	(Collection)
TableCounts	(Collection)
> TableNameToHierarchyGroupTypeRelations	System.Collections.Generic.List`1[OmsDisplayPlugin.DataSetTableNameToHierarchyGroupTypeRelation]
> Tables	OmsDisplayPlugin.PropertyGridCollection`1[OmsDisplayPlugin.DataSetTableConfig]
UseGeneratedData	False
WebTableCounts	(Collection)

Data Set Config: Name of the client data set configuration. This configuration is required to populate data on the OMS Display Plugin displays, which behave differently from common grid tabular displays. The Data Set Config element contains the following configuration options:

- **EnableDataRelations:** If enabled, facilitates some of the filtering operations and generates counts for the filter options.
- **EnableExpressionColumns:** Provides a different method for generating filter counts. This option is deprecated.
- **EnableHistoryCorrectionRefresh:** Set to True to query for history correction data (i.e., cleared and cancelled incidents) on a timer.
- **Filters:** Specifies the filters that can be applied in the Filter By menu in the OMS Display Plugin.



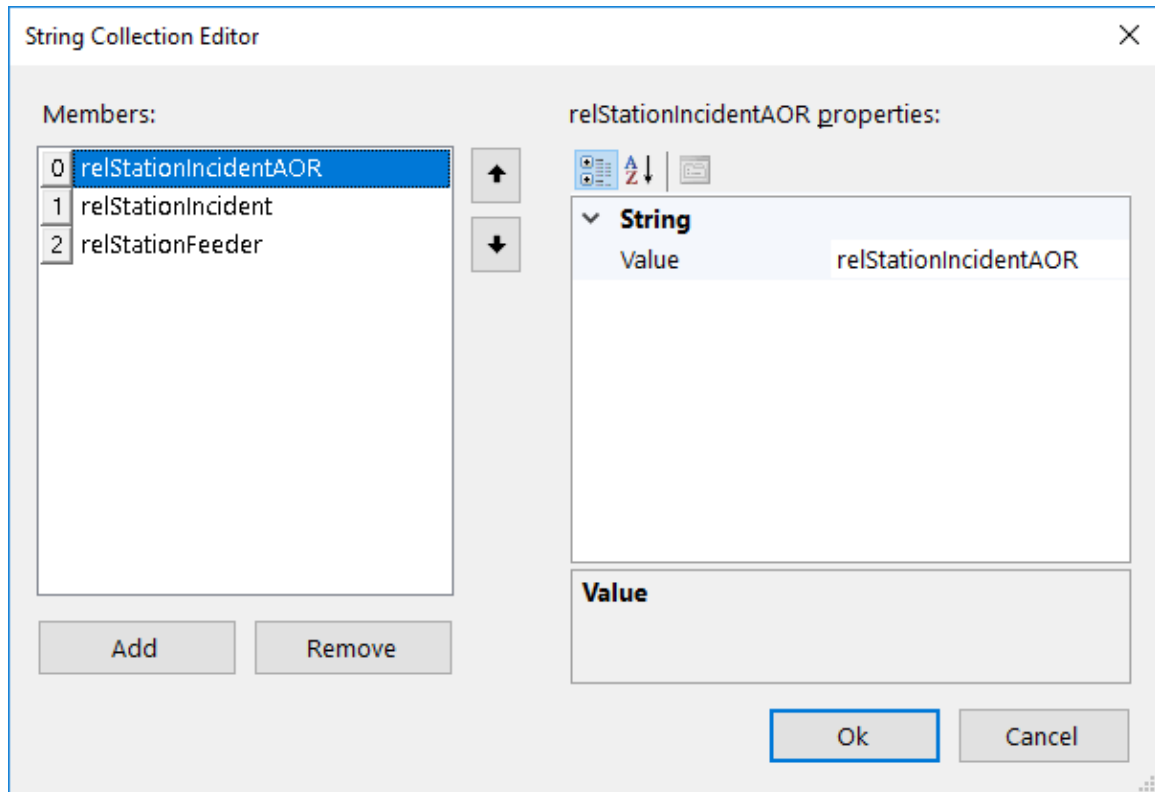
To review and configure the filters, select the Filters item, and click the ellipsis  button at the right. This opens the DataSetFilterConfig Collection Editor dialog box:



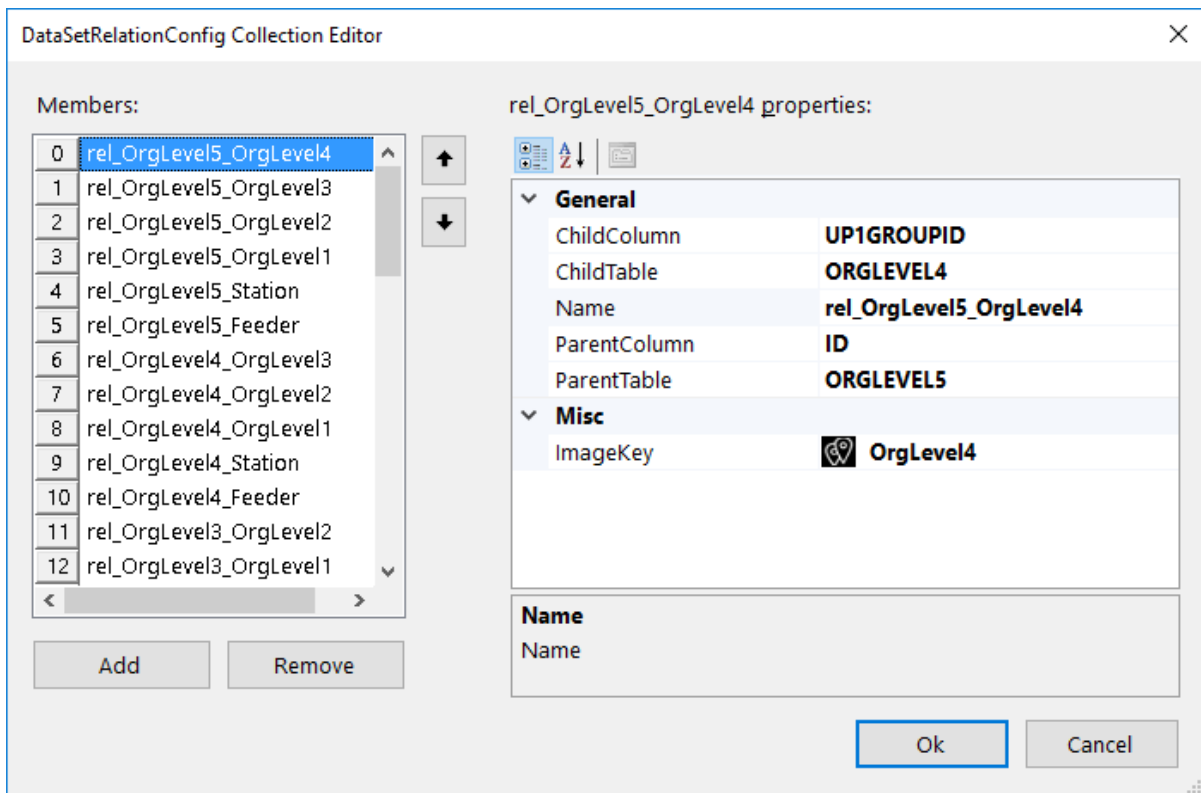
Each filter has the following options:

- **Display Column:** Specifies a column from the parent table to get filter values from.

- **Name:** Sets the filter name to show in the Filter By menu in the OMS Display Plugin.
- **Parent Table:** Specifies the name of the table to get filter values from.
- **Relations:** Defines links between a parent table and child tables. Use relation names defined in the Client Data> Data Set Config> Relations field.



- **Name:** Sets the name of the client data set.
- **RefreshRate:** Sets the time, in seconds, between queries to check for new data.
- **Relations:** Shows a list of defined relation items. Each item defines a child-parent relation between tables. These configurations are required for the Overview tab to function correctly. To open the DataSetRelationConfig Collection Editor dialog box, click the ellipsis  button at the right.



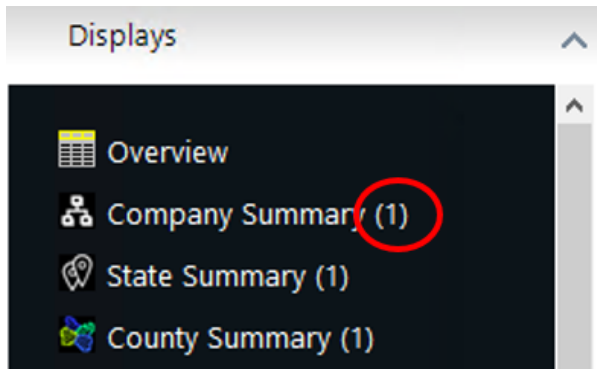
Each item has the following options:

- **Child Column:** Specifies the child table column that contains values to match with the values in the parent column in the parent table.

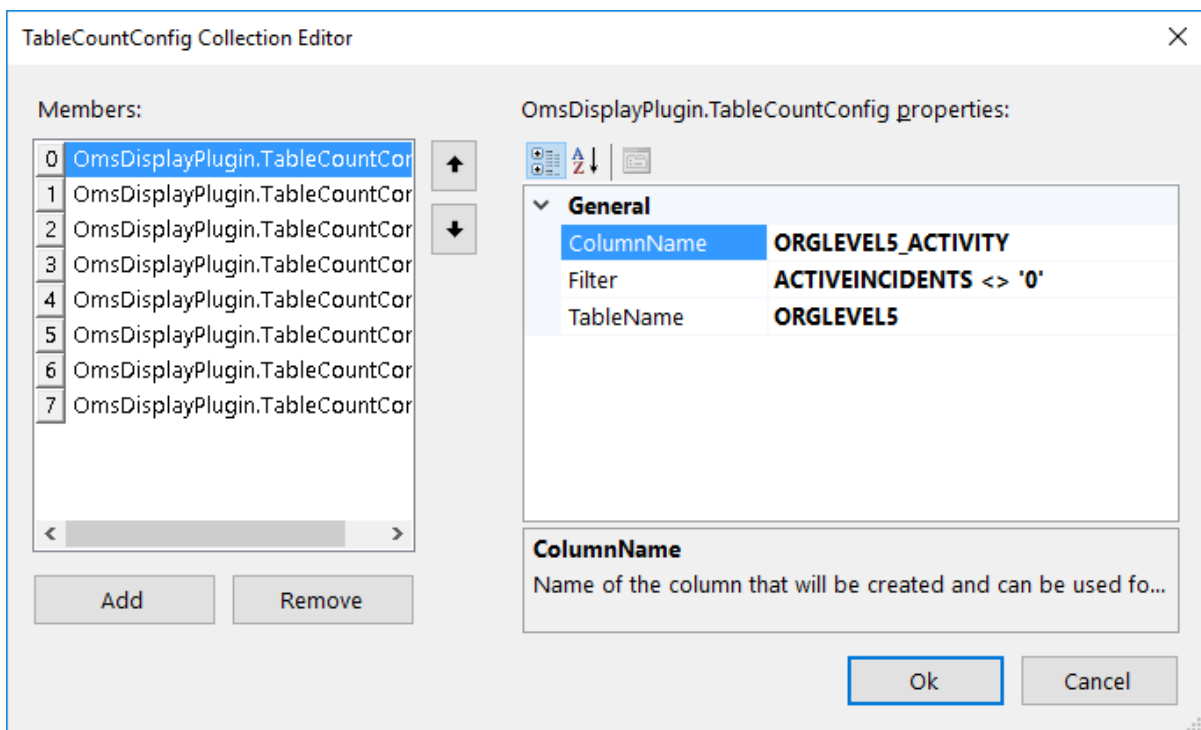
Note

Typically, ChildColumn's value UP1GROUPID corresponds to the highest hierarchy level in a relations group. For example, in a group of rel_OrgLevel4 items (rel_OrgLevel4_OrgLevel3 - rel_OrgLevel4_Feeder), you will use UP1GROUPID for OrgLevel4_OrgLevel3 and UP5GROUPID for rel_OrgLevel4_Feeder.

- **ChildTable:** Specifies the table to be filtered.
- **Name:** Defines the relation name to reference in the Relations settings of the Filter or Overview Configuration> Level Definition group.
- **Parent Column:** Specifies the parent table column containing the values to match with values in the child table column.
- **Parent Table:** Specifies the table with filter values.
- **Image Key:** Contains the image name specified in the ImageList to show for the child table in the filter control.
- **Table Counts:** Defines configuration options for counter items that apply filter conditions (for example, count of areas with incidents).



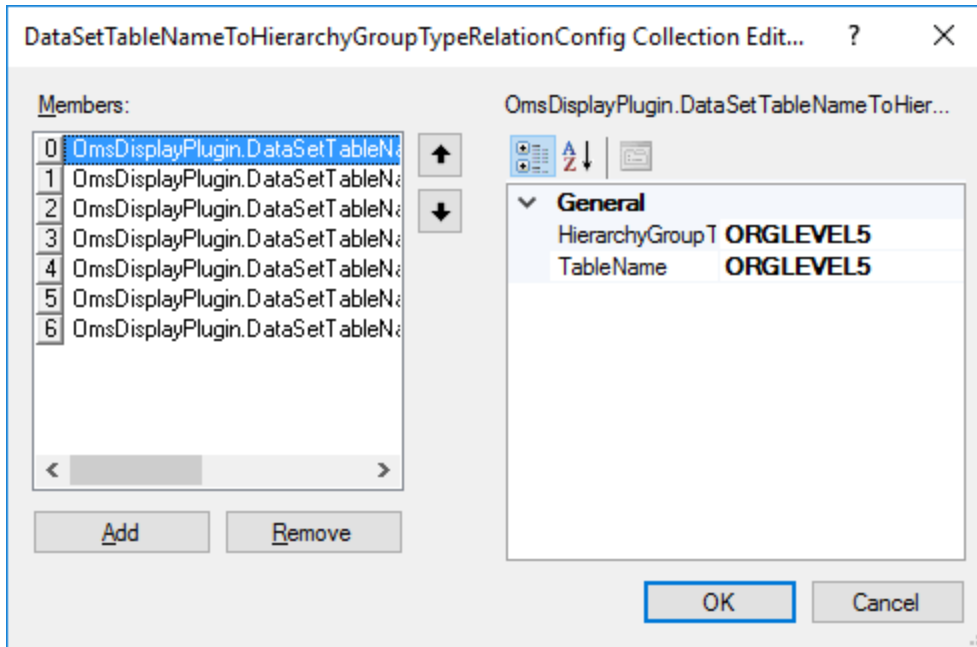
To review and configure the table counter options, click the ellipsis  button at the right. This opens the TableCountConfig Collection Editor dialog box.



Each item has the following options:

- **ColumnName:** Specifies the name of the counter item to use for data binding.
- **Filter:** Specifies the filter to apply to the table.
- **TableName:** Name of the table to filter to produce a filtered count.
- **TableNameToHierarchyGroupTypeRelations:** Configures relationships among client-side data set tables and hierarchy group types.

To open the DataSetTableNameToHierarchyGroupTypeRelationConfig Collection Editor dialog box, click the ellipsis  button at the right.



Each item has the following options:

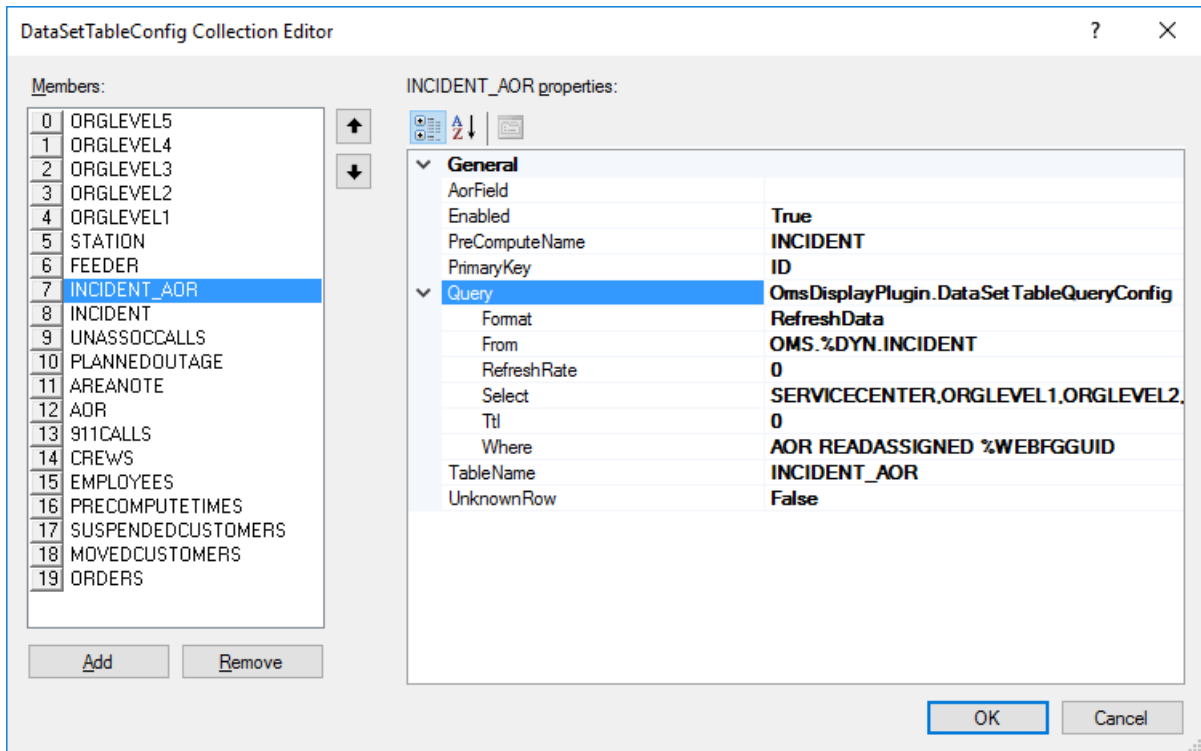
- **Hierarchy Group Type:** The type of the hierarchy group as defined in the code.
- **Table Name:** The name of the client-side data set table that relates to the specified hierarchy group type. The table name is defined by the ClientDataSet> Tables> TableName parameter.

In addition, the TableNameToHierarchyGroupTypeRelations group includes the following parameters:

- **Capacity:** Capacity of relationships among client-side data set tables and hierarchy group types. This field is populated automatically for system purposes.
- **Count:** The number of defined relationships among client-side data set tables and hierarchy group types.
- **Tables:** Shows a list of defined client-side tables that define the client data set. The client data set is required for populating data on OMS Display Plugin grid tabular displays and also for populating configuration items such as filters, counters, and overview tab items. To open the DataSetTableConfig Collection Editor dialog box, click the ellipsis  button at the right.

Note

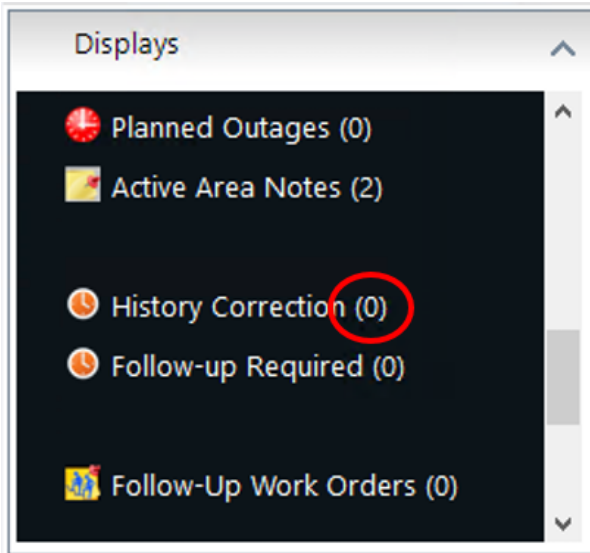
Grid tabular tabs are populated directly from the server and do not require the client data set.



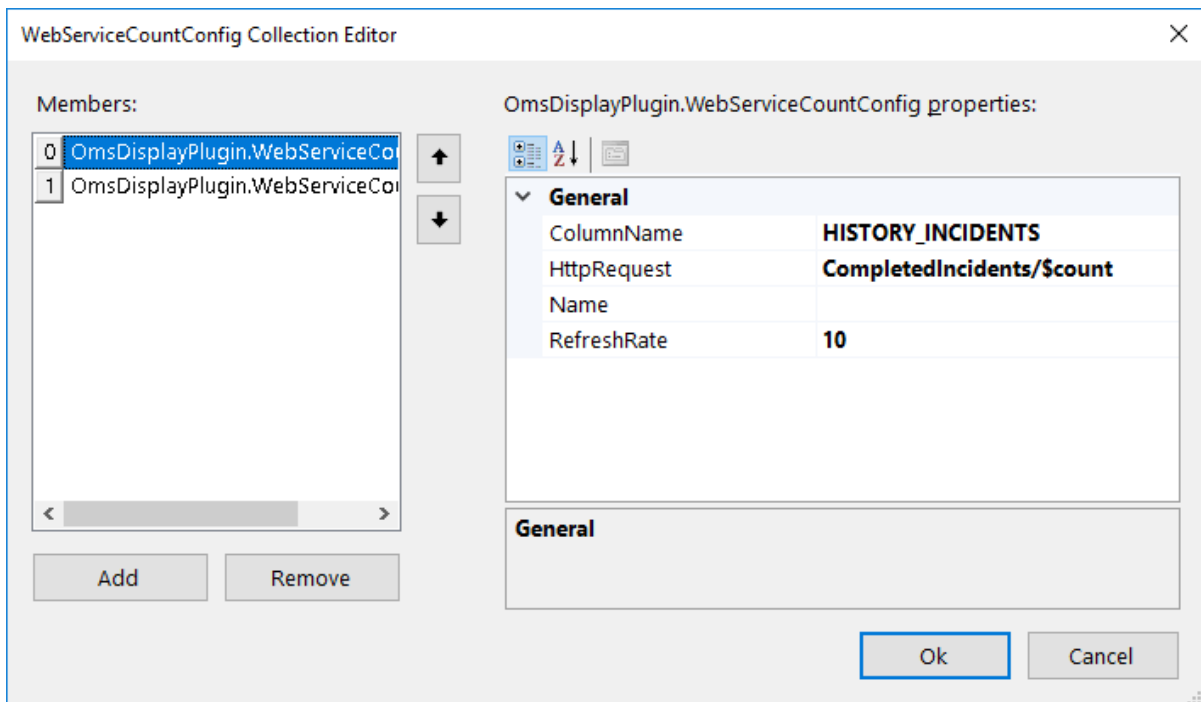
Each item has the following options:

- **AorField:** Specifies the OMS query field that defines the Area of Responsibility required for this table.
- **Enabled:** Specifies whether the table is enabled.
- **PreComputeName:** Name of the table to show on the Server Query Times display in the OMS Display Plugin. Used to check the update time.
- **PrimaryKey:** The key value for the table (should be a unique value).
- **Query:** Name of the query.
 - **Query > Format:** The type of query. Use RefreshData if the object supports incremental refresh, otherwise use DataTable.
 - **Query > From:** Object to query.
 - **Query > RefreshRate:** Time in seconds between queries.
 - **Query > Select:** Fields to query for. OMS Display Plugin relies on precomputed queries, therefore you must specify the required query fields in the Select field both in the grid tabular display file (for example, IncidentListAORGridDisplay.xml) and the associated table (for example, INCIDENT_AOR) to show the data on the display.
 - **Query > Ttl:** TBD.
 - **Query > Where:** Filter string.
- **TableName:** Name of the table for defining relations, filters, and tab displays.
- **UnknownRow:** If true, a row for Unknown is added after the query is received.

- **Use Generated Data:** Specifies whether to use generated data.
- **Web Table Counts:** Defines configuration options for the counter items for OMS data service tables (History Correction and Follow-Up Required).



To review and configure the web service table counter options, click the ellipsis  button at the right. This opens the WebServiceCountConfig Collection Editor dialog box.



Each item has the following options:

- **ColumnName:** Specifies the name of the counter item to be used for data binding.
- **HttpRequest:** Specifies the filter to apply to the data.

- **Name:** Name of the item.
- **RefreshRate:** Time in seconds between queries.

10.3.3. Overview Configuration

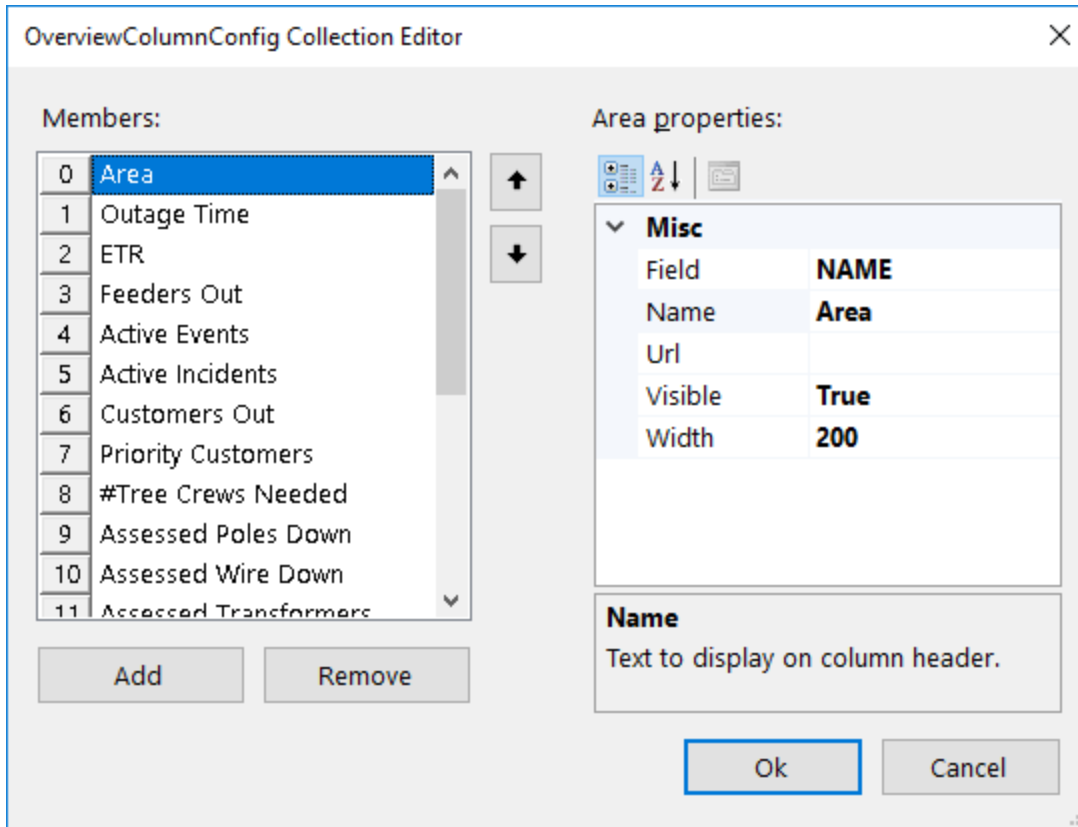
▼ Overview Configuration	OmsDisplayPlugin.OverviewConfig
Column Mapping	(Collection)
Level Definitions	(Collection)
▼ ToolBar	OmsDisplayPlugin.ToolStripConfig
Enabled	True
ImageSize	24
Items	(Collection)

Overview Configuration: Configuration options for the Overview tab in the OMS Display Plugin.

The screenshot shows the OMS Display Plugin interface. On the left, there is a 'Displays' sidebar with a tree view containing: Overview, Company Summary (1), State Summary (1), County Summary (1), City Summary (2), Service Center Summary (2), Station Summary (5), Feeder Summary (8), Incidents (29) with a sub-item 'All Incidents (29)', Unassociated Calls (6), 911 Calls (3), Suspended Customers, and Moved Customers. Below this is an 'Admin Displays' button. The main window has a tab labeled 'Overview'. Inside this tab, there is a table with the following columns: Area, Outage Time, ETR, Feeders Out, Active Events, Active Incidents, and Customers Out. The table contains data for various areas, including GE, OR, LOCKE, SETZER, WA, GORDON, TIDUS, ELIZABTH, GRUMMAN, NEAA, NEAB, and NEPTUNE, with their respective outage times and counts.

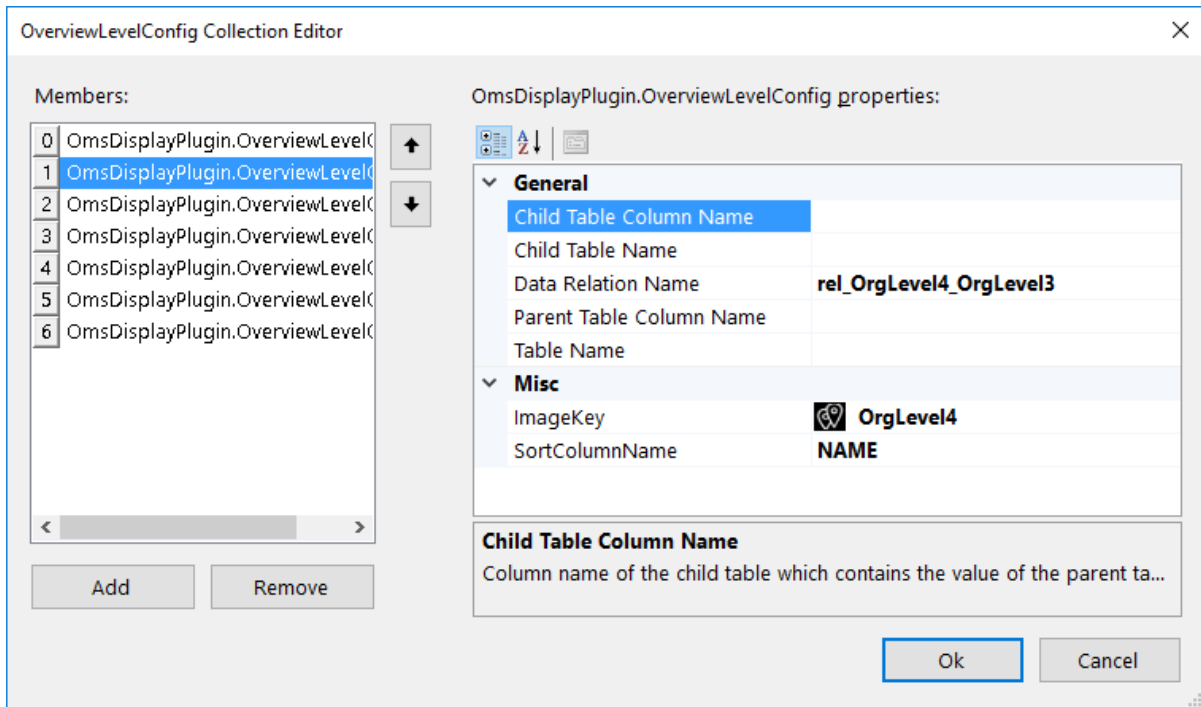
- **Column Mapping:** Column mapping settings for the Overview table.

Clicking the ellipsis  button at the right opens the OverviewColumnConfig Collection Editor dialog box:



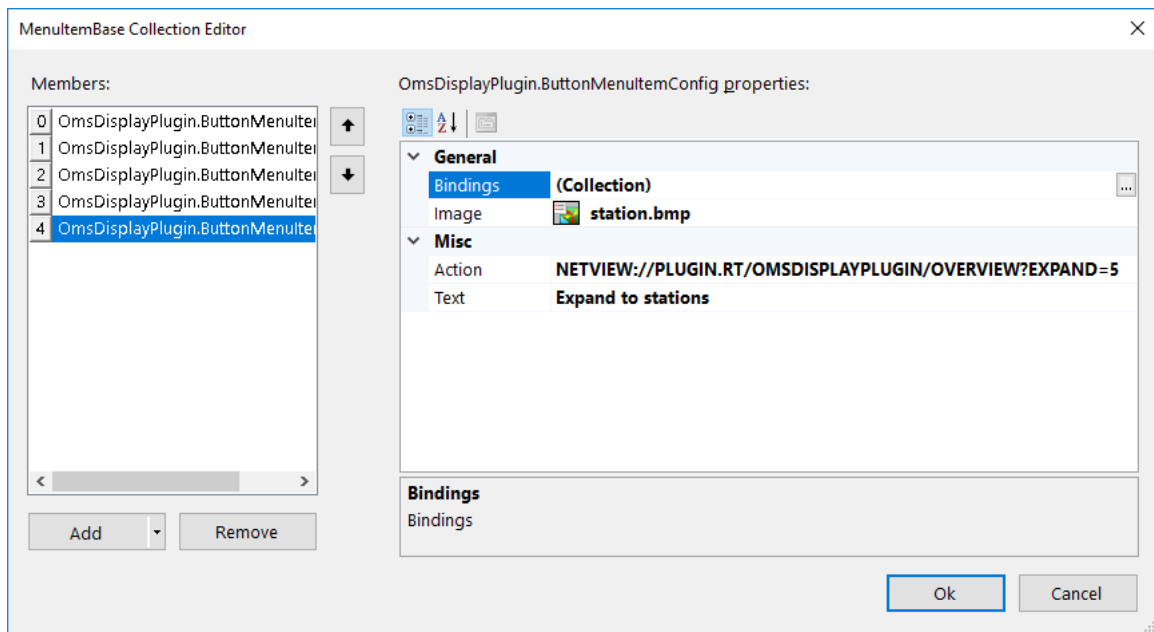
Each item has the following options:

- **Field:** Name of the OMS query field of the HIERARCHYGROUP object to get the value from.
- **Name:** Column name to be shown in the Overview table.
- **URL:** URL to be navigated from this column.
- **Visible:** If true, the column is visible.
- **Width:** Column width.
- **Level Definitions:** Hierarchy level configuration on the Overview tab, in display order. Each level is defined by a data relation or by specifying table, child, and column names. Clicking the ellipsis  button at the right opens the OverviewLevelConfig Collection Editor dialog box:



Each item has the following options:

- **Child Table Column Name:** Column name of the child table that contains the value of the parent table column.
- **Child Table Name:** Name of the child table.
- **Data Relation Name:** Name of the data relation, where parent is a hierarchy level, and child is the level below.
- **Parent Table Column Name:** Column name of the table for this level that contains the key value.
- **Table Name:** Name of the table for this level (parent table).
- **ImageKey:** Image key, defined in the ImageList.
- **SortColumnName:** Name of the column to sort by.
- **Toolbar:** Configuration for the Overview tab toolbar items.
 - **Enabled:** If True, the toolbar is visible on the Active Area Settings tab of the OMS Display Plugin.
 - **ImageSize:** Size for all toolbar buttons.
 - **Items:** Click the ellipsis  button to open the MenuItemBase Collection Editor dialog box.



10.4. Extended Displays

▼ Extended Displays		
Enable Displays expecting Argument		False
Link List expecting Argument		

- **Enable Displays Expecting Argument:** If True, the displays that expect arguments are enabled.
- **Link List Expecting Argument:** The list of links to displays that expect arguments.

10.5. Convert Incident

▼ Convert Incident		
ConvertIncidentPopupHeight		300
ConvertIncidentPopupMinHeight		300
ConvertIncidentPopupMinWidth		580
ConvertIncidentPopupWidth		580

This section contains configuration options for the Convert Incident dialog box (see below image) used to convert outage incidents into non-outage incidents and vice versa.

- **ConvertIncidentPopupHeight:** Specifies the default height of the Convert Incident dialog box.
- **ConvertIncidentPopupMinHeight:** Specifies the minimum height of the Convert Incident dialog box.
- **ConvertIncidentPopupMinWidth:** Specifies the minimum width of the Convert Incident dialog box.
-

ConvertIncidentPopupWidth: Specifies the default width of the Convert Incident dialog box.

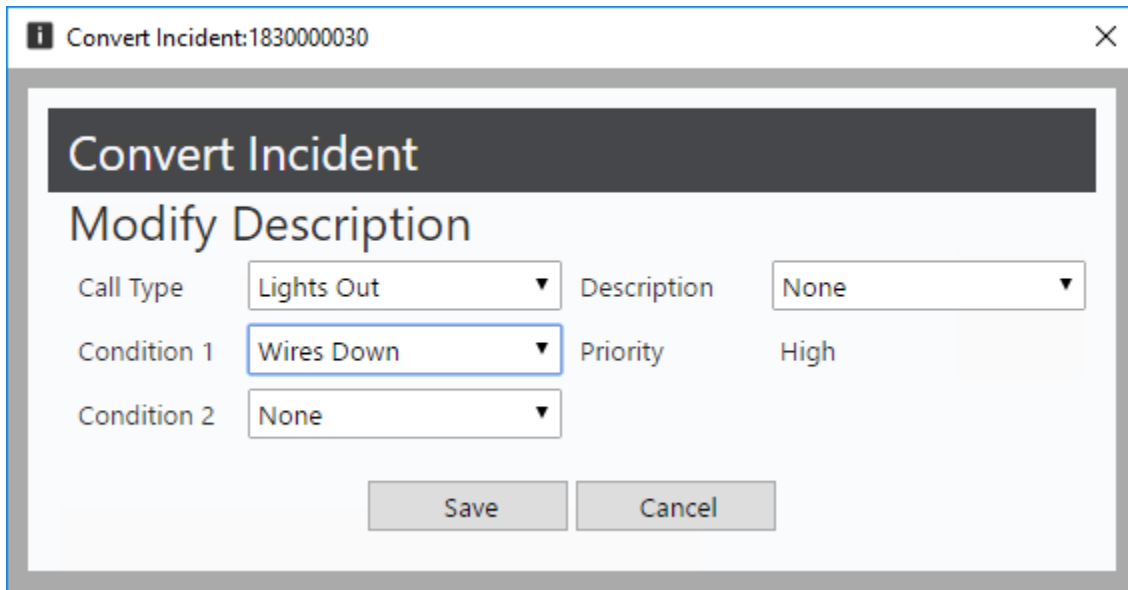


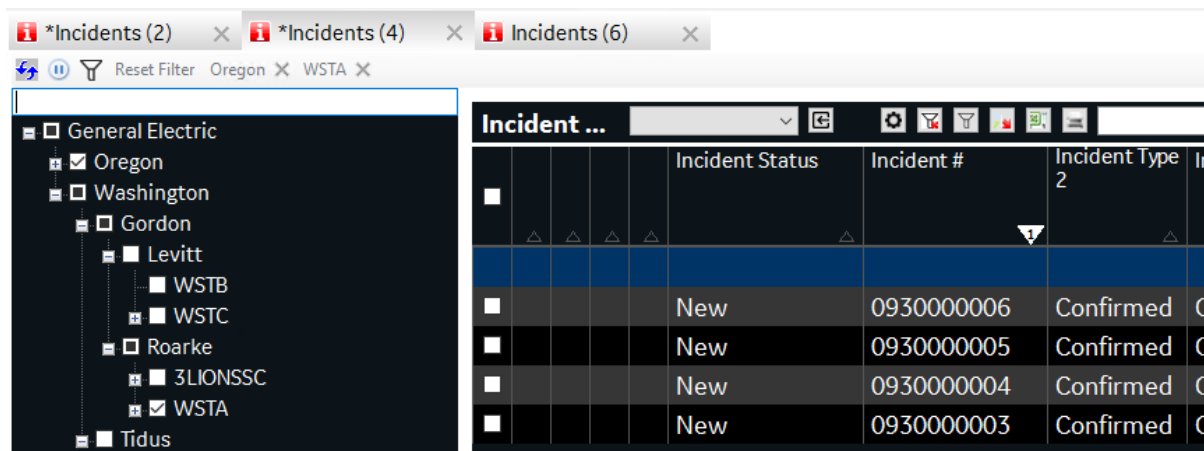
Figure 15. Convert Incident Dialog Box

10.6. General

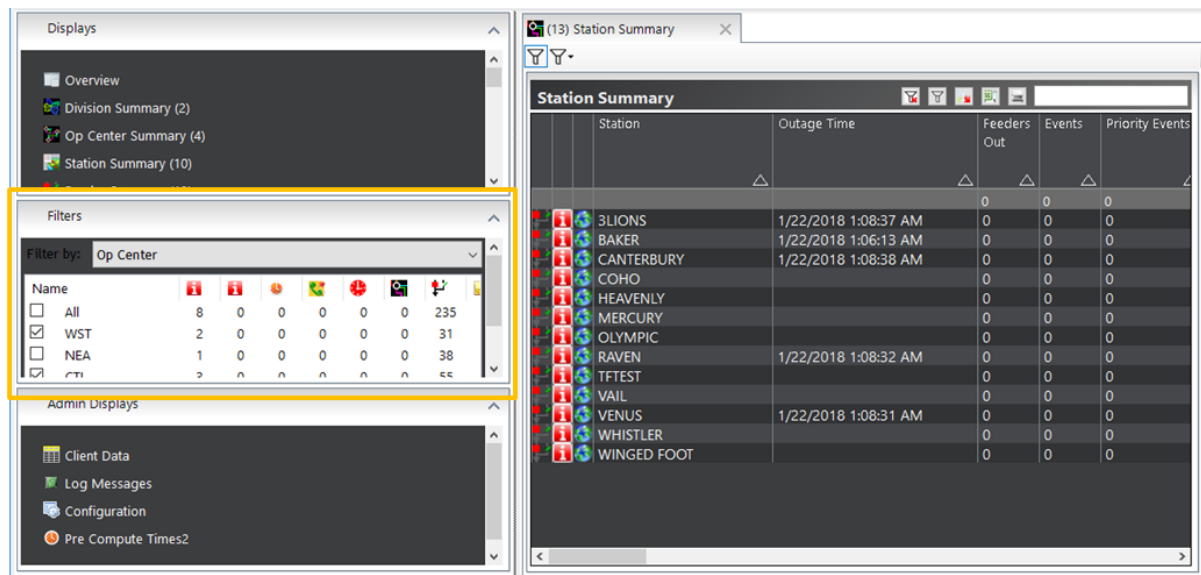
▼ General	
Allow Multiple Filtered Tabs	True
Allow Tab Rearrange	True
Double Buffer Rendering	False
Enable Filter Accordion	False
Enable Internal Notification	False
FilterView Configs	(Collection)
> Geo Tab Tool Bar	OmsDisplayPlugin.ToolStripConfig
Popups	(Collection)
ShowRestorationTabForNonOutageIncidentsDisplayName	True
Startup Tabs	(Collection)
Tab Displays	(Collection)
> Tool Bar	OmsDisplayPlugin.ToolStripConfig

This section contains general OMS display plugin configuration options:

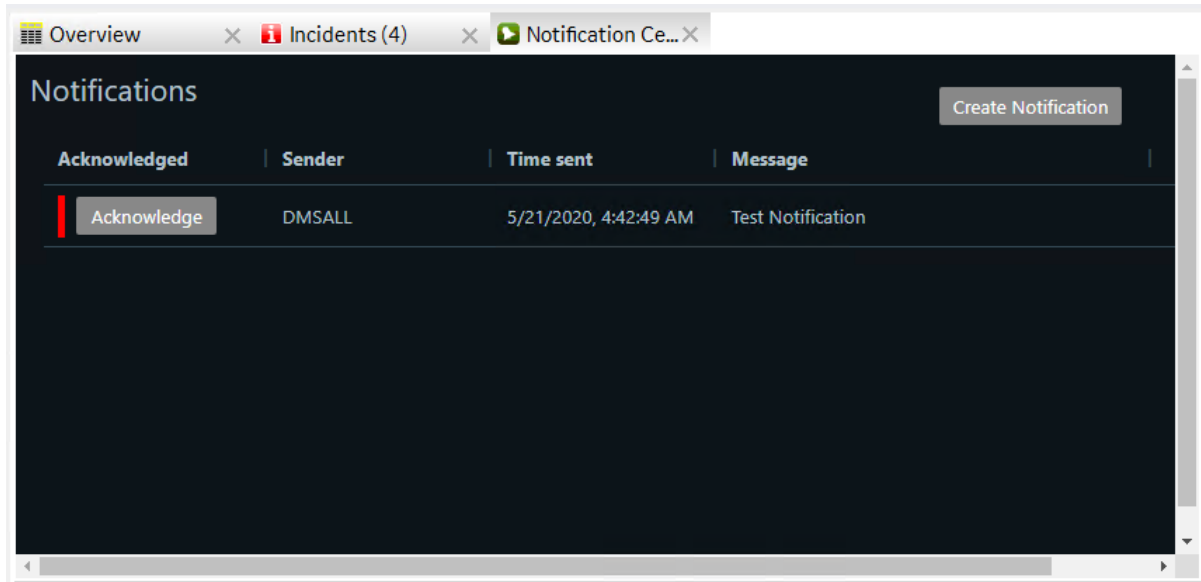
- **AllowMultipleFilteredTabs:** If True, the OMS display plugin can show multiple filtered tabs of the same type (for example, Incidents). If False, the OMS display plugin can show only one tab of the same type, regardless of whether it is filtered or not.



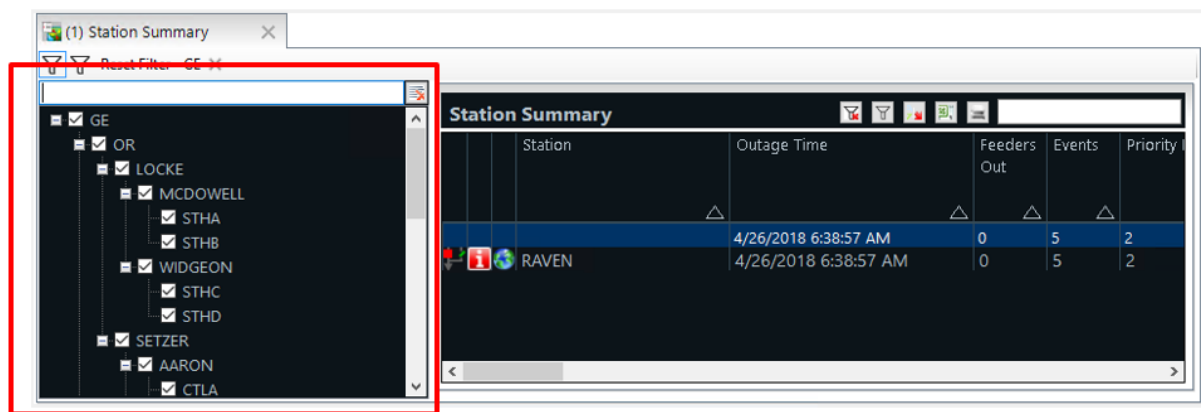
- **AllowTabRearrange:** If True, the tabs in the OMS Display Plugin can be rearranged by drag and drop.
- **DoubleBufferRendering:** If True double buffer rendering is enabled.
- **EnableFilterAccordion:** If True, the Filters control is shown in the navigation pane below the Displays section.



- **EnableInternalNotification:** If True, the Notification Center functionality is enabled in the OMS display plugin.



- **FilterViewConfigs:** Configures hierarchy level filters in OMS displays.



Clicking the ellipsis  button at the right opens the FilterViewConfig Collection Editor dialog box.

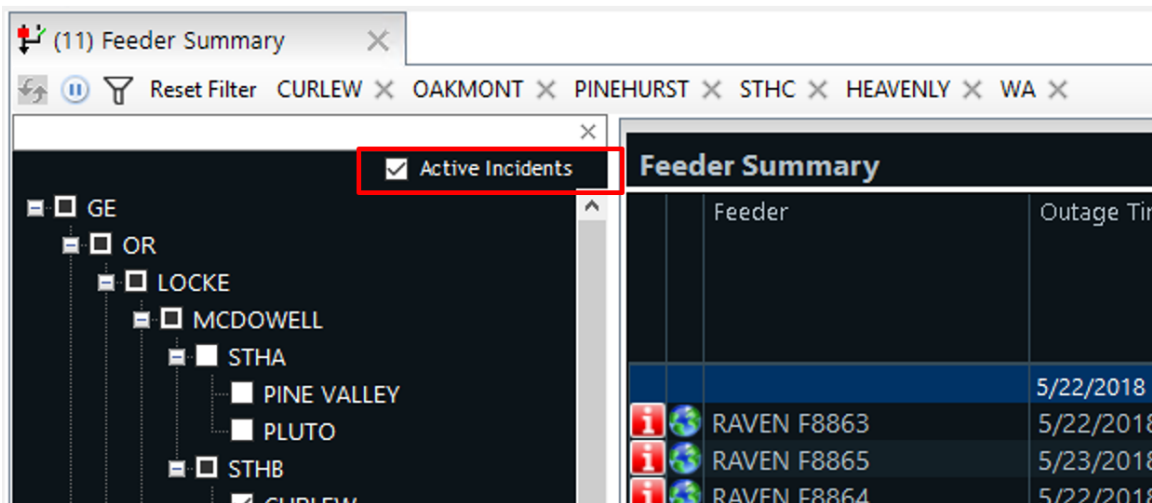
Each item has the following options:

- **Active Incidents Filter:** Specifies the active incidents filter to apply to the table.
- **Lower Most Hierarchy Group Type:** The type of the lowest hierarchy group to be shown in this filter.

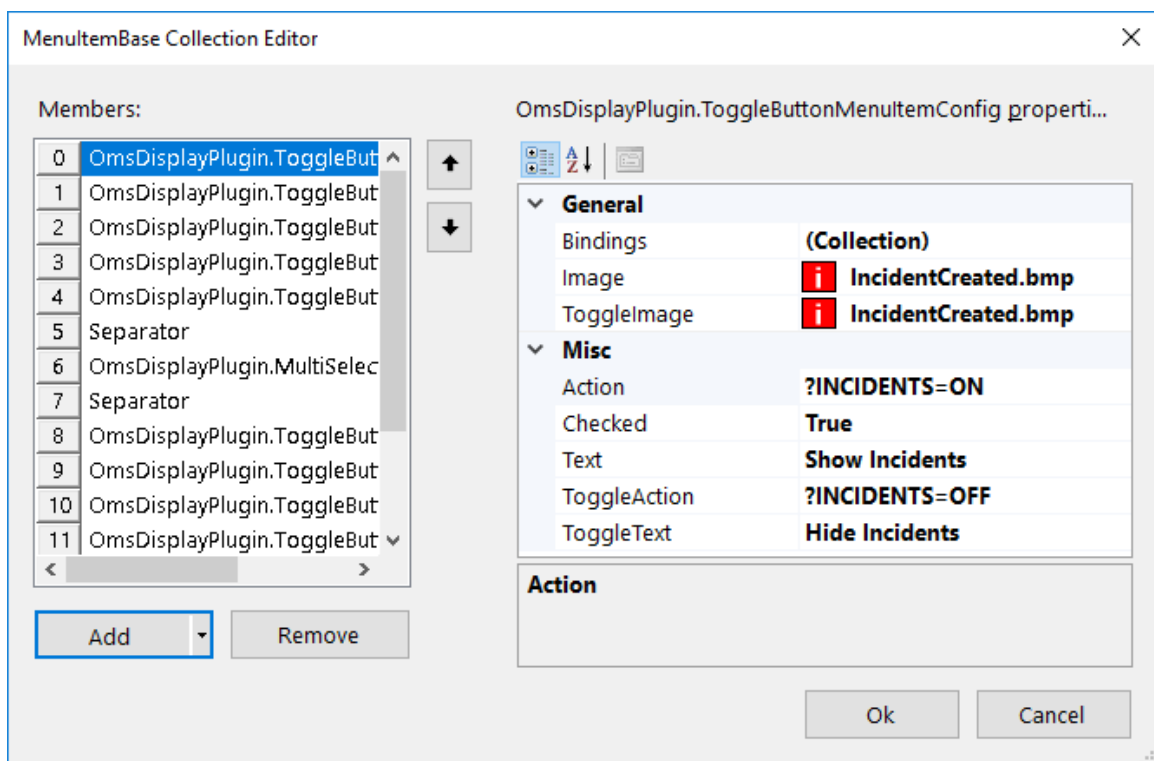
Note

Leave this field blank for the highest hierarchy level item (for example, OrgLevel5 for Company Summary), because it does not have a higher hierarchy level.

- **Name:** The name of the filter view configuration.
- **Use Active Incidents Filter:** If set to True, the Active Incidents check box, which controls showing only areas with incidents, appears in the Filtering Options panel of the associated tabular display.



- **GeoTabToolbar:** Configuration for the geographic overview toolbar items.
 - **Enabled:** If True, the toolbar is visible on the Geographic Overview tab of the OMS Display Plugin.
 - **ImageSize:** Size for all toolbar buttons.
 - **Items:** Click the ellipsis  button to open the MenuItemBase Collection Editor dialog box.

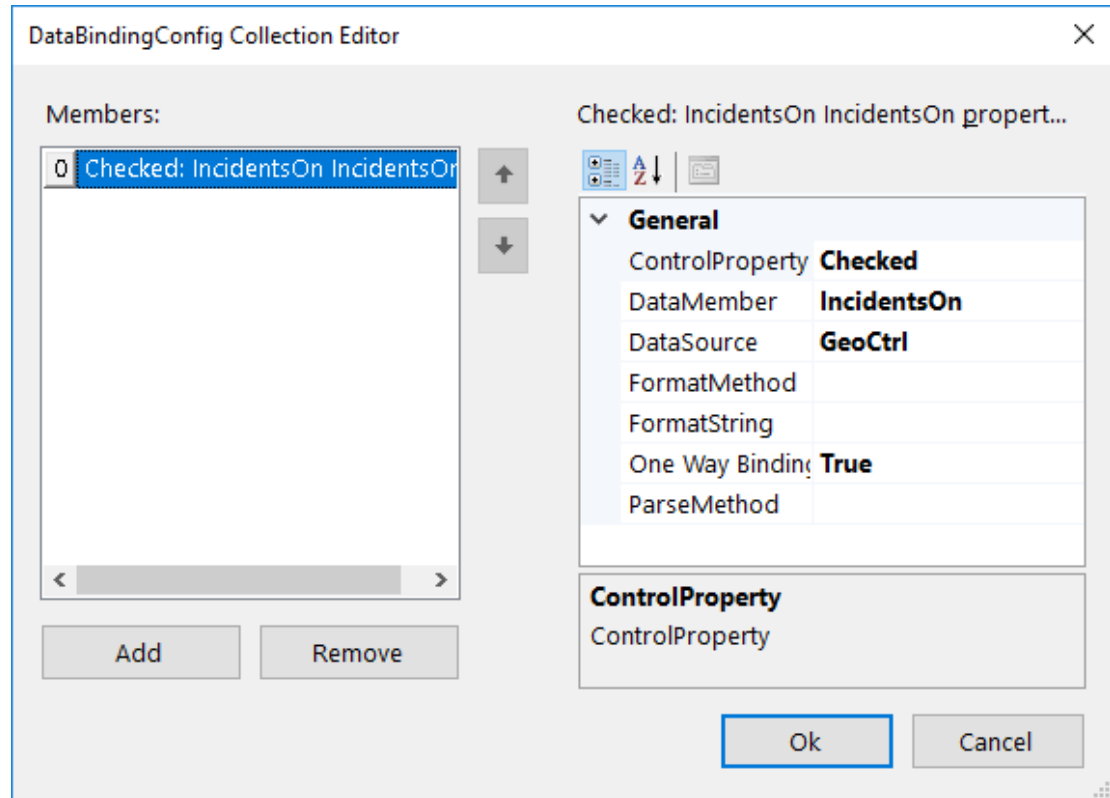


On a toolbar, you can add separators, labels, buttons, containers, toggle buttons, drop-down lists, and drop-down menus.

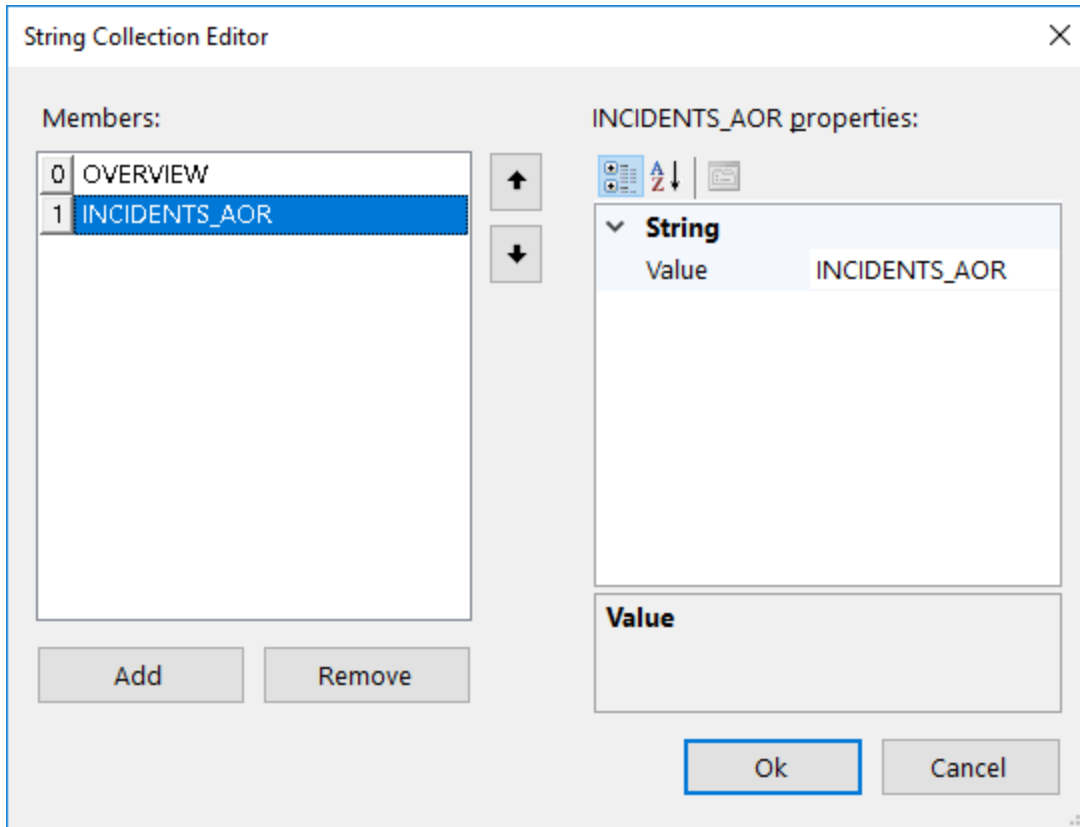
Toolbar items have the following options:

- **Bindings:** Specifies data bindings that are assigned to the link. To open the DataBindingsConfig Collection Editor, click the ellipsis  button next to Bindings.

For toolbar toggle buttons, you need to add a binding for toggling the button, as shown in the below image. For more details about bindings, see [Admin Displays](#).

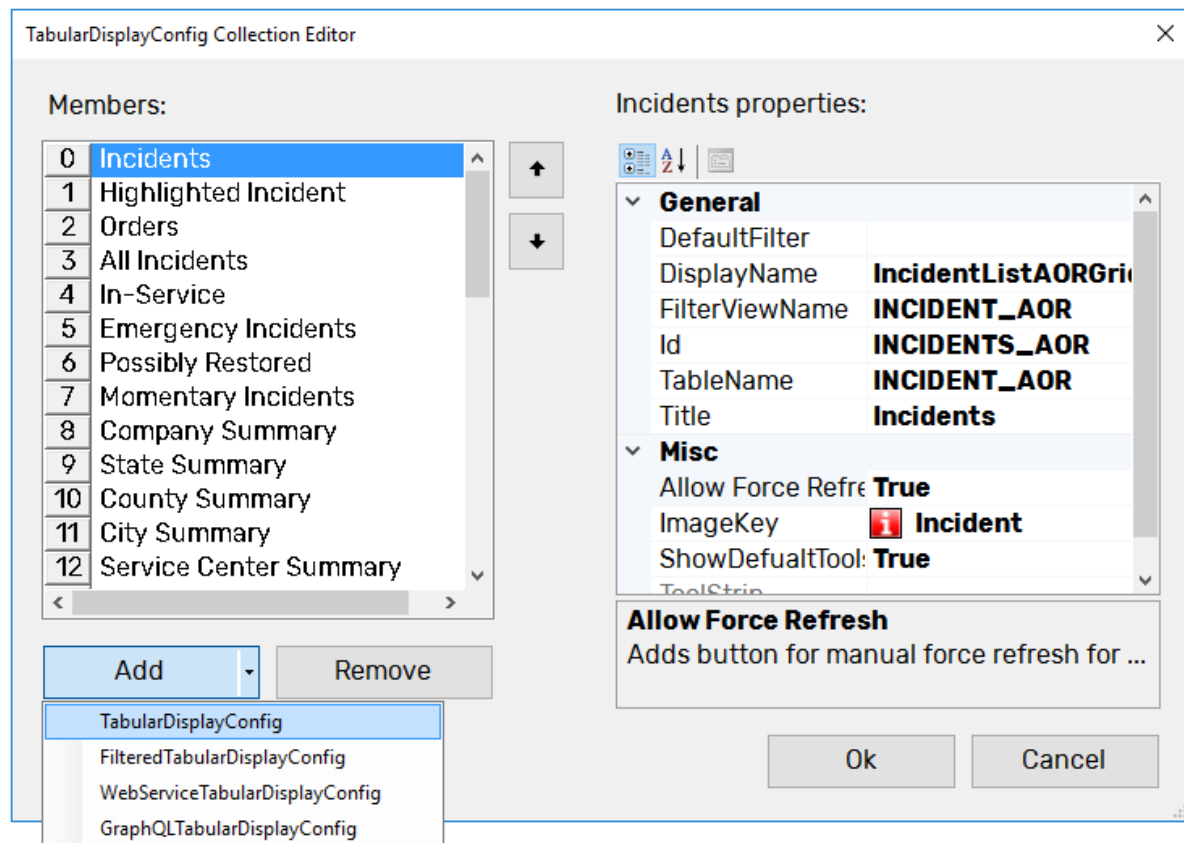


- **Image:** The name and preview of the toolbar button image.
- **ToggleImage:** The name and preview of the toolbar button image for the toggled state.
- **Action:** The URL to call when the toolbar button is clicked.
- **Checked:** If true, the button is clicked by default.
- **Text:** Contains the text to show when the mouse cursor hovers over the toolbar button.
- **ToggleAction:** The URL to call when the toolbar button is clicked.
- **ToggleText:** Contains the text to show when the mouse cursor hovers over the toolbar button when it is clicked.
- **DropDownControl:** Specifies the control for buttons with drop-down lists (for example, filter controls). This control applies to menu items of a container type only.
- **PopUps:** A list of pop-ups to be called by the plugin. For future use.
- **ShowRestorationTabForNonOutageIncidents:** If True, the Restoration tab appears on the Update Incident display for non-outage incidents.
- **StartupTabs:** Specifies tabs that are automatically shown when the OMS Display Plugin first opens. To open the String Collection Editor dialog box, click the ellipsis  button at the right.



Each item has the following option:

- **Value:** The ID of the tab as specified in the ID parameter of the General> TabDisplays group.
- **TabDisplays:** Defines grid tabular-based tabs that can be called by the OMS Display Plugin. To open the TabularDisplayConfig Collection Editor dialog box, click the ellipsis  button at the right.



The following tabular display configuration types are supported:

- **TabularDisplayConfig:** Grid tabular display that receives data from the OMS Server.
- **FilteredTabularDisplayConfig:** Grid tabular display that receives data from the OMS Server. This type of display can also include a filter control.
- **WebServiceTabularDisplayConfig:** Grid tabular display that receives data from the Net Data Server's web service. This type is used for the History Correction and Follow-Up Required displays.
- **GraphQLTabularDisplayConfig:** Grid tabular display that receives data from the Data Correction Interface application's GraphQL interface in GraphQL format. This type is used for the Data Correction display.

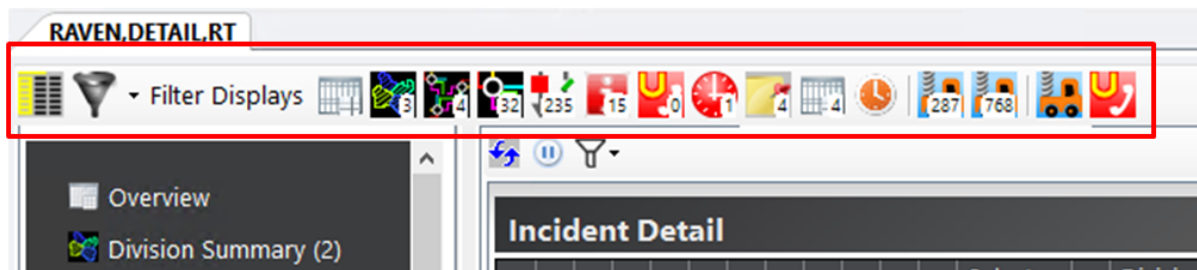
Tabular display items for TabularDisplayConfig and FilteredTabularDisplayConfig types have the following options:

- **DefaultFilter:** Sets the filter to apply to the table when it is first called.
- **DisplayName:** Specifies the name of the grid tabular display file that defines the display.
- **FilterControl:** Specifies the name of the table's filter control. Appears for FilteredTabularDisplayConfig display types.
- **FilterViewName:** Specifies the name of the table's filter configuration from General> FilterViewConfigs.

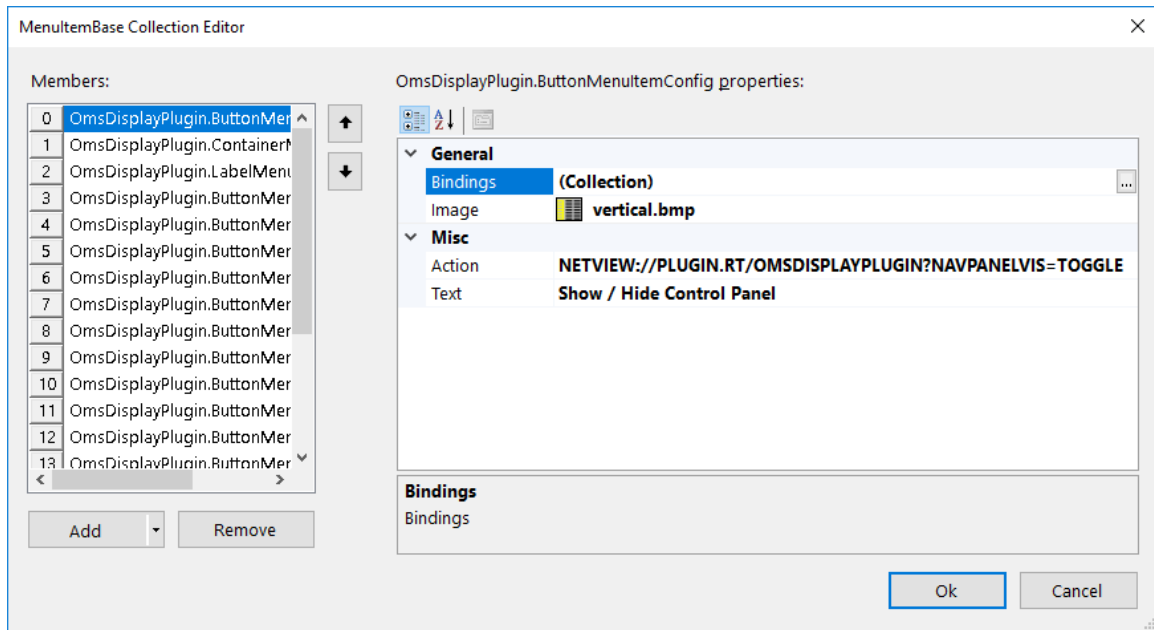
- **ID:** Specifies the ID that is used to navigate to or refer to this table (for example, it is used in the navigation URL specified in the Action string of the Appearance> Display Links group).
- **TableName:** Specifies the table name from ClientDataSet> Tables to provide data to this grid tabular.
- **Title:** Specifies the text to display on the tab.
- **Allow Force Refresh:** Specifies whether the Refresh button is shown on the display. If set to false, the display is refreshed automatically at the configured refresh rate.
- **ImageKey:** Sets the image from the ImageList to show on the tab.
- **ShowDefaultToolStrip:** If True, the default tab toolbar is shown with a Pause/Resume button and a tab filter control.
- **ToolStrip:** Configuration for custom toolbar items.

Tabular display items for WebServiceTabularDisplayConfig (for historical incidents) and GraphQLTabularDisplayConfig (for historical orders) types have the following additional options:

- **Base URI to Entity:** Specifies the endpoint that provides the data to the table. For incidents, the endpoint is the Net Data Server web service's collection. For orders, the endpoint is the Data Correction Interface application's GraphQL endpoint.
- **SumsColumn:** Specifies the name of the column to be added to the sums table for this display.
- **EntityTypename:** Specifies the entity type name.
- **EntityType:** Specifies the entity (object or class) type that defines query fields. For incidents, the entity type is OmsDisplayPlugin.OmsData.Incident. For orders, the entity type is OmsUtilities.IncidentWorkOrder.
- **Operation Name to be Invoked:** The Data Correction Interface application's GraphQL method that is used. Applies to orders only. Supported methods are dataCorrectionData (order data) and dataCorrectionDataCount (order count).
- **Toolbar:** Configuration for the OMS display plugin toolbar items.



- **Enabled:** If True, the toolbar is visible on the OMS display plugin.
- **ImageSize:** Size for all toolbar buttons.
- **Items:** Defines toolbar items. To open the MenuItemBase Collection Editor dialog box, click the ellipsis  button at the right.



Each button menu item has the following options:

- **Bindings:** Specifies data bindings that are assigned to the button. For more details about bindings, see [Admin Displays](#).
- **Image:** The name and preview of the toolbar button image.
- **Action:** The URL to call when the toolbar button is clicked.
- **Text:** Contains text to display when the mouse is hovered over the toolbar button.

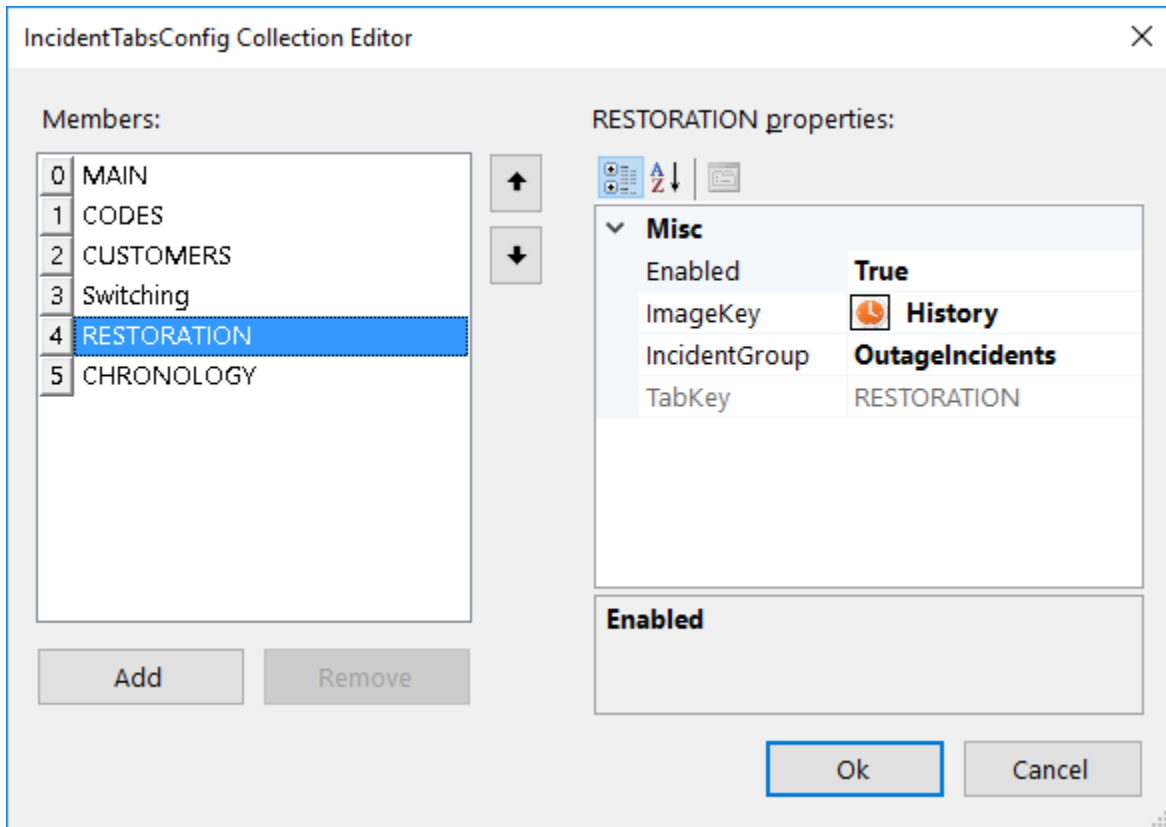
10.7. History Correction

History Correction	
Default query time (hours)	25
Display mode	Tab
History Correction Incident Button Strip Config	
History Correction Tabs Visibility	(Collection)
Max Results	100
MaxCrewSignOnSignOffCorrectionDisplayName	25
Show cancelled incidents by default	True

This section contains configuration options for the History Correction display.

- **Default Query Time (Hours):** Past hours of history to query.
- **Display Mode:** The History Correction window display mode (Tab or Pop-Up).

- **History Correction Tabs Visibility:** Configures the History Correction display tabs. Clicking the ellipsis  button opens the IncidentTabsConfig Collection Editor dialog box.



Each item has the following parameters:

- **Enabled:** If true, this tab is shown on the History Correction display.
- **ImageKey:** Sets the image from the ImageList to show on the tab.
- **IncidentGroup:** Sets the incident type, for which this tab appears on the Update Incident pop-up dialog box (all incident types, outage incidents, or non-outage incidents).
- **TabKey:** ID of the tab.
- **HistoryCorrectionIncidentButtonStripConfig:** The buttons to show along the bottom or top of the pop-up window. For details about the setting, refer to the IncidentButtonStripConfig description in [Update Incident](#).
- **Max Results:** Maximum number of entries to show.
- **Show Cancelled Incidents by Default:** Specifies whether cancelled incidents are shown on the History Correction display by default.

10.8. History Correction Update Incident

▼ History Correction Update Incident	
HistoryCorrectionIncidentToolBarConfig	

This section contains configuration options for the History Correction pop-up, which is used to edit incidents from history correction. The pop-up can have either a toolbar or a set of buttons.

- **HistoryCorrectionIncidentToolBarConfig:** The toolbar buttons to show at the top of the pop-up window.

10.9. Incident PDF Export

Incident PDF Export	
Allow manual geoview screenshot	False
Customer layers	(ALL)
Default background layers	(Collection)
Export Output Folder Path	%IDMS_ROOT%
Export XSL Transform File Path	%IDMS_ROOT%\tabulars\incident2pdf.xsl
Geographic Geometry Name	DETAIL
Geographic View Scale Factor	3
Map Api URL	https://maps.googleapis.com/maps/api/staticmap?center={%latitu
Map Scale Factor	13
Maximum Calls Number	10
Prefer Web Maps	True
Prefer Web Maps on multiple export	True

The following are configuration options for exporting incident properties to a PDF document.

- **Allow Manual Geoview Screenshot:** If set to True, users must manually take an incident location screenshot from the geographic network view. If set to False, the screenshot is taken automatically, with the map centered on the incident.

Note

Taking screenshots manually may be time consuming when exporting multiple incidents to PDFs.

- **Customer Layers:** Specifies the customer layers (ALL, SHOWCRITICALCUST, SHOWNORMALCUST, SHOWNOTCONNECTEDCUST, SHOWACTIVECUST, or NONE) that are shown on the incident location screenshot by default.
- **Default Background Layers:** Specifies the background layers (for example, roads shape, water shape, building footprints shape, boundaries shape, and labels) that are shown on the incident location screenshot by default.

Note

If new background layers are added to the system, the background layers list needs to be manually updated.

- **Export Output Folder Path:** The path to the folder where the PDF documents are created.
- **Export XSL Transform File Path:** The path to the incident2pdf.xsl and order2pdf.xsl files that defines the incident and order PDF conversion template.

- **Geographic Geometry Name:** The name of the geographic geometry that is used in display call-up URLs. The default name is DETAIL. Another common setting is GEOGRAPHIC.
- **Geographic View Scale Factor:** The scaling factor for the incident location in the geographic view.
- **Map API URL:** The map API URL that is used to produce incident location map images based on the input string. For example:

```
https://maps.googleapis.com/maps/api/staticmap?center={%latitude%},{%longitude%}&zoom={%scale%}&size=400x250&markers=color:red|{%latitude%},{%longitude%}
```

Where:

- **Latitude:** Double value that specifies the converted x-coordinate.
- **Longitude:** Double value that specifies the converted y-coordinate.
- **Scale:** Double value that specifies the zoom level for the map image.
- **Size:** Image size in pixels.

For converting system coordinates into Projected Coordinate System (PCS) latitude and longitude, a coordinate converter is required. The default coordinate converter, SampleCoordinateConverter.CoordinateConverter stored in NetUtil.dll, enables conversion based on the North American Datum of 1983 (NAD83) geodetic reference system. It may be not appropriate under specific project circumstances; therefore, a custom converter must be developed.

Note

For workstations without internet connectivity or without access to the specified external maps, the incident location screenshot is taken from the geographic network view.

- **Map Scale Factor:** The scaling factor for the incident location on the web map. Note that the Google Map API does not support fractional numbers such as 10.5.
- **Maximum Calls Number:** The maximum number of calls associated with the incident that can be listed in the PDF file.
- **Prefer Web Maps:** If set to True, the system attempts to take the incident location screenshot from an internet map, and then uses the geographic network view if the internet map is not available. If set to False, the system only takes the incident location screenshot from the geographic network view.
- **Prefer Web Maps Multiple Export:** If set to True, the system attempts to take incident location screenshots from an internet map, and then uses the geographic network view if the internet map is not available. If set to False, the system only takes incident location screenshots from the geographic network view.

For more information about configuring the PDF template, see [Customizing the Incident PDF Export](#).

10.10. Logging

Logging	
EnableLogging	True
EnablePerformanceLogging	False
InMemoryLogMessageCount	5000
LogToFG	True
PerformanceLoggingMin	10

This section contains configuration options for the logging.

- **EnableLogging:** The common log records OMS Display plug-in activities, such as loading a configuration file or opening a display on the Log Messages tab of the Admin Displays group. If True, logging is enabled. If False, logging is disabled.
- **EnablePerformanceLogging:** The performance log records the duration of OMS activity execution on the Log Messages tab of the Admin Displays group. If True, performance logging is enabled. If False, performance logging is disabled.
- **InMemoryLogMessageCount:** The maximum number of messages in the performance log file. When the log file reaches the limit, it is renamed, and a new empty log file is created.
- **LogToFG:** If True, logging to the GE Browser application is enabled.
- **PerformanceLoggingMin:** The minimum elapsed time to log performance counter, in milliseconds.

10.11. Misc

Misc	
FileName	D:\Eterra\Distribution\FantasyIsland\config\client\OmsDisplayPluginConfig.xml

This section contains the following option:

- **FileName:** Full path to the OMS Display Plugin file.

10.12. OMS Data Service

OMS Data Service	
Auto Find OMS Data Service	False
Data Service Url A	http://localhost:8800/Oms
Data Service Url B	http://localhost:8801/Oms
Data Service Url C	http://localhost:8802/Oms
Data Service Url D	http://localhost:8803/Oms

Data service URL is used by the OMS client to connect to the OMS data service on the Net Data server, which is necessary for some features, such as history correction.

- **Auto Find OMS Data Service:** If True, the OMS client automatically selects the URL on the Net Data server to connect to. If false, the OMS client uses the URLs specified by the Data Service

URL parameters.

For redundancy purposes, the OMS client supports four data service URLs:

- Data Service URL A
- Data Service URL B
- Data Service URL C
- Data Service URL D

The URL format is `http://<host_name>:<port_number>/OMS`.

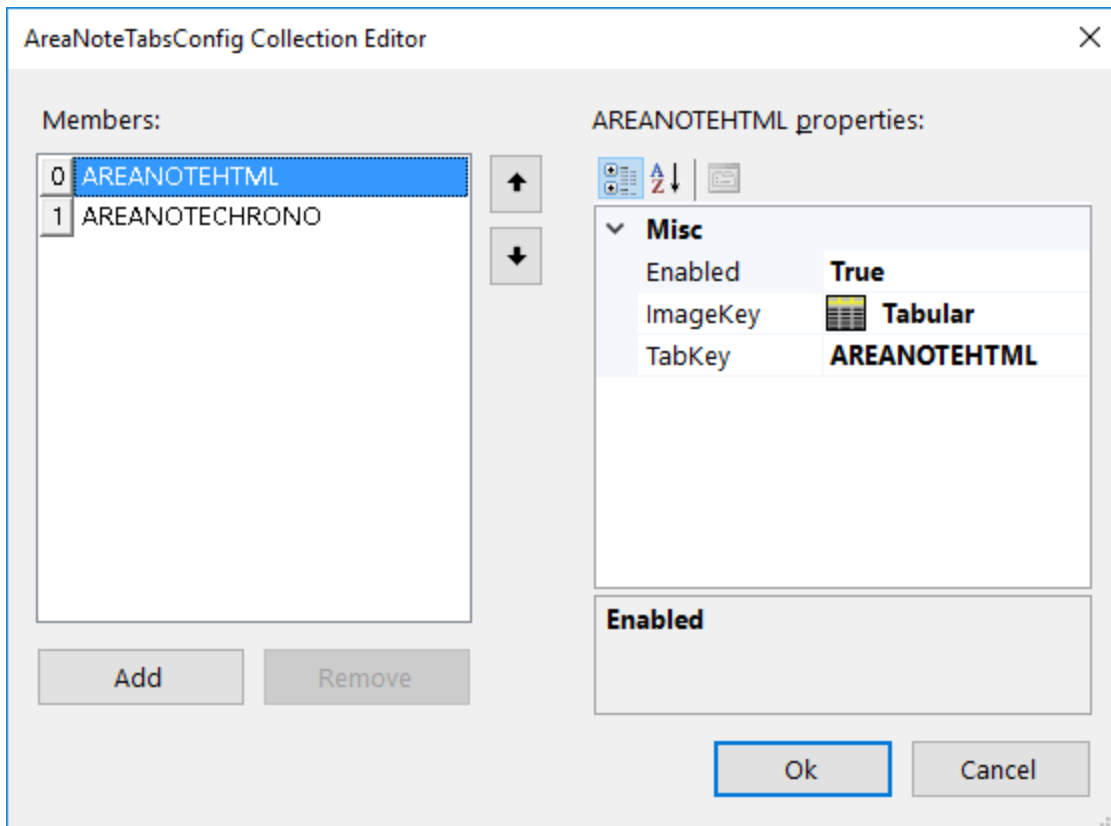
If a data service URL can't be accessed, the OMS client will try to connect to the next URL in the list, and so on.

10.13. Update Area Settings

Update Area Note	
UpdateAreaNotePopupHeight	600
UpdateAreaNotePopupMinHeight	600
UpdateAreaNotePopupMinWidth	850
UpdateAreaNotePopupWidth	850
UpdateAreaNoteTabs	(Collection)

This section contains configuration options for the Update Area Settings display (see below image) used to create and modify area settings.

- **UpdateAreaNotePopupHeight:** Specifies the default height of the Update Area Settings display.
- **UpdateAreaNotePopupMinHeight:** Specifies the minimum height of the Update Area Settings display.
- **UpdateAreaNotePopupMinWidth:** Specifies the minimum width of the Update Area Settings display.
- **UpdateAreaNotePopupWidth:** Specifies the default width of the Update Area Settings display.
- **UpdateAreaNoteTabs:** Configures the Update Area Settings display tabs. Clicking the ellipsis  button opens the AreaNoteTabsConfig Collection Editor dialog box.



Each item has the following parameters:

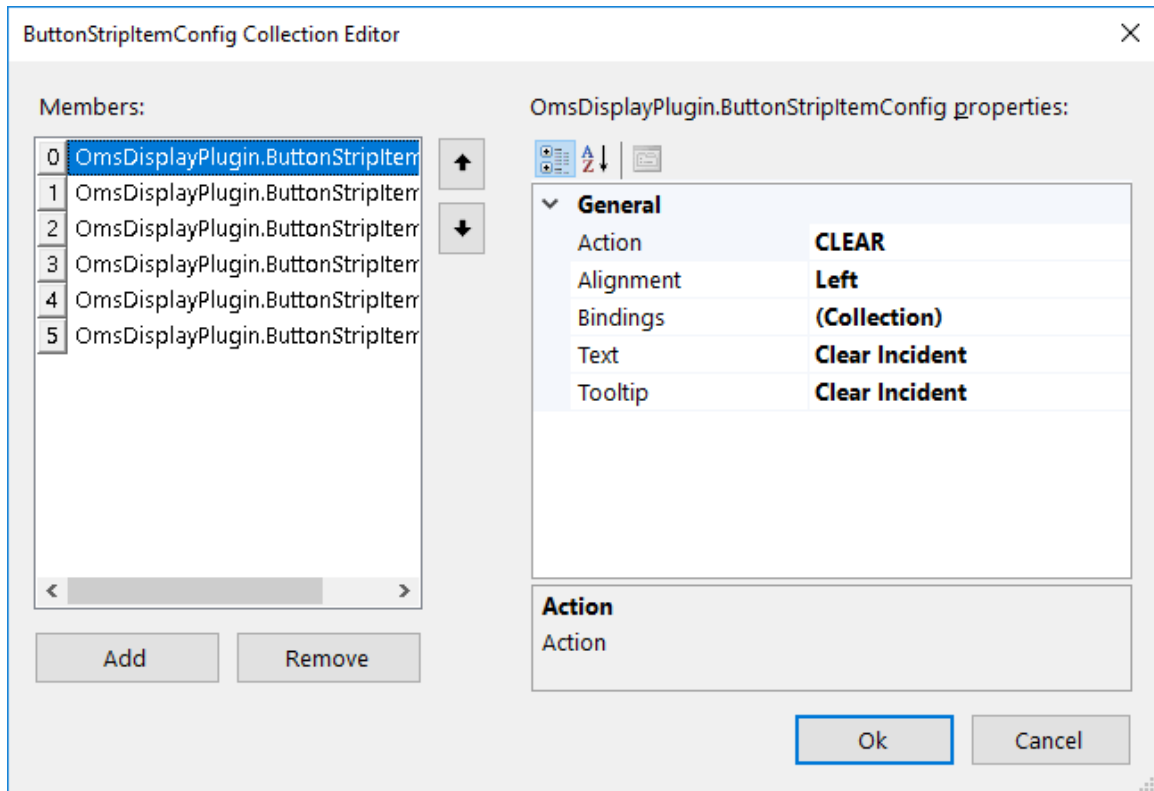
- **Enabled:** If true, this tab appears on the Update Area Settings display.
- **ImageKey:** Sets the image from the ImageList to appear on the tab.
- **TabKey:** ID of the tab.

Figure 16. Update Area Settings Display

10.14. Update Incident

This section contains configuration options for the Update Incident display used to edit incidents. The display can have either a toolbar or a set of buttons.

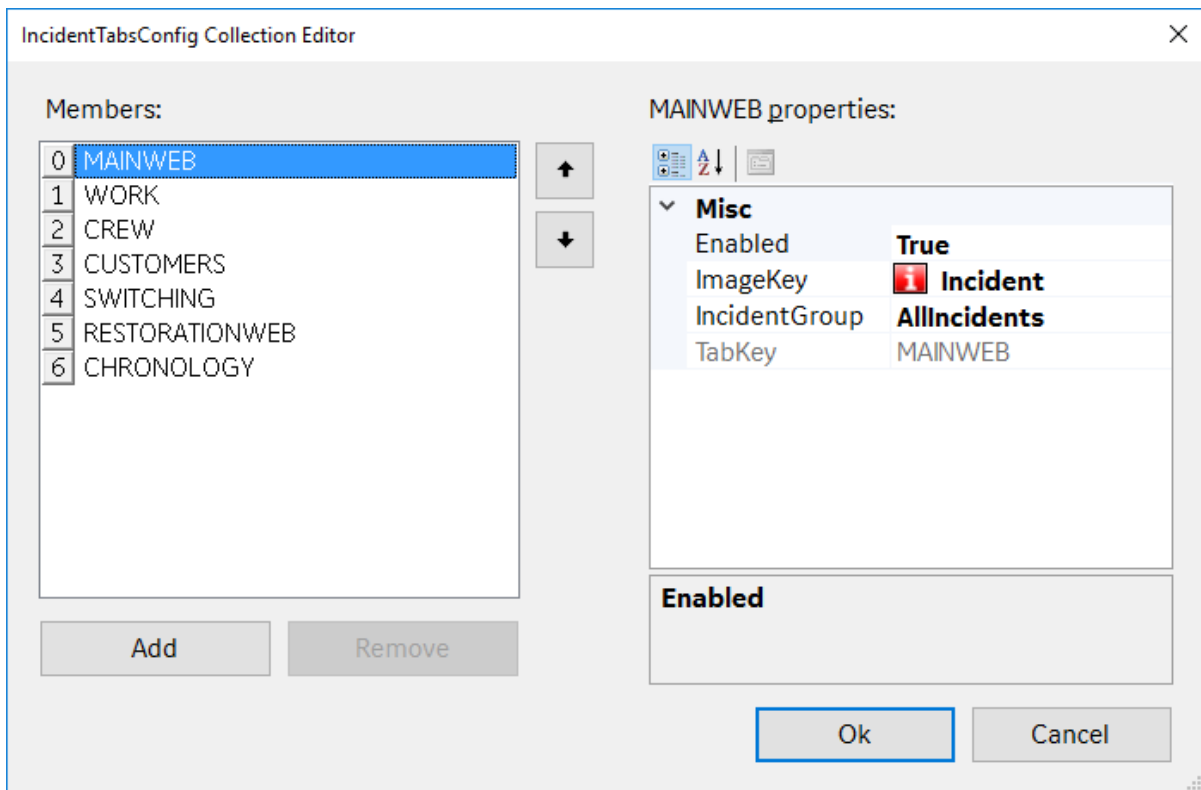
- **AllowMultipleIncidentTabs:** If True, multiple Update Incident displays can be open at the same time.
- **BlockPredictionImage:** The image for the Block/Unblock Prediction button.
- **ConfirmStateImage:** The image for the Confirm Device State button.
- **EditableSystemAffected:** If True, the System Affected field on the Properties tab of the Update Incident display can be updated by the user.
- **EnableSwoLink:** If True, the Switching Orders field is shown on the Properties tab of the Update Incident display.
- **IncidentButtonStripConfig:** The buttons to show along the bottom or top of the display window. Click to open the configuration options.
 - **ButtonHeight:** Height of the buttons.
 - **Enabled:** If true, the button strip is displayed.
 - **Items:** Click to open the ButtonStripItemConfig Collection Editor dialog box.



- **Action:** URL or action to call when the button is clicked.
- **Alignment:** Specifies left or right alignment. Buttons are added in the order they are shown, and there can be both left and right aligned buttons.
- **Bindings:** Data bindings with the button properties, such as disabling a button by binding to the Enabled property.
- **Text:** Button text to display.

- **Tooltip:** Tooltip to display.
 - **Location:** Location of the button strip (top or bottom).
- **IncidentTabStyle:** Specifies the Update Incident display to use normal tabs or an accordion control.
- **IncidentToolBarConfig:** Specifies the toolbar buttons to appear at the top of the display window.
 - **Enabled:** If true, the toolbar is displayed.
 - **ImageSize:** Size of toolbar icons.
 - **Items:** Clicking the ellipsis  button opens the MenuItemBase Collection Editor dialog box.
- **LocateCrewCmd:** Specifies the command for navigating to the primary crew from the Update Incident display.
- **LocateDeviceCmd:** Specifies the command for navigating to the device of record from the Update Incident display.
- **LocateIncidentCmd:** Specifies the command for navigating to the incident from the Update Incident display.
- **LocateLastOpenCmd:** Specifies the command for navigating to the last open device from the Update Incident display.
- **NewIncidentTabMode:** Defines whether the Update Incident display is shown as a pop-up or a tab.
- **ShowCMI:** Controls the display of the CMI field on the Properties tab of the Update Incident display.
- **ShowFirstResponderNeeded:** Controls whether the First Responder Needed field is shown on the Properties tab of the Update Incident display.
- **ShowWorkOrderID:** Controls whether the Work Order ID field is shown on the Properties tab of the Update Incident display.
- **TopIncidentButtonStripConfig:** Defines the appearance of the toolbar with buttons (Acquire Lock, Clear Incident, Cancel Incident, and Assign to WFM).
 - **ButtonHeight:** Button height.
 - **Enabled:** If true, the toolbar is displayed.
 - **Items:** Clicking the ellipsis  button opens the ButtonStripItemConfig Collection Editor.
 - **Location:** Location of the toolbar (Top or Bottom).
- **UpdateIncidentPopupHeight:** Specifies the default height of the Update Incident display.
- **UpdateIncidentPopupMinHeight:** Specifies the minimum height of the Update Incident display.
- **UpdateIncidentPopupMinWidth:** Specifies the minimum width of the Update Incident display.
- **UpdateIncidentPopupWidth:** Specifies the default width of the Update Incident display.
- **UpdateIncidentTabs:** Configures the Update Incident tabs. Clicking the ellipsis  button

opens the IncidentTabsConfig Collection Editor dialog box.



Each item has the following parameters:

- **Enabled:** If true, this tab appears on the Update Incident display.
- **ImageKey:** Sets the image from the ImageList to show on the tab.
- **IncidentGroup:** Sets the incident type, for which this tab appears on the Update Incident display (all incident types, outage incidents, or non-outage incidents).
- **TabKey:** ID of the tab.

Note

To use the old-style Properties tab on the Update Incident display, manually set the TabKey_ReadOnly property value to MAIN in the OmsDisplayPlugin.xml file in the IncidentTabsConfig node.

- **Work Model Service API:** Sets the Work Model Event Handler Service endpoint.
- **Work Model TLS Enabled:** Specifies whether the HTTPS TLS security is enabled for the Work Model Event Handler Service connection for web clients and web socket clients.
- **Work Model TLS Ignore Remote Name Mismatch:** Specifies whether the remote server certificate name mismatch is ignored. If set to True, the remote server certificate name can be different from the URL host name that is used to connect to the server.
- **Work Model TLS Subject Name:** Specifies the certificate subject name that is used when

searching for client certificates to send to the Work Model Event Handler Service.

- **Work Model Web Socket Connection Retry Delay:** Specifies the time period in seconds to wait before re-initiating a web socket connection if the connection had not been established.
- **Work Model Web Socket For Task Transition Only:** Specifies whether the Work Model Event Handler Service web socket is only used for task status updates, because the device status update results in a device status change. If set to True, the request is sent through the web socket to the Work Model Event Handler Service, and the device state change request is sent back to the client instead of the OMS Interface Server. If set to False, the Work Model Event Handler Service web socket can be used for other operations, such as updating the work order status, assigning a crew, and adding a task.
- **Work Model Web Socket Keep Alive Interval in Minutes:** Time period in minutes that the Work Model Event Handler Service web socket waits without receiving an appropriate message from the remote before disconnecting. Setting this value to zero disables the keep-alive logic.
- **Work Model Web Socket Request Timeout in Seconds:** Time period in seconds to wait for the client response for an HTTP request.
- **Work Model Web Socket Url:** Sets the Work Model Event Handler Service web socket client URL.

10.15. Update Work Order

▼ Update Work Order	
Allow Multiple Order Worksheet Tabs	False
New Order Worksheet Mode	Popup
> Order Worksheet Button Strip Config	OmsDisplayPlugin.ButtonStripConfig
Order Worksheet Popup Height	850
Order Worksheet Popup Min Height	800
Order Worksheet Popup Min Width	1650
Order Worksheet Popup Width	1700
Order Worksheet Tabs	
> Order Worksheet Tool Bar Config	OmsDisplayPlugin.ToolStripConfig
Show Crew Statuses In Crew Display	True
> Top Order Worksheet Button Strip Config	OmsDisplayPlugin.ButtonStripConfig
Work Order Tab Style Display Name	Tabs

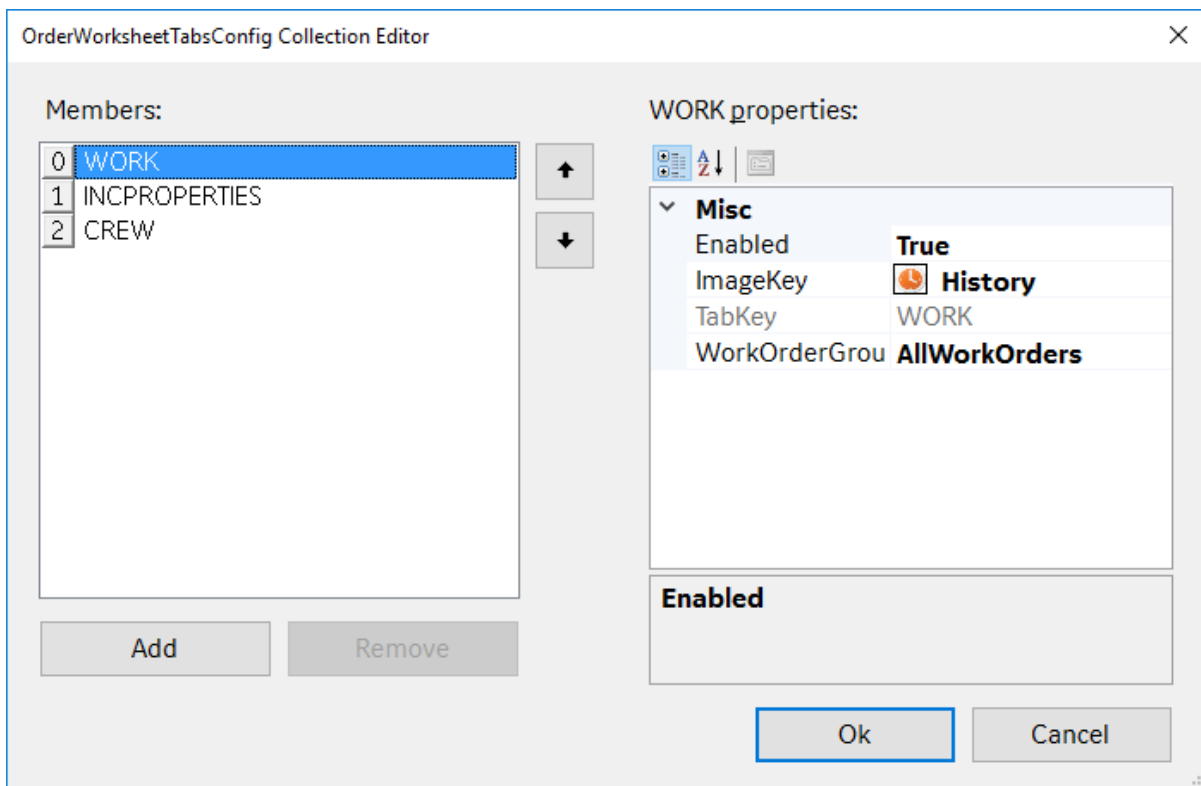
Note

Order functionality requires integration with the Enterprise OMS module.

This section contains configuration options for the Order Worksheet display used to edit orders. The display can have either a toolbar or a set of buttons.

- **AllowMultipleWorksheetTabs:** If True, multiple Order Worksheet displays can be open at the same time.

- **NewOrderWorksheetMode:** Defines whether the Order Worksheet display is shown as a pop-up or a tab.
- **OrderWorksheetButtonStripConfig:** Defines the appearance of the toolbar with buttons.
 - **ButtonHeight:** Height of the buttons.
 - **Enabled:** If true, the toolbar is displayed.
 - **Items:** Clicking the ellipsis  button opens the ButtonStripItemConfig Collection Editor dialog box.
 - **Location:** Location of the toolbar (Top or Bottom).
- **OrderWorksheetPopupHeight:** Specifies the default height of the Order Worksheet display.
- **OrderWorksheetPopupMinHeight:** Specifies the minimum height of the Order Worksheet display.
- **OrderWorksheetPopupMinWidth:** Specifies the minimum width of the Order Worksheet display.
- **OrderWorksheetPopupWidth:** Specifies the default width of the Order Worksheet display.
- **OrderWorksheetTabs:** Configures the Order Worksheet tabs. Clicking the ellipsis  button opens the WorkOrderTabsConfig Collection Editor dialog box.



Each item has the following parameters:

- **Enabled:** If true, this tab appears on the Order Worksheet display.
- **ImageKey:** Sets the image from the ImageList to show on the tab.

- **TabKey:** ID of the tab.
- **WorkOrderGroup:** Sets the work order type, for which this tab appears on the Order Worksheet display.
- **WorkOrderToolBarConfig:** Specifies the toolbar buttons to appear at the top of the display window.
 - **Enabled:** If true, the toolbar is displayed.
 - **ImageSize:** Size of toolbar icons.
 - **Items:** Clicking the ellipsis  button opens the MenuItemBase Collection Editor dialog box.
- **ShowCrewStatusesInCrewDisplay:** Specifies whether the Crew Shift Status, Availability Status, and Mobile Communication Status fields are visible in the crew display on the Order Worksheet.
 - **Enabled:** If true, the crew statuses are displayed.
- **TopOrderWorksheetButtonStripConfig:** Defines the appearance of the toolbar with buttons.
 - **ButtonHeight:** Height of the buttons.
 - **Enabled:** If true, the toolbar is displayed.
 - **Items:** Clicking the ellipsis  button opens the ButtonStripItemConfig Collection Editor dialog box.
 - **Location:** Location of the toolbar (Top or Bottom).
- **WorkOrderTabStyle:** Specifies whether the Order Worksheet display uses normal tabs or an accordion control.

11. Customizing the Incident PDF Export

You can export incident properties in PDF. The PDF generation process is as follows:

1. Incident export XML and XSLT documents are fed to an XSLT engine.

Note

The incident export XML file is generated and used internally. It is not saved in the file system. For an example of the file, see [Sample Incident Export XML File](#).

2. Using the rules defined in the XSLT document, the XSLT engine transforms the XML document to an XSL-FO (XSL Formatting Objects) document.
3. The XSL-FO document, which is usually stored only in memory, is fed to an FO engine.
4. The FO engine uses formatting rules defined in the XSL-FO document to render a printable document as a PDF file.

The below image shows the general Incident to PDF export workflow.

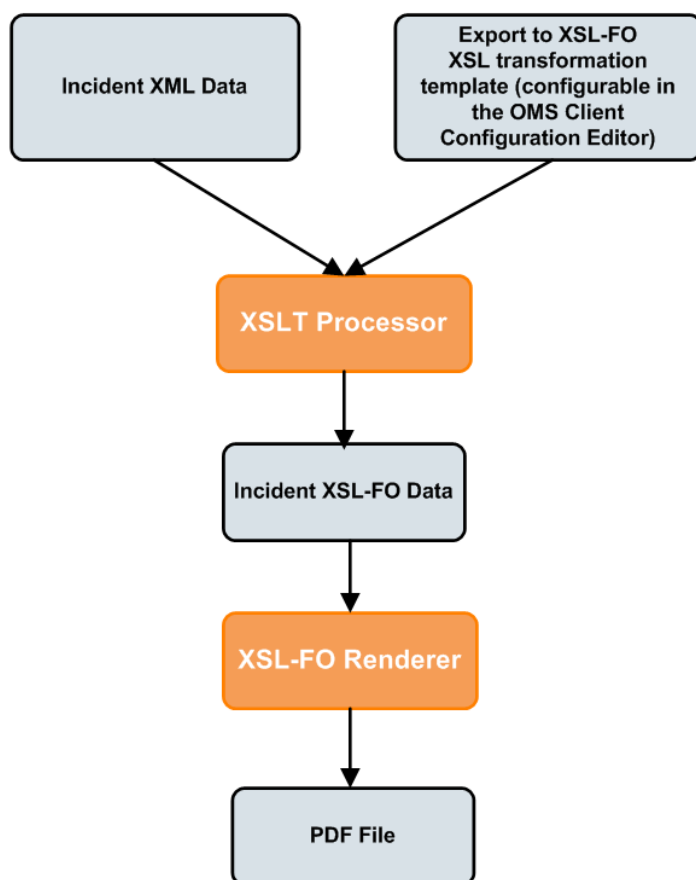


Figure 17. Incident to PDF Export Customization Diagram

The incident PDF conversion template is defined in the `incident2pdf.xml` configuration file. You can

create or edit XSL files using text editors or XSL and XSL-FO graphical editors with "What You See is What You Get" (WYSIWYG) functionality.

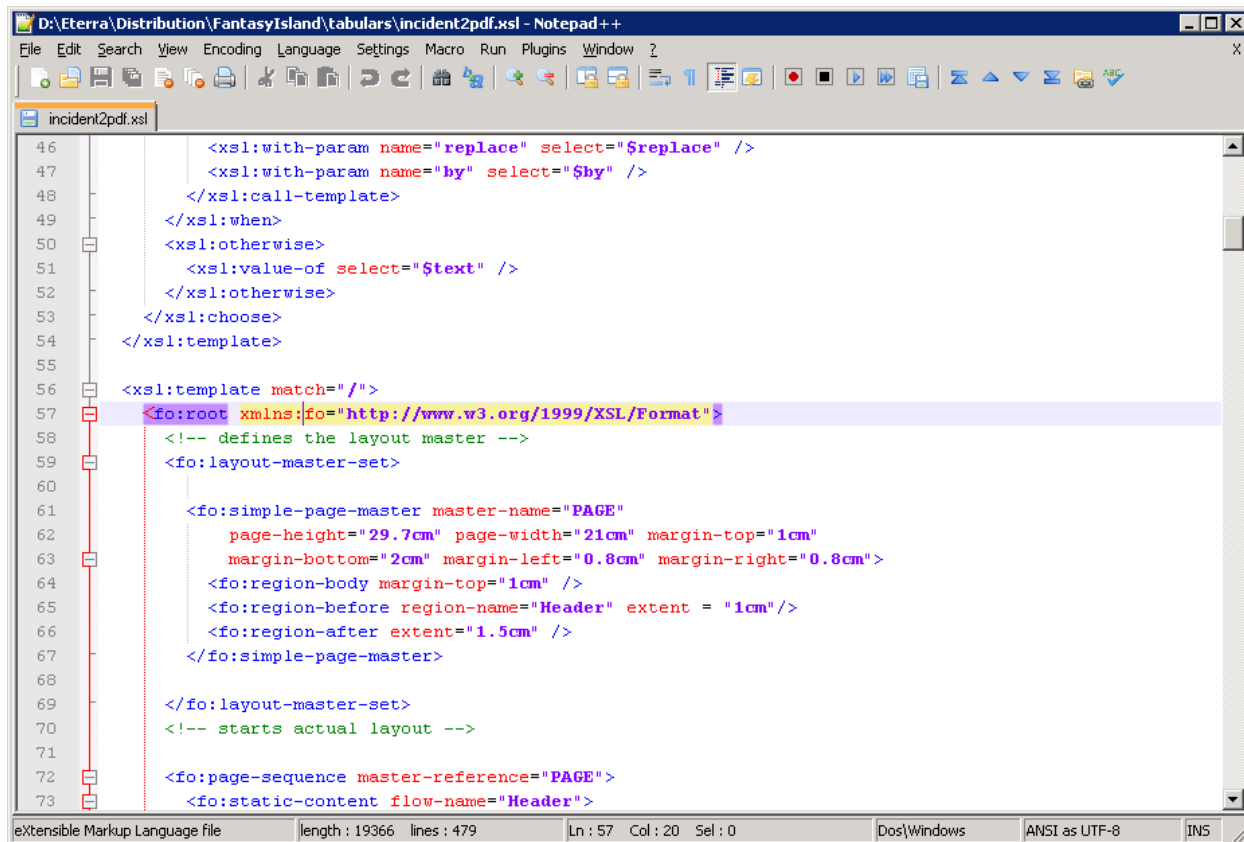


Figure 18. incident2pdf.xsl Configuration File

In the incident2pdf.xsl configuration file, elements in the `xmlns:fo` namespace define the PDF layout. For example:

- **fo:layout-master-set:** Specifies the name, height, and width of the master page, and specifies the margins and orientation (portrait or landscape) of the entire page. (For more information, refer to <https://www.webucator.com/xml-training/course/introduction-xsl-fo/>.)
- **fo:block:** Specifies the formatting of paragraphs, titles, headlines, figure captions, and table captions. (For more information, refer to http://w3schools.sinsixx.com/xslfo/xslfo_blocks.asp.htm.)
- **fo:inline:** Applies formatting to text within a block. (For more information, refer to <https://www.webucator.com/xml-training/course/introduction-xsl-fo/>.)
- **fo:table:** Defines a table. Common attributes include table-layout (fixed, auto, or inherit), border-width, table columns and cells, and background color. (For more information, refer to <https://www.webucator.com/xml-training/course/introduction-xsl-fo/>.)

Elements in the `xmlns:xsl` namespace define transformations of the original incident XML file. For example:

- **xsl:value-of:** Inserts a text value. The `select` attribute contains an XPath expression. (For more

information, refer to https://www.w3schools.com/xml/xsl_value_of.asp.)

i Note

The incident field names must be defined in uppercase (as they are defined in the XML file). For the list of available fields, refer to [Sample Incident Export XML File](#).

- **xsl:if:** Applies a conditional test against the content of the XML file. (For more information, refer to http://www.w3schools.com/xml/xsl_if.asp.)

Incident Number: 2140000022

08/02/2017 03:46:08

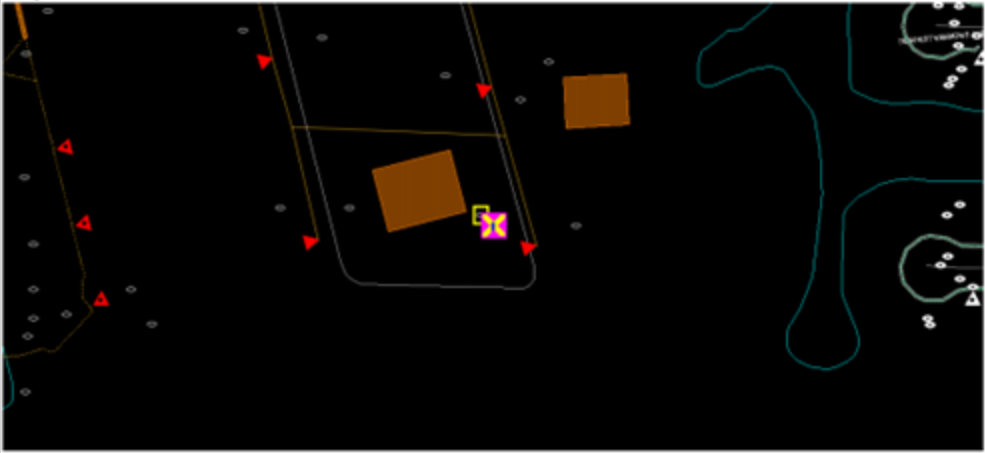
Summary				
Duration: 0.0 h	Start Time: 08/02/2017 3:44 AM	Crews: Larive [5290] - AUTO		
Calls: 1	First Call: 08/02/2017 3:44 AM			
Affected: 1	End Time:			
Priority: 0	ETR: 08/02/2017 6:00 AM			
Device				
Device Name: 6-7313-4799-0-7	Division: NORTH	Feeder: F8666		
ID: 22067493	Op Center: WST	Phase: A		
Type: Transformer	Substation: COHO	Responsible System: Overhead Distribution		
State: Closed				
Damage Assessment				
Poles Down: 1	Transformers: 3	Wire Down: 2		
Remarks: Operator Remarks				
Dispatcher Comments:				
Crew Comments: Crew Comments				
Map:				
				
Calls				
Name	Account #	Address	Phone Number	Comments
Myers Freddie O	555047920	9234 CHERRY SW	(555) 567-2453	

Figure 19. Incident Details PDF Document

11.1. Default PDF Export Template

The default PDF template defined in the `incident2pdf.xml` configuration file includes the following sections:

- **Header:** Defines the header of the incident detail document. Contains the following fields:
 - Incident Number
 - Incident Export Time Stamp

The document header section is formatted using the `<fo:static-content>` element. It is shown as a header on each document page. The height of the header section is formatted by the `<fo:region-before>` element.

- **Table 1:** Provides incident details and the map. Contains the following fields:
 - Summary
 - Duration
 - Start Time
 - Crews
 - Calls
 - First Call
 - Affected
 - End Time
 - Priority
 - ETR
 - Device
 - Device Name
 - Division
 - Feeder
 - ID
 - Service Center
 - Phase
 - Type
 - Substation
 - System Affected
 - State
 - Damage Assessment
 - Poles Down

- Transformers
- Wire Down
- Remarks
- Dispatcher Comments
- Crew Comments
- Map

The Table 1 section is formatted using a table (<fo:table>) with 3 columns. Column widths are set by the corresponding <fo:table-column> elements. Column captions are set by the <fo:table-cell> elements. Header cells (for example, Summary and Device) are formatted using the number-columns-spanned field.

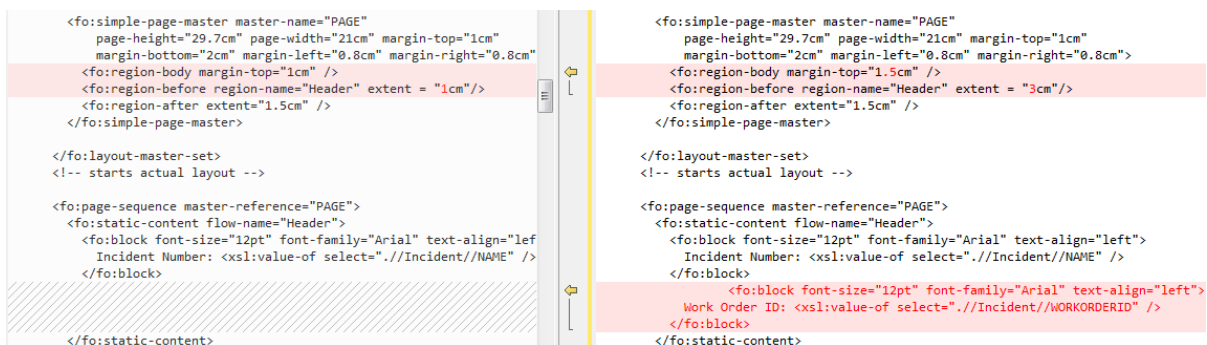
• Table 2: Calls

The Table 2 section is formatted using a table (<fo:table>) with 5 columns. The data for separate calls is populated from the incident export XML file using the <xsl:for-each> element.

11.2. Adding Header Fields to the PDF Export Template

To add a header field in the Incident to PDF conversion template:

1. In the PDF export template file (incident2pdf.xsl), increase the height of the header section from 1 cm to 3 cm using the <region-before> element. This allocates a new header field.



2. Increase the margin between the header section and the next table from 1 cm to 1.5 cm using the <fo:region-body margin-top> element.
3. In the <fo:static-content> element, add a new row using the <fo:block> sub-element.
4. Define the name and value of the new header field as follows:

```
Work Order ID: <xsl:value-of select="//Incident//WORKORDERID">
```

The label is defined by the plain text "Work Order ID:". The value is defined by <xsl:value-of select="//Incident//WORKORDERID", where WORKORDERID is the name of the corresponding field in the incident export XML file. For the list of all incident fields, see [Sample Incident Export XML](#)

File.

5. Export the incident details to a PDF and verify the new field.

The Work Order ID field appears in the PDF next to the Incident Number field.

Incident Number: 0010000006		
Work Order ID: WO1234		
Summary		
Duration: 23.8 h	Start Time: 04/01/2017 10:34 AM	Crews: Thompson (1184), Dodge (1101)
Calls: 2	First Call: 04/01/2017 2:10 PM	

11.3. Adding Column Fields to the PDF Export Template

To add a column in the Incident to PDF conversion template:

1. In the PDF export template file (incident2pdf.xml), add a new row in the table definition or modify an existing empty row.

In the example below, the empty field next to the First Call field is modified to show Work Order ID.

<pre><fo:table-row> <fo:table-cell border-style="solid" padding="1mm"> <fo:block text-align="left"> <fo:inline font-weight="bold">Calls: </fo:inline> <xsl:value-of select="//Incident//CALLCOUNT" /> </fo:block> </fo:table-cell> <fo:table-cell border-style="solid" padding="1mm"> <fo:block text-align="left"> <fo:inline font-weight="bold">First Call: </fo:inline> <xsl:value-of select="locdt:GetDateTime(//Incident//FIRSTCALL)" /> </fo:block> </fo:table-cell> <fo:table-cell border-style="solid" padding="1mm"> <fo:block text-align="left"> <fo:inline font-weight="bold">Work Order ID: </fo:inline> <xsl:value-of select="//Incident//WORKORDERID" /> </fo:block> </fo:table-cell> </fo:table-row></pre>	<pre><fo:table-row> <fo:table-cell border-style="solid" padding="1mm"> <fo:block text-align="left"> <fo:inline font-weight="bold">Calls: </fo:inline> <xsl:value-of select="//Incident//CALLCOUNT" /> </fo:block> </fo:table-cell> <fo:table-cell border-style="solid" padding="1mm"> <fo:block text-align="left"> <fo:inline font-weight="bold">First Call: </fo:inline> <xsl:value-of select="locdt:GetDateTime(//Incident//FIRSTCALL)" /> </fo:block> </fo:table-cell> <fo:table-cell border-style="solid" padding="1mm"> <fo:block text-align="left"> <fo:inline font-weight="bold">Work Order ID: </fo:inline> <xsl:value-of select="//Incident//WORKORDERID" /> </fo:block> </fo:table-cell> </fo:table-row></pre>
--	--

The bold formatting of the field label is defined by the `<fo:inline font-weight="bold">` element.

2. Export the incident details to a PDF and verify the new field.

The Work Order ID field appears in the PDF file next to the First Call field.

Summary		
Duration: 24.0 h	Start Time: 04/01/2017 10:34 AM	Crews: Thompson (1184), Dodge (1101)
Calls: 2	First Call: 04/01/2017 2:10 PM	Work Order ID: WO1234
Affected: 1342	End Time:	
Priority: 0	ETR: 04/01/2017 5:30 PM	

The conversion engine wraps text at the location of spaces to fit the defined column width. To avoid truncating values that do not contain spaces and exceed the specified column width, use the following custom formatting. This places hidden spaces after each symbol in the value. Note that only UTF-8 fonts (for example, Arial) are supported.

```
<xsl:variable name="myVar">
  <xsl:call-template name="zero_width_space_1">
    <xsl:with-param name="data" select="//Incident//OPENDEVICEID"/>
  </xsl:call-template>
</xsl:variable>
<xsl:call-template name="string-replace-all">
  <xsl:with-param name="text" select="$myVar" />
  <xsl:with-param name="replace" select="'&#8203; &#8203;'" />
  <xsl:with-param name="by" select="' '" />
</xsl:call-template>
```

11.4. Sample Incident Export XML File

This section provides a sample incident export XML file. Note that this file is generated and used internally. It is not available for review or debugging.

Any of the incident-related fields, except for those highlighted in yellow, can be used in the PDF export template.

```
<ExportData>
  <Incident>
    <GUID>5c10265a-8080-46ba-bead-dc5aa62ea822</GUID>
    <NAME>0740000005</NAME>

    <ASSOCRESTORATIONSTEPS>System.Collections.ObjectModel.ReadOnlyCollection`1[OmsUtilities.AssociateRestorationStep]</ASSOCRESTORATIONSTEPS>
    <PT>{X=932643.8, Y=811575.1}</PT>
    <PROTECTIVESTATION>RAVEN</PROTECTIVESTATION>
    <PROTECTIVEDEVICEID>17824971</PROTECTIVEDEVICEID>
    <PROTECTIVEDEVICENAME>6-7512-2630-0-1</PROTECTIVEDEVICENAME>
    <CLEARED>False</CLEARED>
    <STARTTIME>5247937977197210926</STARTTIME>
    <ACKNOWLEDGEDTIME>4611686018427387904</ACKNOWLEDGEDTIME>
    <FIRSTCALLTIME>5247937977197210926</FIRSTCALLTIME>
    <CALLCOUNT>16</CALLCOUNT>
    <AMICOUNT>0</AMICOUNT>
```

```

<AMIOUTCOUNT>0</AMIOUTCOUNT>
<AMIINCOUNT>0</AMIINCOUNT>
<OVERRIDEPREDICTEDSTABLE>True</OVERRIDEPREDICTEDSTABLE>
<ISPREDICTEDSTABLE>False</ISPREDICTEDSTABLE>
<STATION>RAVEN</STATION>
<FEEDERNAME>F8864</FEEDERNAME>
<CANCELLED>False</CANCELLED>
<REPREDICTIONNEEDED>False</REPREDICTIONNEEDED>
<RECEIVEDCOMPLETIONFROMWFM>False</RECEIVEDCOMPLETIONFROMWFM>
<POINTOFFFAILURE></POINTOFFFAILURE>
<OPENDEVICEPT>{X=932593.8, Y=811525.1}</OPENDEVICEPT>
<OPENDEVICEPTX>932593.8</OPENDEVICEPTX>
<OPENDEVICEPTY>811525.1</OPENDEVICEPTY>
<OPENDEVICESTATION>RAVEN</OPENDEVICESTATION>
<OPENDEVICEID>17824971</OPENDEVICEID>
<OPENDEVICENAME>6-7512-2630-0-1</OPENDEVICENAME>
<OPENDEVICETYPE>16</OPENDEVICETYPE>
<OPENSWITCHTYPE>-1</OPENSWITCHTYPE>
<CREWASSIGNMENTFAILED>False</CREWASSIGNMENTFAILED>
<EXCLUDEFROMSTATISTICS>False</EXCLUDEFROMSTATISTICS>
<WORKORDERID></WORKORDERID>
<PRIORITYCALLSCOUNT>0</PRIORITYCALLSCOUNT>
<OPENDEVICEIDSPHASEA></OPENDEVICEIDSPHASEA>
<OPENDEVICEIDSPHASEB>17824971</OPENDEVICEIDSPHASEB>
<OPENDEVICEIDSPHASEC></OPENDEVICEIDSPHASEC>
<CLOSEDDEVICEIDSPHASEA></CLOSEDDEVICEIDSPHASEA>
<CLOSEDDEVICEIDSPHASEB></CLOSEDDEVICEIDSPHASEB>
<CLOSEDDEVICEIDSPHASEC></CLOSEDDEVICEIDSPHASEC>
<ARMSSTATUS>10</ARMSSTATUS>
<INCIDENTSTATUS>3</INCIDENTSTATUS>
<CLEARTIME>4611686018427387904</CLEARTIME>
<CLEARINGOPERATOR></CLEARINGOPERATOR>
<ASSISTANCENEEEDED>False</ASSISTANCENEEEDED>
<ENDTIME>4611686018427387904</ENDTIME>
<LASTAMIINTIMEREPORTED>4611686018427387904</LASTAMIINTIMEREPORTED>
<OPENDEVICEISSCADADEVICE>False</OPENDEVICEISSCADADEVICE>
<HASMEASUREMENT>False</HASMEASUREMENT>
<HASINTERNAL>False</HASINTERNAL>
<HASBEENCONFIRMED>False</HASBEENCONFIRMED>
<EMSNAME>RAVEN</EMSNAME>
<MEASNAME></MEASNAME>
<AMICALLCOUNT>0</AMICALLCOUNT>
<CASENOTE>301</CASENOTE>
<CALCULATEDCASENOTE>301</CALCULATEDCASENOTE>
<MANUALCASENOTEOVERRIDE>False</MANUALCASENOTEOVERRIDE>
<ISDEVICEOFRECORDOPEN>False</ISDEVICEOFRECORDOPEN>
<FROMCALLBACK>False</FROMCALLBACK>
<ORIGINALDEVICENAME>6-7512-2630-0-1</ORIGINALDEVICENAME>
<ORIGINALDEVICEID>17824971</ORIGINALDEVICEID>
<ORIGINALDEVICETYPE>16</ORIGINALDEVICETYPE>
<ORIGINALSWITCHTYPE>-1</ORIGINALSWITCHTYPE>
<OUTAGEISLANDCRC>System.Collections.Generic.HashSet`1[System.UInt32]</OUTAGEISLANDCRC>
<ASSIGNEDCREWS>NetUtil.AssignedCrewsList</ASSIGNEDCREWS>

```

```

<DIVISION>NORTH</DIVISION>
<AOR>RAVENSW</AOR>
<OPCENTER>WST</OPCENTER>
<INCIDENTTYPE>2</INCIDENTTYPE>
<PRIORITYCODE>0</PRIORITYCODE>
<OMSDATASERVICE></OMSDATASERVICE>
<NOOFPOLESDOWN>1</NOOFPOLESDOWN>
<NOSERVICESDOWN>4</NOSERVICESDOWN>
<NOOFTRANSFORMERS>3</NOOFTRANSFORMERS>
<CONDUCTORSPAN>2</CONDUCTORSPAN>
<FOLLOWUPREQUIRED>False</FOLLOWUPREQUIRED>
<CALLBACKTYPE>2</CALLBACKTYPE>
<INACCESSIBLE>False</INACCESSIBLE>
<PHASEOUT>B</PHASEOUT>
<OUTAGEDEVICETYPE>Transformer</OUTAGEDEVICETYPE>
<ACTIONTAKEN>0</ACTIONTAKEN>
<MATERIAL>0</MATERIAL>
<OUTAGETYPE>0</OUTAGETYPE>
<CURRENTCUSTOMER>24</CURRENTCUSTOMER>
<CURRENTCRITICALCUSTOMER>0</CURRENTCRITICALCUSTOMER>
<DISPATCHERCOMMENTS>dispatcher comments</DISPATCHERCOMMENTS>
<CREWCOMMENTS>Crew text</CREWCOMMENTS>
<REMARKS>remarks text</REMARKS>
<CREATEDAFTERTHEFACT>False</CREATEDAFTERTHEFACT>
<CASENOTEREMARKS></CASENOTEREMARKS>
<LINECREWNEEDED>False</LINECREWNEEDED>
<TREECREWNEEDED>False</TREECREWNEEDED>
<FIRSTRESPONDERNEEDED>False</FIRSTRESPONDERNEEDED>
<EVENTOFINTEREST>False</EVENTOFINTEREST>
<REASON>0</REASON>
<NOTROUBLEFOUND>False</NOTROUBLEFOUND>
<CANCELORDER>False</CANCELORDER>
<MANUALETROVERRIDE>False</MANUALETROVERRIDE>
<ETR>5247938084434662482</ETR>
<CALCULATEDETR>5247938084434662482</CALCULATEDETR>
<CUSTOMERMODIFIEDTIME>5247937977322543458</CUSTOMERMODIFIEDTIME>
<LOCKING></LOCKING>
<KICKEDOUTLOCK></KICKEDOUTLOCK>
<CREATED>5247937977320173221</CREATED>
<MODIFIED>4611686018427387904</MODIFIED>
<ARMSORDERSTATUS>Unassign</ARMSORDERSTATUS>
<SERVICEPOINTS>System.Collections.Generic.List`1[OmsUtilities.ServicePointInfo]</SERVICEPOINTS>

<AFFECTEDCUSTOMERS>System.Collections.ObjectModel.ReadOnlyCollection`1[OmsUtilities.AffectedCustomer]<
/AFFECTEDCUSTOMERS>

<ASSOCSWITCHINGSTEPS>System.Collections.ObjectModel.ReadOnlyCollection`1[OmsUtilities.AssociateSwitchi
ngStep]</ASSOCSWITCHINGSTEPS>
  <UNASSOCCALLS>System.Collections.Generic.List`1[NetUtil.CallMessage]</UNASSOCCALLS>
  <MY911CALLS>System.Collections.Generic.List`1[NetUtil.CallMessage]</MY911CALLS>
  <CALLBACKSENTCUSTOMERS></CALLBACKSENTCUSTOMERS>

<ASSESSMENTLOCATIONS>System.Collections.Generic.Dictionary`2[System.String,OmsUtilities.AssessmentLoca

```

```

tionInfo]</ASSESSMENTLOCATIONS>
  <CENTERPT>{X=0, Y=0}</CENTERPT>
  <PROTECTIVEPT>{X=932593.8, Y=811525.1}</PROTECTIVEPT>
  <RELATEDINCIDENTID></RELATEDINCIDENTID>
  <DELETED>False</DELETED>
  <CALLBACKSENT>0</CALLBACKSENT>
  <PINGSENT>0</PINGSENT>
  <INCIDENTCOUNTER>16</INCIDENTCOUNTER>
  <AFFECTEDCUSTOMERSCOUNT>24</AFFECTEDCUSTOMERSCOUNT>
</Incident>
<Calls>
  <Call>
    <ACCOUNTNUMBER>555065528</ACCOUNTNUMBER>
    <PRIORITYCODE>0</PRIORITYCODE>
    <CUSTOMERKEY>555065528555065528555065528</CUSTOMERKEY>
    <X>932663.80</X>
    <Y>811562.90</Y>
    <CRITICALACCOUNT>00</CRITICALACCOUNT>
    <CALLTYPE>0</CALLTYPE>
    <CONDITION_1>0</CONDITION_1>
    <CONDITION_2>0</CONDITION_2>
    <DESCRIPTION>0</DESCRIPTION>
    <PRIORITY>0</PRIORITY>
    <CUSTOMERNAME>Kleenex Manufacturing Plant</CUSTOMERNAME>
    <AREACODE>555</AREACODE>
    <PHONEPREFIX>590</PHONEPREFIX>
    <PHONESUFFIX>2554</PHONESUFFIX>
    <ADDRESS>8115 MOUNTAIN W</ADDRESS>
    <PRIORITYCALL>False</PRIORITYCALL>
    <PHONECALL>True</PHONECALL>
    <CALLTIME>03/15/2017 17:31:40</CALLTIME>
    <COMMENT></COMMENT>
    <AMROUTAGEREPORT>False</AMROUTAGEREPORT>
    <AMRPOWERONREPORT>False</AMRPOWERONREPORT>
    <LASTAMRREPORTTIME>01/01/0001 00:00:00</LASTAMRREPORTTIME>
    <CONFIRMEDOUT>False</CONFIRMEDOUT>
    <PINGSTATUS></PINGSTATUS>
  </Call>
</Calls>
<Crews>
  <Crew>
    <CREWDYNSTATUS>5</CREWDYNSTATUS>
    <INCIDENTID>0740000005</INCIDENTID>
    <ID>FI1502</ID>
    <RADIONUMBER>1101</RADIONUMBER>
    <AUTOMATICDISPATCHID>11101</AUTOMATICDISPATCHID>
    <ASSIGNEDPT>{X=932543.8, Y=811675.1}</ASSIGNEDPT>
    <CURRENTLOCATION>{X=932543.8, Y=811675.1}</CURRENTLOCATION>
    <NAME>Dodge (1101)</NAME>
    <CREWTYPE>First Responder</CREWTYPE>
    <CREWTYPENAME></CREWTYPENAME>
    <HEADQUARTERS></HEADQUARTERS>
    <STATUS>Assigned</STATUS>
  </Crew>
</Crews>

```

```
<ISAVAILABLE>True</ISAVAILABLE>  
<INCIDENTIDS>0740000005</INCIDENTIDS>  
<EMPLOYEECOUNT>1</EMPLOYEECOUNT>  
<MDTFLAG>False</MDTFLAG>  
<ISTEMPEVALUATOR>False</ISTEMPEVALUATOR>  
<ISEVALUATOR>False</ISEVALUATOR>  
</Crew>  
</Crews>  
</ExportData>
```

12. Customizing the Order PDF Export

You can export order properties in PDF. The PDF generation process is as follows:

1. Order export XML and XSLT documents are fed to an XSLT engine.

Note

The order export XML file is generated and used internally. It is not saved in the file system. For an example of the file, see [Sample Incident Export XML File](#).

2. Using the rules defined in the XSLT document, the XSLT engine transforms the XML document to an XSL-FO (XSL Formatting Objects) document.
3. The XSL-FO document, which is usually stored only in memory, is fed to an FO engine.
4. The FO engine uses formatting rules defined in the XSL-FO document to render a printable document as a PDF file.

The below figure shows the general order to PDF export workflow.

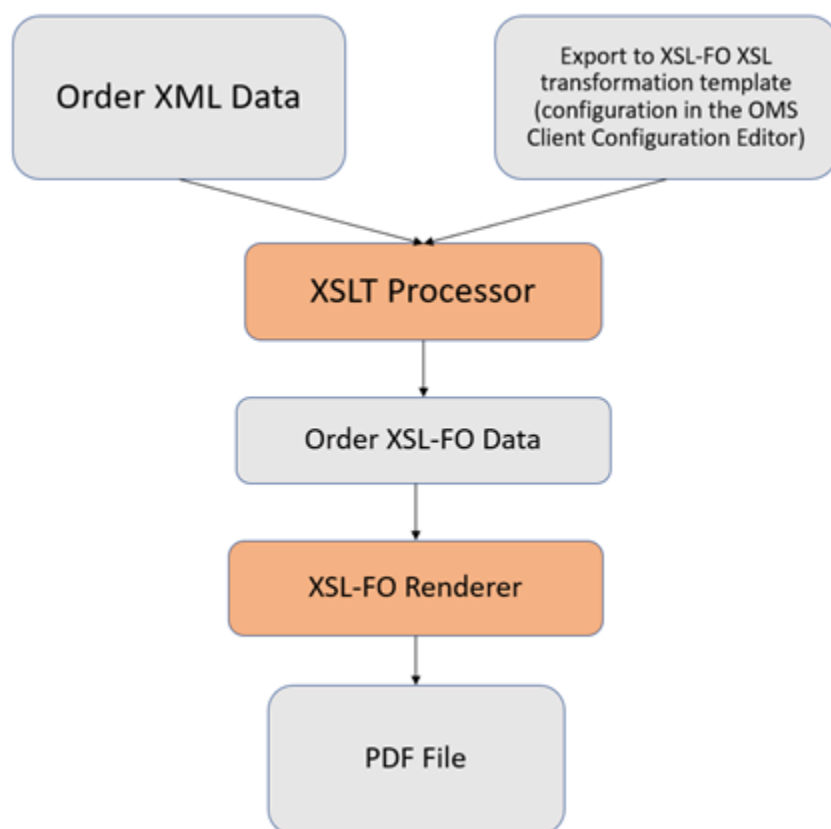


Figure 20. Order to PDF Export Customization Diagram

The order PDF conversion template is defined in the order2pdf.xsl configuration file. You can create or edit XSL files using text editors or XSL and XSL-FO graphical editors with “What You See is What

You Get” (WYSIWYG) functionality.

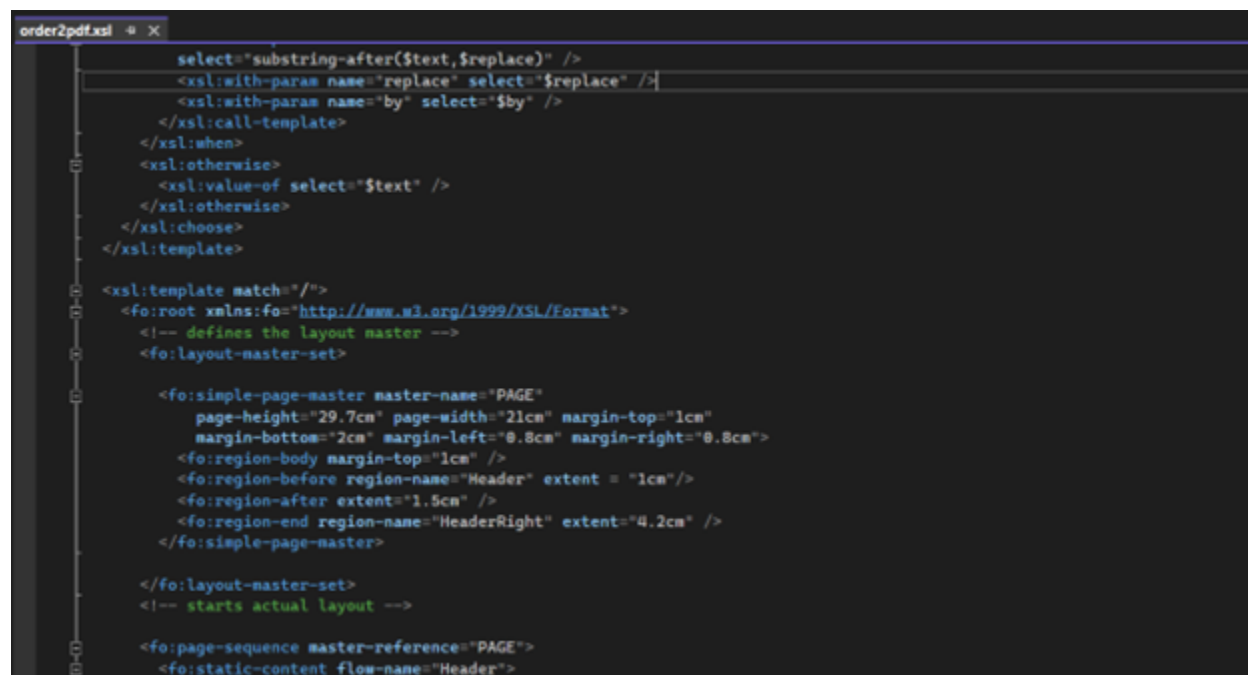


Figure 21. order2pdf.xml Configuration File

In the order2pdf.xml configuration file, elements in the xmlns:fo namespace define the PDF layout. For example:

- **fo:layout-master-set:** Specifies the name, height, and width of the master page, and specifies the margins and orientation (portrait or landscape) of the entire page. (For more information, refer to <https://www.webucator.com/xml-training/course/introduction-xsl-fo/>.)
- **fo:block:** Specifies the formatting of paragraphs, titles, headlines, figure captions, and table captions. (For more information, refer to http://w3schools.sinsixx.com/xslfo/xslfo_blocks.asp.htm.)
- **fo:inline:** Applies formatting to text within a block. (For more information, refer to <https://www.webucator.com/xml-training/course/introduction-xsl-fo/>.)
- **fo:table:** Defines a table. Common attributes include table-layout (fixed, auto, or inherit), border-width, table columns and cells, and background color. (For more information, refer to <https://www.webucator.com/xml-training/course/introduction-xsl-fo/>.)

Elements in the xmlns:xsl namespace define transformations of the original order XML file. For example:

- **xsl:value-of:** Inserts a text value. The “select” attribute contains an XPath expression. (For more information, refer to https://www.w3schools.com/xml/xsl_value_of.asp.)

Note

The incident field names must be defined in uppercase (as they are defined in the XML file). For the list of available fields, refer to [Sample Incident Export XML File](#).

- **xsl:if:** Applies a conditional test against the content of the XML file. (For more information, refer to http://www.w3schools.com/xml/xsl_if.asp.)

Order Name: 3320000001-1

11/28/2023 12:52 PM

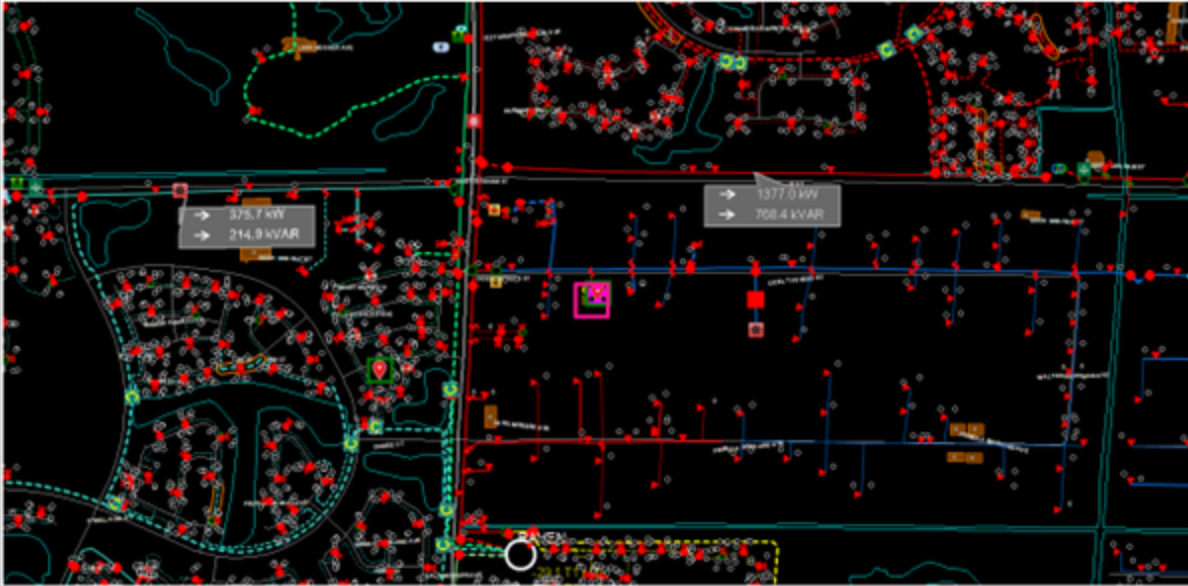
Summary				
Duration: 0.7 h	Start Time: 11/28/2023 12:07 PM	Order Crews:		
Calls: 0	First Call:			
Affected: 2	End Time:			
Priority: 0	ETR: 11/28/2023 12:40 PM			
Device				
Device Name: 6-7512-9972-0-0	Division: ROARKE	Feeder: F6533		
ID: 22057363	Service Center: 3LIONSSC	Phase: C		
Type: Transformer	Substation: PEBBLE BEACH	Responsible System:		
State: Closed				
Damage Assessment				
Poles Down: 0	Transformers: 0	Conductor Span: 0		
Dispatcher Comments:				
Crew Comments:				
Map: 				
Calls				
Name	Account #	Address	Phone Number	Comments

Figure 22. Order Details PDF Document

12.1. Default PDF Export Template

The default PDF template defined in the order2pdf.xml configuration file includes the following sections:

- **Header:** Defines the header of the order detail document. Contains the following fields:
 - Order Name
 - Incident Export Time Stamp

The document header section is formatted using the <fo:static-content> element. It is shown as a header on each document page. The height of the header section is formatted by the <fo:region-before> element.

- Table 1: Provides incident details and the map. Contains the following fields:
 - Summary
 - Duration
 - Start Time
 - Crews
 - Calls
 - First Call
 - Affected
 - End Time
 - Priority
 - ETR
 - Device
 - Device Name
 - Division
 - Feeder
 - ID
 - Service Center
 - Phase
 - Type
 - Substation
 - System Affected
 - State
 - Damage Assessment
 - Poles Down

- Transformers
- Wire Down
- Remarks
- Dispatcher Comments
- Crew Comments
- Map

The Table 1 section is formatted using a table (<fo:table>) with 3 columns. Column widths are set by the corresponding <fo:table-column> elements. Column captions are set by the <fo:table-cell> elements. Header cells (for example, Summary and Device) are formatted using the “number-columns-spanned” field.

- Table 2: Calls

The Table 2 section is formatted using a table (<fo:table>) with 5 columns. The data for separate calls is populated from the order export XML file using the <xsl:for-each> element.

12.2. Adding Header Fields to the PDF Export Template

To add a header field in the Order to PDF conversion template:

1. In the PDF export template file (order2pdf.xml), increase the height of the header section from 1 cm to 3 cm using the <region-before> element. This allocates a new header field.



2. Increase the margin between the header section and the next table from 1 cm to 1.5 cm using the <fo:region-body margin-top> element.
3. In the <fo:static-content> element, add a new row using the <fo:block> sub-element.
4. Define the name and value of the new header field as follows:

INCIDENT NAME: <xsl:value-of select="."/Incident//NAME" />

The label is defined by the plain text “Incident Name:”. The value is defined by <xsl:value-of select="."/Incident//NAME" where Incident is the corresponding field and NAME is the corresponding sub-field in the order export XML file. For the list of all incident fields, see [Sample](#)

Incident Export XML File.

5. Export the order details to a PDF and verify the new field.

The Incident Name field appears in the PDF next to the Order Name field.

Order Name: 3320000001-1

Incident Name: 3320000001

Summary	
Duration: 46.8 h	Start Time: 11/28/2023 7:33 PM
Calls: 1	First Call: 11/28/2023 7:33 PM

12.3. Adding Column Fields to the PDF Export Template

To add a column in the Order to PDF conversion template:

1. In the PDF export template file (order2pdf.xml), add a new row in the table definition or modify an existing empty row.

In the example below, the empty field next to the First Call field is modified to show Work Order ID.

<pre><fo:table-row> <fo:table-cell border-style="solid" padding="1em"> <fo:block text-align="left"> <fo:inline font-weight="bold">Calls: </fo:inline> <xsl:value-of select="//Incident/CALLCOUNT" /> </fo:block> </fo:table-cell> <fo:table-cell border-style="solid" padding="1em"> <fo:block text-align="left"> <fo:inline font-weight="bold">First Call: </fo:inline> <xsl:value-of select="local:getDateTime(//Incident/FIRSTCALLTIME, 'MM/dd/yyyy h:mm tt')"/> </fo:block> </fo:table-cell> <fo:table-cell border-style="solid" padding="1em"> <fo:block text-align="left"> </fo:block> </fo:table-cell> </fo:table-row></pre>	<pre><fo:table-row> <fo:table-cell border-style="solid" padding="1em"> <fo:block text-align="left"> <fo:inline font-weight="bold">Calls: </fo:inline> <xsl:value-of select="//Incident/CALLCOUNT" /> </fo:block> </fo:table-cell> <fo:table-cell border-style="solid" padding="1em"> <fo:block text-align="left"> <fo:inline font-weight="bold">First Call: </fo:inline> <xsl:value-of select="local:getDateTime(//Incident/FIRSTCALLTIME, 'MM/dd/yyyy h:mm tt')"/> </fo:block> </fo:table-cell> <fo:table-cell border-style="solid" padding="1em"> <fo:block text-align="left"> <fo:inline font-weight="bold">Order Type: </fo:inline> <xsl:value-of select="//Order/ORDERTYPE" /> </fo:block> </fo:table-cell> </fo:table-row></pre>
---	--

The bold formatting of the field label is defined by the `<fo:inline font-weight>` element.

2. Export the Order details to a PDF and verify the new field.

The Order Type field appears in the PDF file next to the First Call field.

Summary		
Duration: 47.0 h	Start Time: 11/28/2023 7:33 PM	Order Crews:
Calls: 1	First Call: 11/28/2023 7:33 PM	Order Type: RESTORATION
Affected: 1	End Time:	
Priority: 0	ETR: 11/29/2023 1:00 AM	

The conversion engine wraps text at the location of spaces to fit the defined column width. To avoid

truncating values that do not contain spaces and exceed the specified column width, use the following custom formatting. This places hidden spaces after each symbol in the value. Note that only UTF-8 fonts (for example, Arial) are supported.

```
<xsl:variable name="myVar">
  <xsl:call-template name="zero_width_space_1">
    <xsl:with-param name="data" select="//Incident//OPENDEVICEID"/>
  </xsl:call-template>
</xsl:variable>
<xsl:call-template name="string-replace-all">
  <xsl:with-param name="text" select="$myVar" />
  <xsl:with-param name="replace" select="'&#8203; &#8203;'" />
  <xsl:with-param name="by" select="' '" />
</xsl:call-template>
```

12.4. Sample Order Export XML File

This section provides a sample order export XML file. Note that this file is generated and used internally. It is not available for review or debugging.

Any of the incident-related fields, except for those highlighted in yellow, can be used in the PDF export template.

```
<ExportData>
  <Incident>
    <CALLBACKSENTCUSTOMERS/>
    <ISMOVEUP>False</ISMOVEUP>
    <OPENDEVICEIDSPHASEA/>
    <OPENDEVICEIDSPHASEB/>
    <OPENDEVICEIDSPHASEC>22057363</OPENDEVICEIDSPHASEC>
    <CLOSEDDEVICEIDSPHASEA/>
    <CLOSEDDEVICEIDSPHASEB/>
    <CLOSEDDEVICEIDSPHASEC/>
    <CLEARTIME>4611686018427387904</CLEARTIME>
    <STARTTIME>5250053719142151707</STARTTIME>
    <STARTTIMEOFFSET>638367700714763803</STARTTIMEOFFSET>
    <ENDTIME>4611686018427387904</ENDTIME>
    <ENDTIMEOFFSET>0</ENDTIMEOFFSET>
    <HASBEENCONFIRMED>False</HASBEENCONFIRMED>
    <UNDOCONFIRMED>False</UNDOCONFIRMED>
    <INCIDENTTYPE>2</INCIDENTTYPE>
    <HASANUPSTREAMRECLOSINGDISABLEDEDEVICE>False</HASANUPSTREAMRECLOSINGDISABLEDEDEVICE>
    <UPSTREAMRECLOSABLEDEVICES/>
    <IMPLEMENTATIONINCIDENT>Outage: 3320000008</IMPLEMENTATIONINCIDENT>
    <UNASSOCCALLS>System.Collections.Generic.List`1[NetUtil.CallMessage]</UNASSOCCALLS>
    <MY911CALLS>System.Collections.Generic.List`1[NetUtil.CallMessage]</MY911CALLS>
    <NONOUTAGECALLS>System.Collections.Generic.List`1[NetUtil.CallMessage]</NONOUTAGECALLS>

  <AFFECTEDCUSTOMERS>System.Collections.ObjectModel.ReadOnlyCollection`1[OmsUtilities.AffectedCustomer]<
    /AFFECTEDCUSTOMERS>
```

```

<DISTINCTTROUBLECODES/>
<SEQNUMBER>8</SEQNUMBER>
<DAILYSEQNUMBER>1</DAILYSEQNUMBER>
<GUID>0b5ca275-f8bf-40db-b699-9c07ca650f08</GUID>
<NAME>3320000008</NAME>
<PLANNEDOUTAGEID/>
<INCIDENTTYPE2>0</INCIDENTTYPE2>
<OUTAGESTATUS>0</OUTAGESTATUS>
<INCIDENTSUBTYPE>0</INCIDENTSUBTYPE>
<INCIDENTSTATUS>0</INCIDENTSTATUS>
<ISSINGLECUSTOMERTYPE>False</ISSINGLECUSTOMERTYPE>
<ISOUTAGEINCIDENT>True</ISOUTAGEINCIDENT>
<DISABLEITRGENERATION>False</DISABLEITRGENERATION>
<PT>{X=936313.4, Y=813800.7}</PT>
<INCIDENTPTX>936313.4</INCIDENTPTX>
<INCIDENTPTY>813800.7</INCIDENTPTY>
<DEVICEX>936263.4</DEVICEX>
<DEVICEY>813750.7</DEVICEY>
<CANCELLED>False</CANCELLED>
<CLEARED>False</CLEARED>
<CALLSHAVEBEENCLEARED>False</CALLSHAVEBEENCLEARED>
<CLEARINGOPERATOR/>
<STATUSSTARTTIME>5250053719142151707</STATUSSTARTTIME>
<STATUSSTARTTIMEOFFSET>638367700714763803</STATUSSTARTTIMEOFFSET>
<ACKNOWLEDGEDTIME>4611686018427387904</ACKNOWLEDGEDTIME>
<FIRSTCALLTIME>4611686018427387904</FIRSTCALLTIME>
<FIRSTCALLCUSTOMER>555097201555097201555097201</FIRSTCALLCUSTOMER>
<FIRSTCALLCUSTOMERADDRESS>9363 WILLOW PKWY</FIRSTCALLCUSTOMERADDRESS>
<FIRSTCALLCOMMENT/>
<FIRSTCREWTIME>4611686018427387904</FIRSTCREWTIME>
<ETR>5250053738427387904</ETR>

<MANUALETR>4611686018427387904</MANUALETR>
<ETRSOURCE/>
<AREANOTEETRSOURCE/>
<CALCULATEDETRSOURCE/>
<CALCULATEDETR>4611686018427387904</CALCULATEDETR>
<ITR>5250053738427387904</ITR>
<CUSTOMERMODIFIEDTIME>5250053719142578539</CUSTOMERMODIFIEDTIME>
<ETRSEQUENCENUMBER>0</ETRSEQUENCENUMBER>
<STATION>PEBBLE BEACH</STATION>
<FEEDERID>8168047</FEEDERID>
<FEEDERNAME>F6533</FEEDERNAME>
<OPERATEDSTATION>PEBBLE BEACH</OPERATEDSTATION>
<OPERATEDFEEDERNAME>F6533</OPERATEDFEEDERNAME>
<SWITCHINGSOURCE>0</SWITCHINGSOURCE>
<ISTHIRDPARTY>False</ISTHIRDPARTY>

```

```

<ASSOCRESTORATIONSTEPS>System.Collections.ObjectModel.ReadOnlyCollection`1[OmsUtilities.AssociateRestorationStep]</ASSOCRESTORATIONSTEPS>
<CREWASSIGNMENTFAILED>False</CREWASSIGNMENTFAILED>
<ASSIGNEDCREWS>NetUtil.AssignedCrewsList</ASSIGNEDCREWS>
<WORKORDERID/>

```



```

<LINKEDORDERIDS>d64ea5dc-2d61-4d22-a25c-9889461e798c</LINKEDORDERIDS>
<ISDEVICEOFRECORDOPEN>False</ISDEVICEOFRECORDOPEN>
<ARMSSTATUS>10</ARMSSTATUS>
<RECEIVEDCOMPLETIONFROMWFM>False</RECEIVEDCOMPLETIONFROMWFM>
<WFMRESTORATIONTIME>4611686018427387904</WFMRESTORATIONTIME>
<CASNOTE>301</CASNOTE>
<CALCULATEDCASNOTE>301</CALCULATEDCASNOTE>
<MANUALCASNOTEOVERRIDE>False</MANUALCASNOTEOVERRIDE>
<AREANOTECODE>0</AREANOTECODE>
<STORMNAME/>
<DYNAREANOTEWORKINGID/>
<AREANOTESTARTTIME>4611686018427387904</AREANOTESTARTTIME>
<AREANOTEITRDISABLE>False</AREANOTEITRDISABLE>
<AREANOTEETR>4611686018427387904</AREANOTEETR>
<DISPATCHERCOMMENTS/>
<CREWCOMMENTS/>
<REMARKS/>
<CREATEDAFTERTHEFACT>False</CREATEDAFTERTHEFACT>
<CASNOTEREMARKS/>
<LINECREWNEEDED>False</LINECREWNEEDED>
<TREECREWNEEDED>False</TREECREWNEEDED>
<FIRSTRESPONDERNEEDED>False</FIRSTRESPONDERNEEDED>
<EVALUATORNEEDED>False</EVALUATORNEEDED>
<EVENTOFINTEREST>False</EVENTOFINTEREST>
<REASON>0</REASON>
<NOTROUBLEFOUND>False</NOTROUBLEFOUND>
<CANCELORDER>False</CANCELORDER>
<SUPPRESSNOTIFICATION>False</SUPPRESSNOTIFICATION>
<SUPPRESSNOTIFICATIONREASON>0</SUPPRESSNOTIFICATIONREASON>
<SYSTEMMODEAREANOTE>0</SYSTEMMODEAREANOTE>
<NOOFPOLESDOWN>0</NOOFPOLESDOWN>
<NOFCROSSARMS>0</NOFCROSSARMS>
<NOFTREETRIMSPANS>0</NOFTREETRIMSPANS>
<NOSERVICESDOWN>0</NOSERVICESDOWN>

<NOOFTRANSFORMERS>0</NOOFTRANSFORMERS>
<CONDUCTORSPAN>0</CONDUCTORSPAN>
<FOLLOWUPREQUIRED>False</FOLLOWUPREQUIRED>
<PINGINPROGRESS>False</PINGINPROGRESS>
<INACCESSIBLE>False</INACCESSIBLE>
<CALLBACKTYPE>2</CALLBACKTYPE>
<PHASEOUT>C</PHASEOUT>
<OUTAGEDEVICETYPE>Transformer</OUTAGEDEVICETYPE>
<OPENDEVICEPHASEIND>False</OPENDEVICEPHASEIND>
<ACTIONTAKEN>0</ACTIONTAKEN>
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<SYSTEMAFFECTED>0</SYSTEMAFFECTED>
<FAILUREMODE>0</FAILUREMODE>
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<ANIMALS>0</ANIMALS>
<PLANTS>0</PLANTS>

```

```
<OUTAGETYPE>0</OUTAGETYPE>
<MATERIAL>0</MATERIAL>
<FAILEDEQUIPMENT>0</FAILEDEQUIPMENT>
```

```
<REGIONLEVELS>System.Collections.Generic.List`1[OmsUtilities.InterfaceModels_V2.Common.RegionLevel]</R
EGIONLEVELS>
```

```
<ASSOCIATEDASSESSMENTLOCATIONS/>
<PROTECTIVESTATION>PEBBLE BEACH</PROTECTIVESTATION>
<PROTECTIVEDEVICEID>22057363</PROTECTIVEDEVICEID>
<PROTECTIVEDEVICENAME>6-7512-9972-0-0</PROTECTIVEDEVICENAME>
<CALLTYPE>0</CALLTYPE>
<CONDITION_1>0</CONDITION_1>
<CONDITION_2>0</CONDITION_2>
<CALLDDESCRIPTION>0</CALLDDESCRIPTION>
<CALLCOUNT>0</CALLCOUNT>
<AMICOUNT>0</AMICOUNT>
<AMIOUTCOUNT>0</AMIOUTCOUNT>
<AMIINCOUNT>0</AMIINCOUNT>
<DISABLEPREDICTEDSTABLETIMER>False</DISABLEPREDICTEDSTABLETIMER>
<ISPREDICTEDSTABLE>True</ISPREDICTEDSTABLE>
<INCIDENTCLEARATTEMPTFAILED>False</INCIDENTCLEARATTEMPTFAILED>
<POINTOFFFAILURE/>
<OPENDEVICEPT>{X=936263.4, Y=813750.7}</OPENDEVICEPT>
<DEVICESIZE>25.00 kVA</DEVICESIZE>
<OPENDEVICEPTX>936263.4</OPENDEVICEPTX>
<OPENDEVICEPTY>813750.7</OPENDEVICEPTY>
<OPENDEVICESTATION>PEBBLE BEACH</OPENDEVICESTATION>
<OPENDEVICEID>22057363</OPENDEVICEID>
<ISOVERHEAD>True</ISOVERHEAD>
<OPENDEVICENAME>6-7512-9972-0-0</OPENDEVICENAME>
<SWITCHINGORDERID/>
<SWITCHINGORDERNAME/>
<OPENDEVICETYPE>16</OPENDEVICETYPE>
<OPENSWITCHTYPE>-1</OPENSWITCHTYPE>
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<CRITICALCUSTOMERCOUNT4>0</CRITICALCUSTOMERCOUNT4>
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<EXCLUDEFROMSTATISTICS>False</EXCLUDEFROMSTATISTICS>
<CMI>436</CMI>
<PRIORITYCALLSCOUNT>0</PRIORITYCALLSCOUNT>
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<LASTAMIINTIMEREPORTED>4611686018427387904</LASTAMIINTIMEREPORTED>
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```

```

<MIDPOINTSWITCHING>False</MIDPOINTSWITCHING>
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<OPENDEVICEVOLTAGETYPE>False</OPENDEVICEVOLTAGETYPE>
<CALLBACKSENT>0</CALLBACKSENT>
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<HASINTERNAL>False</HASINTERNAL>
<EMSNAME>PBLBEACH</EMSNAME>
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<ORIGINALSWITCHTYPE>-1</ORIGINALSWITCHTYPE>
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<WORKORG2>DSP_HIGH</WORKORG2>
<WORKORG3>DSP_HIGH</WORKORG3>
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<SERVICECENTER>3LIONSSC</SERVICECENTER>
<SERVICECENTERNAME>3LIONSSC</SERVICECENTERNAME>
<OPERATEDSERVICECENTER>3LIONSSC</OPERATEDSERVICECENTER>
<LOCATIONDESCRIPTION>5500 West St, Roarke</LOCATIONDESCRIPTION>
<STATE>Washington</STATE>
<COUNTY>Gordon</COUNTY>
<MUNICIPALITY>Roarke</MUNICIPALITY>
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<RANKINGMANUALOVERRIDE>0</RANKINGMANUALOVERRIDE>
<RANKINGOVERRIDDEN>False</RANKINGOVERRIDDEN>

<ISPREDICTEDDEVICEWRONG>False</ISPREDICTEDDEVICEWRONG>
<RANKING>12</RANKING>
<FRP>-1</FRP>
<REPREDICTIONNEEDED>False</REPREDICTIONNEEDED>
<TOTALCUSTOMERSAFFECTED>2</TOTALCUSTOMERSAFFECTED>
<CALLAGGREGATIONCOUNTS>0,0,0,0,0,0,0,0,0</CALLAGGREGATIONCOUNTS>
<CURRENTCUSTOMER>2</CURRENTCUSTOMER>
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<CURRENTCRITICALCUSTOMER2>0</CURRENTCRITICALCUSTOMER2>
<CURRENTCRITICALCUSTOMER3>0</CURRENTCRITICALCUSTOMER3>
<CURRENTCRITICALCUSTOMER4>0</CURRENTCRITICALCUSTOMER4>

```

```

<CURRENTCRITICALCUSTOMERS>0</CURRENTCRITICALCUSTOMERS>
<SERVICEINDEX>3</SERVICEINDEX>
<SAFETYINDEX>2</SAFETYINDEX>
<HAZARDOUSCUSTOMER>False</HAZARDOUSCUSTOMER>
<WASTEDTRIP>False</WASTEDTRIP>
<PUBLICINFLICTEDDAMAGEINDICATOR>False</PUBLICINFLICTEDDAMAGEINDICATOR>
<FOLLOWUPCOMPLETEDINDICATOR>False</FOLLOWUPCOMPLETEDINDICATOR>
<TOTALCONNECTEDKVASERVED>13284.5</TOTALCONNECTEDKVASERVED>
<TOTALCONNECTEDKVAINTERRUPTED>25</TOTALCONNECTEDKVAINTERRUPTED>
<CALCULATECONNECTEDKVA>False</CALCULATECONNECTEDKVA>
<SYSTEMCONTRIBUTING>0</SYSTEMCONTRIBUTING>
<CONTRIBUTINGCAUSE>0</CONTRIBUTINGCAUSE>
<FAULTLOCATIONSOURCE>0</FAULTLOCATIONSOURCE>
<DISTRIBUTIONLOCATIONNUMBER>0</DISTRIBUTIONLOCATIONNUMBER>
<FAULTLATITUDE>0</FAULTLATITUDE>
<FAULTLONGITUDE>0</FAULTLONGITUDE>
<ISFAULTLOCATIONMANUALLYOVERRIDEN>False</ISFAULTLOCATIONMANUALLYOVERRIDEN>
<FAULTID/>
<HISTORYCORRECTIONDEPARTMENT/>
<HISTORYCORRECTIONREASON/>
<HISTORYCORRECTIONAPPROVEDBY/>
<HISTORYCORRECTIONAPPROVEDDATE>4611686018427387904</HISTORYCORRECTIONAPPROVEDDATE>
<REASONGROUP>0</REASONGROUP>
<FIELDCONFIRMED>False</FIELDCONFIRMED>
<OUTAGEISLANDCRCS>System.Collections.Generic.HashSet`1[System.UInt32]</OUTAGEISLANDCRCS>
<ETA>4611686018427387904</ETA>
<ETACREW/>
<CUSTOMFIELDS/>
<CREATED>5250053719142151707</CREATED>
<MODIFIED>5250053719218231015</MODIFIED>
<SERVICEPOINTS>System.Collections.Generic.List`1[OmsUtilities.ServicePointInfo]</SERVICEPOINTS>
<CENTERPT>{X=0, Y=0}</CENTERPT>
<PROTECTIVEPT>{X=936263.4, Y=813750.7}</PROTECTIVEPT>
<DELETED>False</DELETED>
<_FIRSTCREWTIME>4611686018427387904</_FIRSTCREWTIME>
<_WORKQUEUES>0</_WORKQUEUES>

```

```

<ASSOCSWITCHINGSTEPS>System.Collections.ObjectModel.ReadOnlyCollection`1[OmsUtilities.AssociateSwitchingStep]</ASSOCSWITCHINGSTEPS>

```

```

<ARMSORDERSTATUS>10</ARMSORDERSTATUS>
<PINGSENT>0</PINGSENT>
<AFFECTEDCUSTOMERSCOUNT>2</AFFECTEDCUSTOMERSCOUNT>
<CREATEFILETIME>-8585004025014767419</CREATEFILETIME>
</Incident>
<Calls/>
<Crews/>
<Order>
<INCIDENT>Outage: 3320000008</INCIDENT>
<X>936313.4</X>
<Y>813800.7</Y>
<LOCATIONX>936313.4375</LOCATIONX>
<LOCATIONY>813800.6875</LOCATIONY>

```

```

<GROUPSEQ>1</GROUPSEQ>
<ORDERID>d64ea5dc-2d61-4d22-a25c-9889461e798c</ORDERID>
<ORDERNAME>3320000001-1</ORDERNAME>
<ORDERTYPE>RESTORATION</ORDERTYPE>
<ORDERSTATUS>NEW</ORDERSTATUS>
<PROJECTID/>
<INCIDENTID>0b5ca275-f8bf-40db-b699-9c07ca650f08</INCIDENTID>
<DESCRIPTION/>
<DEVICEID>22057363</DEVICEID>
<SERVICECENTER>3LIONSSC</SERVICECENTER>
<OWNERID/>
<ORGANIZATION>DSP_HIGH</ORGANIZATION>
<ORGANIZATION2/>
<ORGANIZATION3/>
<HASDEPENDENT>False</HASDEPENDENT>
<HASPREREQUISITE>False</HASPREREQUISITE>
<CREATETIME>2023-11-28T17:07:51.4760000+00:00</CREATETIME>
<CANCELLEDTIME/>
<CLOSEDTIME/>
<FIELDCOMPLETETIME/>
<CREATEDBY>ADMS</CREATEDBY>
<ASSIGNTIME/>
<DISPATCHTIME/>
<ENROUTETIME/>
<ARRIVALTIME/>
<RELEASETIME/>
<UPDATEDTIME/>
<PRIORITY>0</PRIORITY>
<SUMMARYCREWSTATUS>UNASSIGNED</SUMMARYCREWSTATUS>
<SUSPENDEDTIME>4611686018427387904</SUSPENDEDTIME>
<SUSPENDEDREASON/>
<CREWS>System.Collections.Generic.List`1[WorkModelRestClientOpenApi.Crew]</CREWS>
<DISPATCHERCOMMENT/>
<CREWCOMMENT/>
<LOCATIONDESCRIPTION/>
<SUMMARYTASKSTATUS>UNASSIGNED</SUMMARYTASKSTATUS>
<LEADCREWID/>
<HASLEADCREW>False</HASLEADCREW>
<LEADCREWSTATUS/>
<LEADCREWHASCOMPLETEDWORK>False</LEADCREWHASCOMPLETEDWORK>
<OrderCrews/>
</Order>
</ExportData>

```

13. Configuring Management Areas Hierarchy

The default Fantasy Island model uses a five-level management areas hierarchy: Company - State - County - City - Service Center. This chapter describes how to configure the OMS display plugin for a custom multi-level management areas hierarchy.

Note

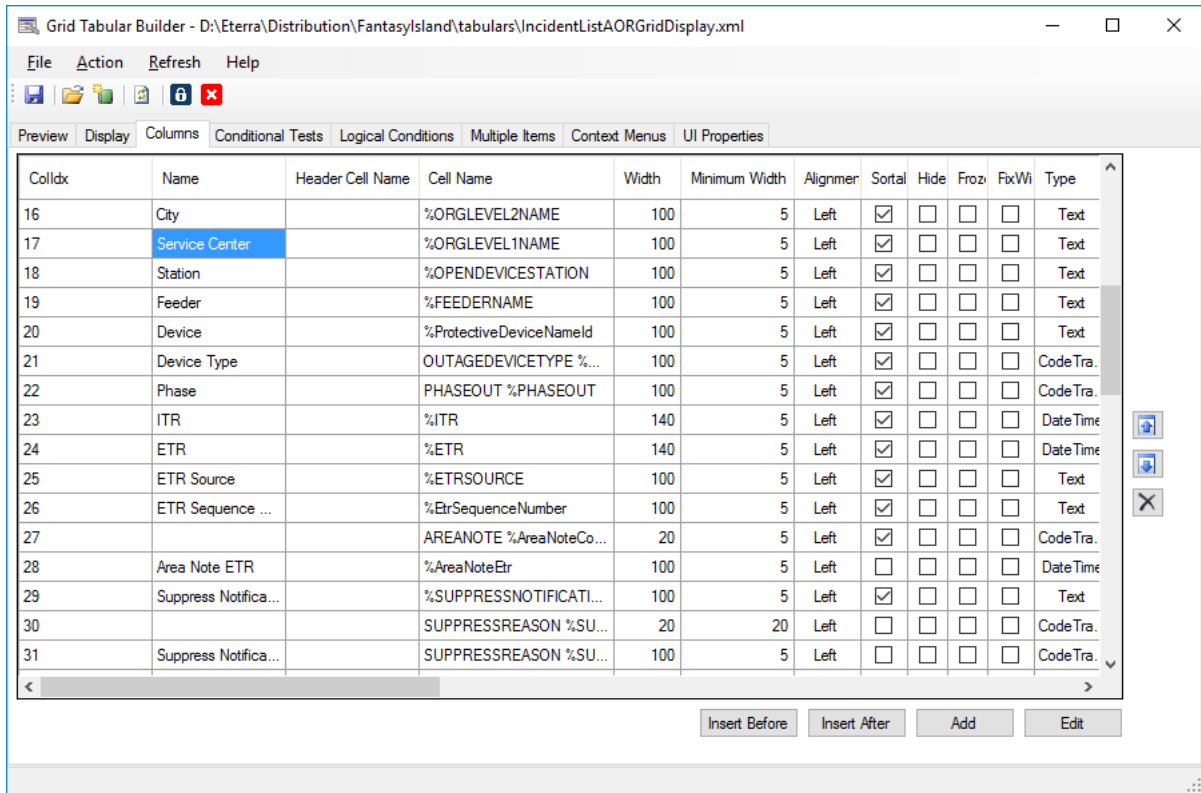
Configuring the management areas hierarchy also requires modifying the customer model, Archive database, and other components. For more details about configuring the management areas hierarchy, refer to the *Customer List Compiler User Guide* and the *ADMS Software Installation and Maintenance Guide*.

13.1. Preparing Grid Tabular Displays

For a custom management areas hierarchy, you need to prepare a set of grid tabular displays that show outage management data. Typically, it is best to modify the default displays, such as by updating display names, column labels, and other properties as appropriate.

13.1.1. Preparing the Incident Detail Display

1. Open the Grid Tabular Builder.
2. Open the `IncidentListAORGridDisplay.xml` file, which defines the Incident Detail grid tabular display.
3. Update the column labels for hierarchy levels.



For details about using the Grid Tabular Builder, refer to the *Grid Tabular Builder User Guide*.

4. Update column queries for hierarchy levels.

Use ORGLEVEL1NAME - ORGLEVEL9NAME query fields to query for hierarchy level labels. ORGLEVEL1NAME stands for the lowest logical hierarchy level, which is service center.

The following query fields are used for the default five-level hierarchy:

- Company = %ORGLEVEL5NAME
- State = %ORGLEVEL4NAME
- County = %ORGLEVEL3NAME
- City = %ORGLEVEL2NAME
- Service Center = %ORGLEVEL1NAME

5. Save your configurations.

The example below shows the City and Service Center hierarchy levels on the Incident Details display.

Grid Tabular Builder - D:\Eterra\Distribution\FantasyIsland\tabulars\OrgLevel1SummaryGridDisplay.xml

File Action Refresh Help

Preview Display Columns Conditional Tests Logical Conditions Multiple Items Context Menus UI Properties

ColIdx	Name	Header Cell Name	Cell Name	Width	Minimum Width	Alignmer	St
0				20	5	Left	[
1				20	5	Left	[
2				20	5	Left	[
3	Service Center		%NAME	175	5	Left	[
4	Outage Time		%EARLIESTOUTAGETIME	200	5	Left	[
5	Latest ITR/ETR		%LATESTETR	200	5	Left	[

5. Save your configurations.

The example below shows the Service Center Summary display.

(3) Service Center Sum... X

Reset Filter General Electric X

Service Center Summary

	Service Center	Outage Time	Latest ITR/ETR	Feeders Out	Events	Incidents	Line Crews Needed Count
		9/24/2018 4:12:00 AM	10/2/2018 5:00:00 AM	0	2	8	0
	3LIONSSC	9/26/2018 2:22:59 AM	9/26/2018 4:30:00 AM	0	0	1	0
	WSTC	9/24/2018 4:12:00 AM	9/28/2018 11:59:00 PM	0	1	6	0
	NEAB	10/2/2018 2:59:29 AM	10/2/2018 5:00:00 AM	0	1	1	0

6. Prepare summary displays for other management areas hierarchy levels using default displays.

Note

The default configuration includes summary displays for 5 management areas hierarchy levels only. For management areas hierarchies with 6 levels and more, create summary displays for level 6 and later based on OrgLevel15SummaryGridDisplay.xml. Additionally, update the "From" clause and add navigation to lower summary displays.

13.1.3. Preparing Other OMS Displays

Update management areas hierarchy labels in the following OMS displays.

- 911Call_GridDisplay.xml
-

CallEntry_CustomerCall_GridDisplay.xml

- CallEntry_CustomerSearchResults_GridDisplay.xml
- CallEntry_UnassociatedCall_GridDisplay.xml
- ChronologySystem_GridDisplay.xml
- CrewAreaGridDisplay.xml
- DeviceOperationsSwitchingGridDisplay.xml
- FeederSummaryForGridDisplay.xml
- FeederSummary_GridDisplay.xml
- HierarchyGroupGridDisplay.xml
- HistoryCorrectionIncidentGridDisplay.xml
- HistoryCorrectionMergeIncidentGridDisplay.xml
- IncidentGridDisplay.xml
- IncidentListAORGridDisplay.xml
- IncidentListForGridDisplay.xml
- IncidentListGridDisplay.xml
- IncidentLocking_GridDisplay.xml
- MovedCustomers_GridDisplay.xml
- OrgLevel1SummaryForGridDisplay.xml
- OrgLevel1SummaryGridDisplay.xml
- OrgLevel2SummaryForGridDisplay.xml
- OrgLevel2SummaryGridDisplay.xml
- OrgLevel3SummaryForGridDisplay.xml
- OrgLevel3SummaryGridDisplay.xml
- OrgLevel4SummaryForGridDisplay.xml
- OrgLevel4SummaryGridDisplay.xml
- RestorationStepsGridDisplay.xml
- SuspendedCustomers_GridDisplay.xml
- UnassociatedCall_GridDisplay.xml

13.2. Modifying the OMS Display Plugin Configuration

For a custom management areas hierarchy, you need to modify the `OmsDisplayPluginConfiguration.xml` file.

The following tips may help you customize the OMS display plugin for a multi-level management

areas hierarchy based on the default five-level Fantasy Island management areas hierarchy:

- Use a text editor (for example, Notepad++) to copy and paste nodes directly in the `OmsDisplayPluginConfig.xml` file. Update the file by modifying elements for the highest level management areas hierarchy (for example, `OrgLevel5`) first.
- When configuring a five-level management areas hierarchy, update labels and links with new names.
- When configuring a four-level management areas hierarchy, remove all configuration items with `OrgLevel5`. Clear the `LowerMostHierarchyGroupType` value for the `OrgLevel4` filter.
- When configuring a six-level management areas hierarchy, add configuration items for `OrgLevel6` similar to the ones for `OrgLevel5`.

Typically, you need to copy entire `OrgLevel5` configuration item groups, make another copy of configuration items for the lowest hierarchy element (`OrgLevel1` or `Feeder`, depending on the configuration), update `OrgLevel5` values with `OrgLevel6`, and then update configuration items for the lowest hierarchy element.

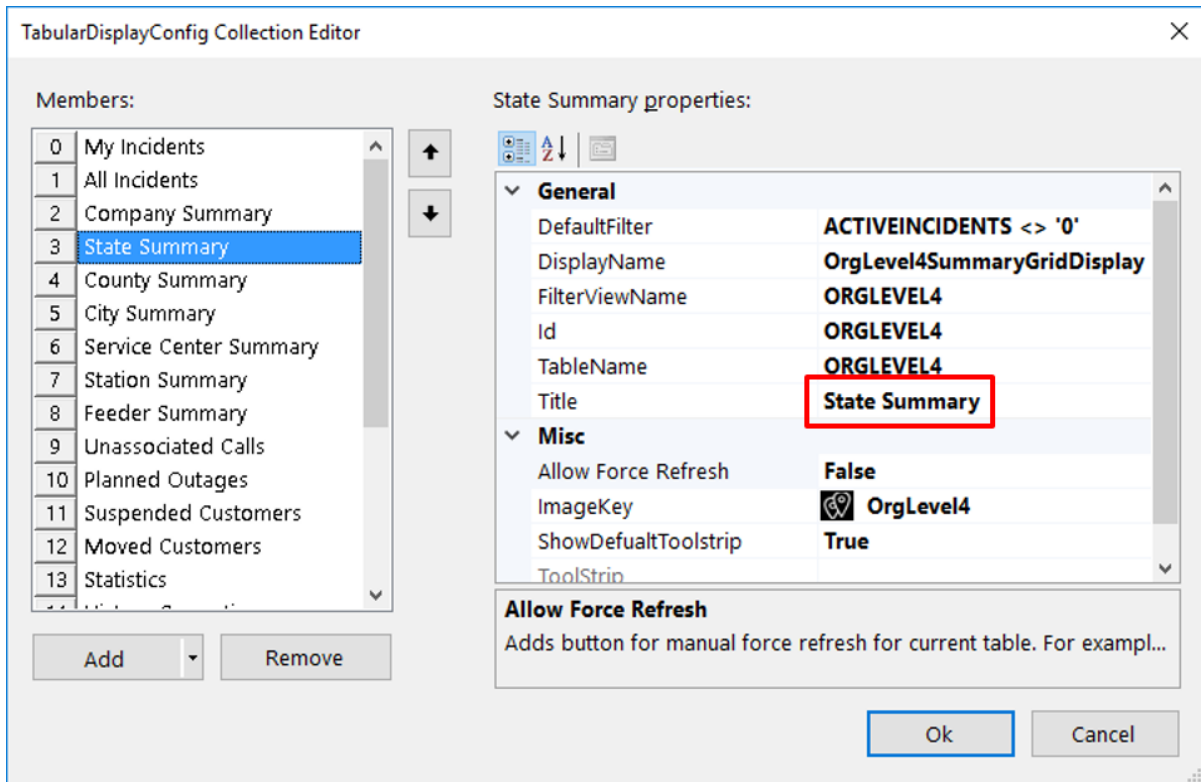
- When updating the `OmsDisplayPluginConfig.xml` file using a text editor, search by the default management areas hierarchy level values (for example, `ORGLEVEL5`, `State`, and `City`) to make sure that all the required configurations are updated.

13.2.1. Configuring a Five-Level Management Areas Hierarchy

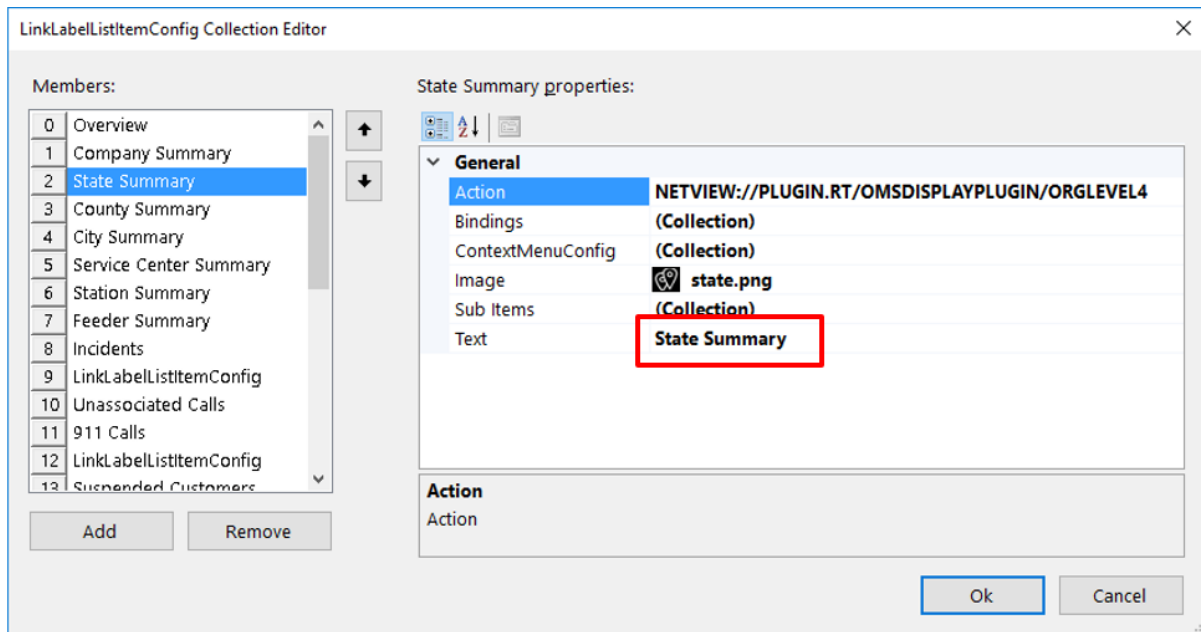
For a custom five-level management areas hierarchy, update the labels in the default five-level management areas hierarchy configuration.

Modify the default `OmsDisplayPluginConfig.xml` as follows:

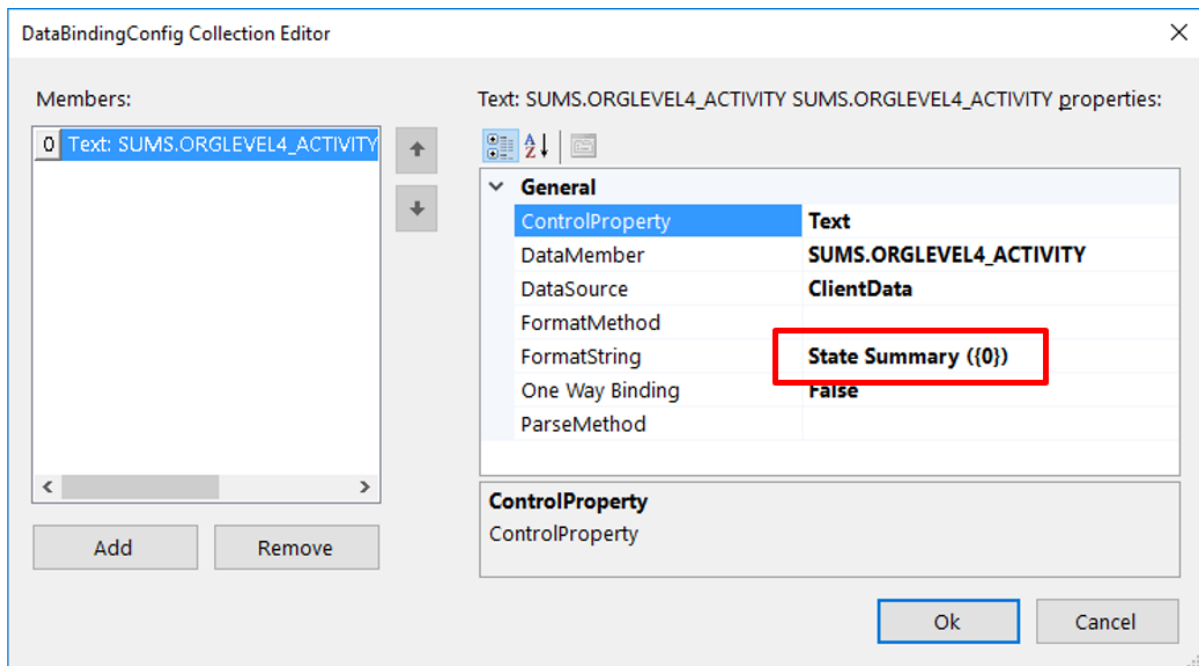
1. Open the OMS Client Configuration Editor dialog box.
2. On the Plugins tab, select `OmsDisplayPlugin`.
3. Update the titles of tabs for hierarchy levels, using the `TabDisplays` item in the General group.



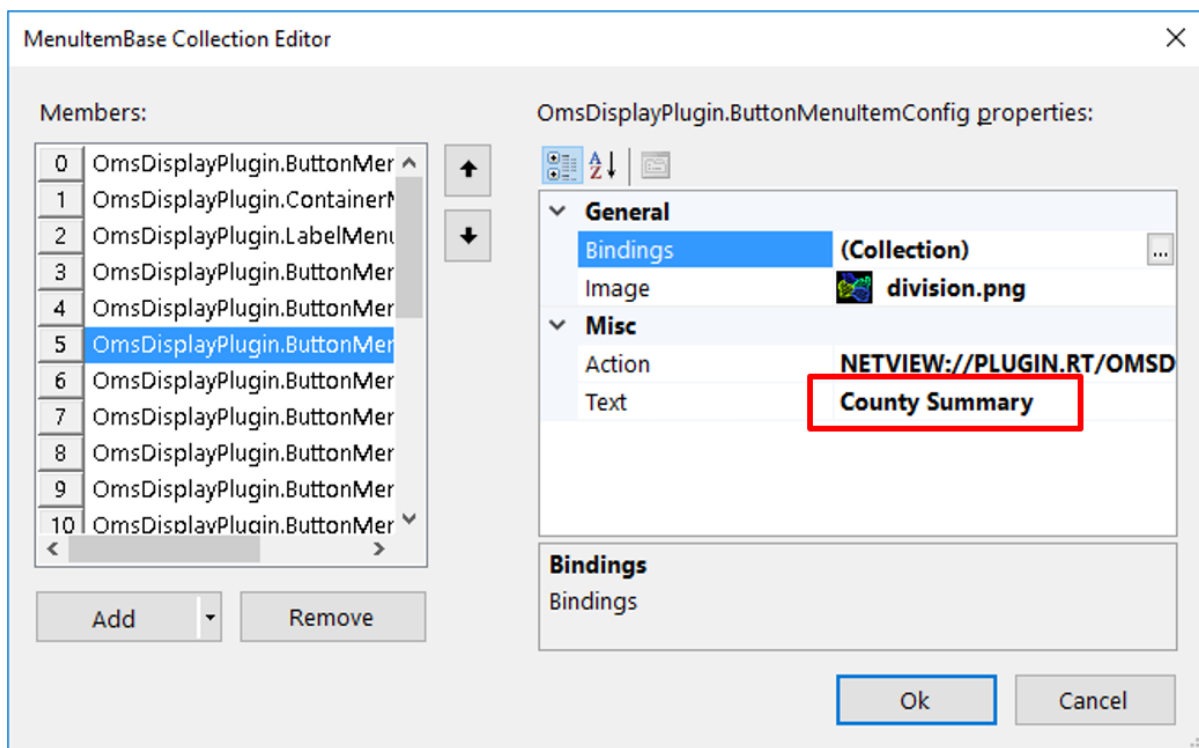
- Update the labels of control panel links for hierarchy level tabs, using the Display Links item in the Appearance group.



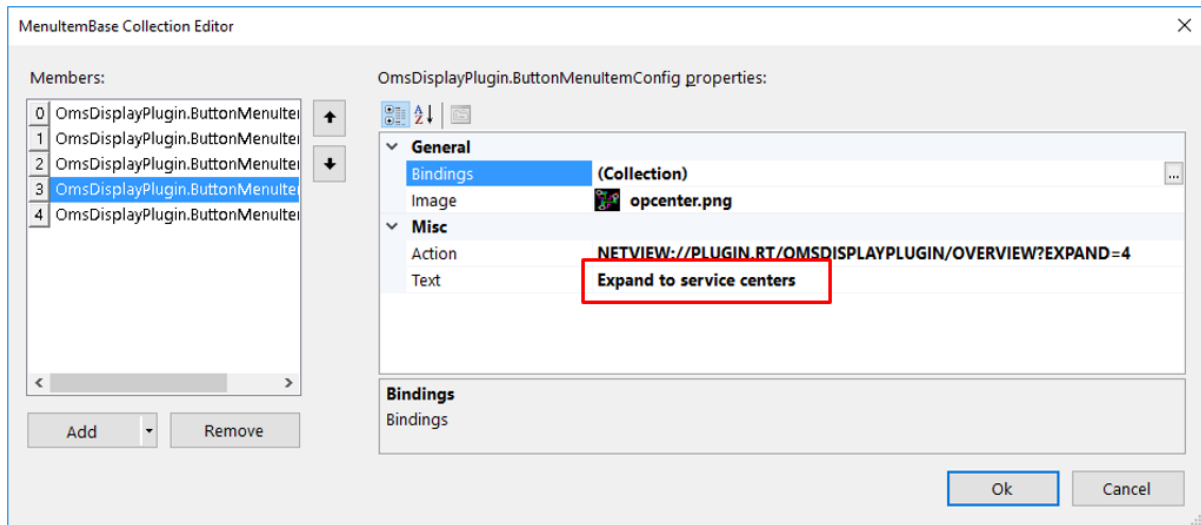
Also update the labels in the binding configurations.



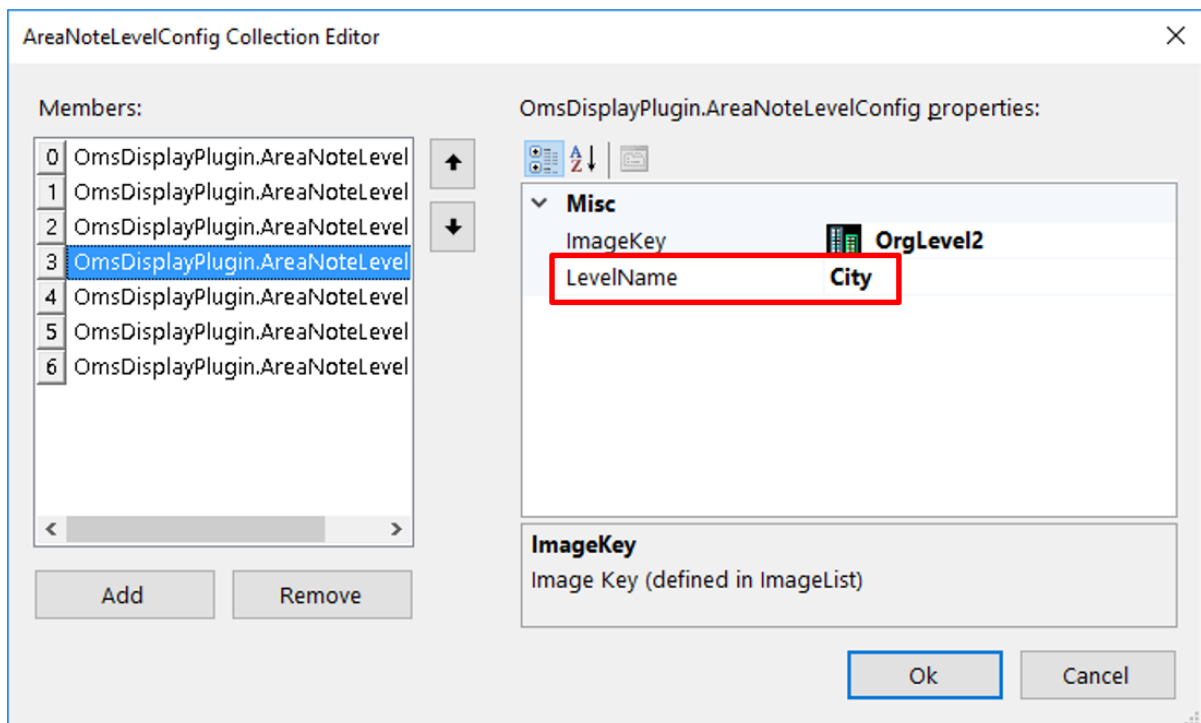
- Update the tooltips for the OMS display plugin toolbar buttons, using the Toolbar item in the General group.



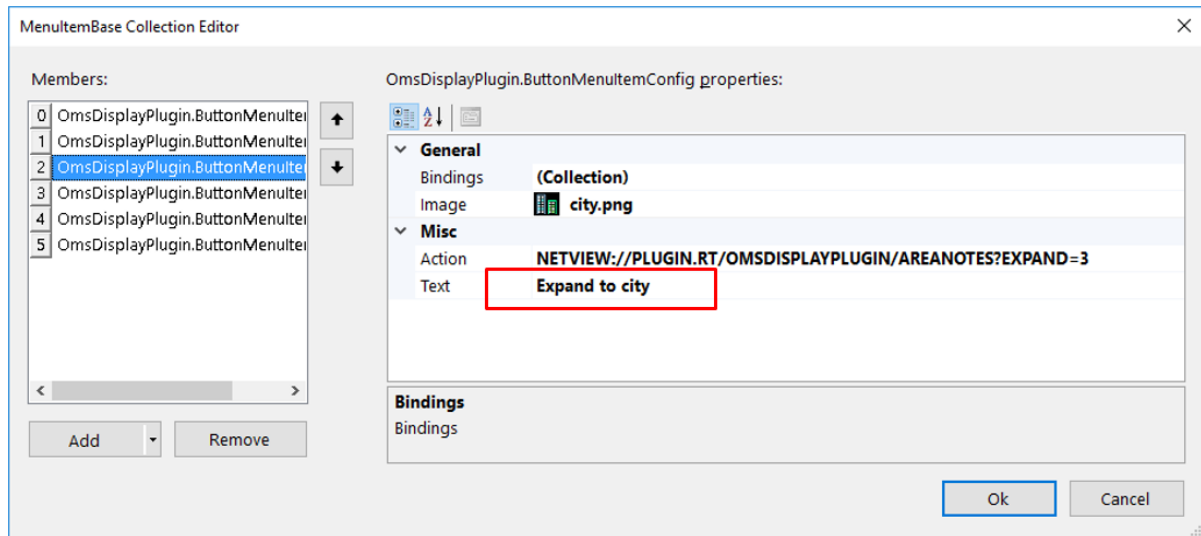
- Update the tooltips for the Overview display toolbar buttons, using the Toolbar item in the Overview Configuration group.



7. Update the management areas hierarchy level names for the Active Area Settings display, using the Level Definitions item in the Client Data's Area Settings Configuration group.

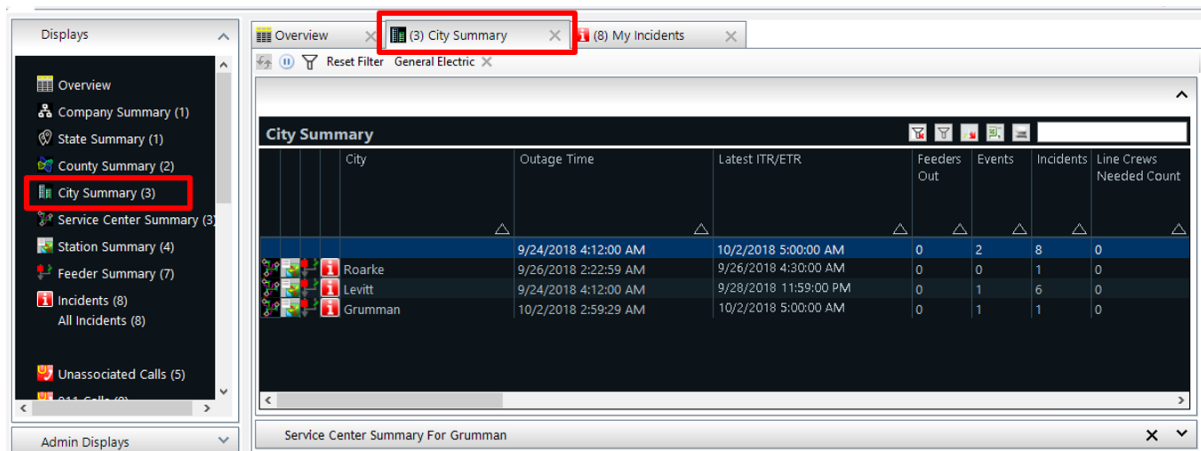


8. Update the tooltips for the Active Area Settings display toolbar buttons, using the Toolbar item in the Client Data's Area Settings Configuration group.



9. Save your configurations.

The example below shows the City Summary tab on the OMS display plugin.

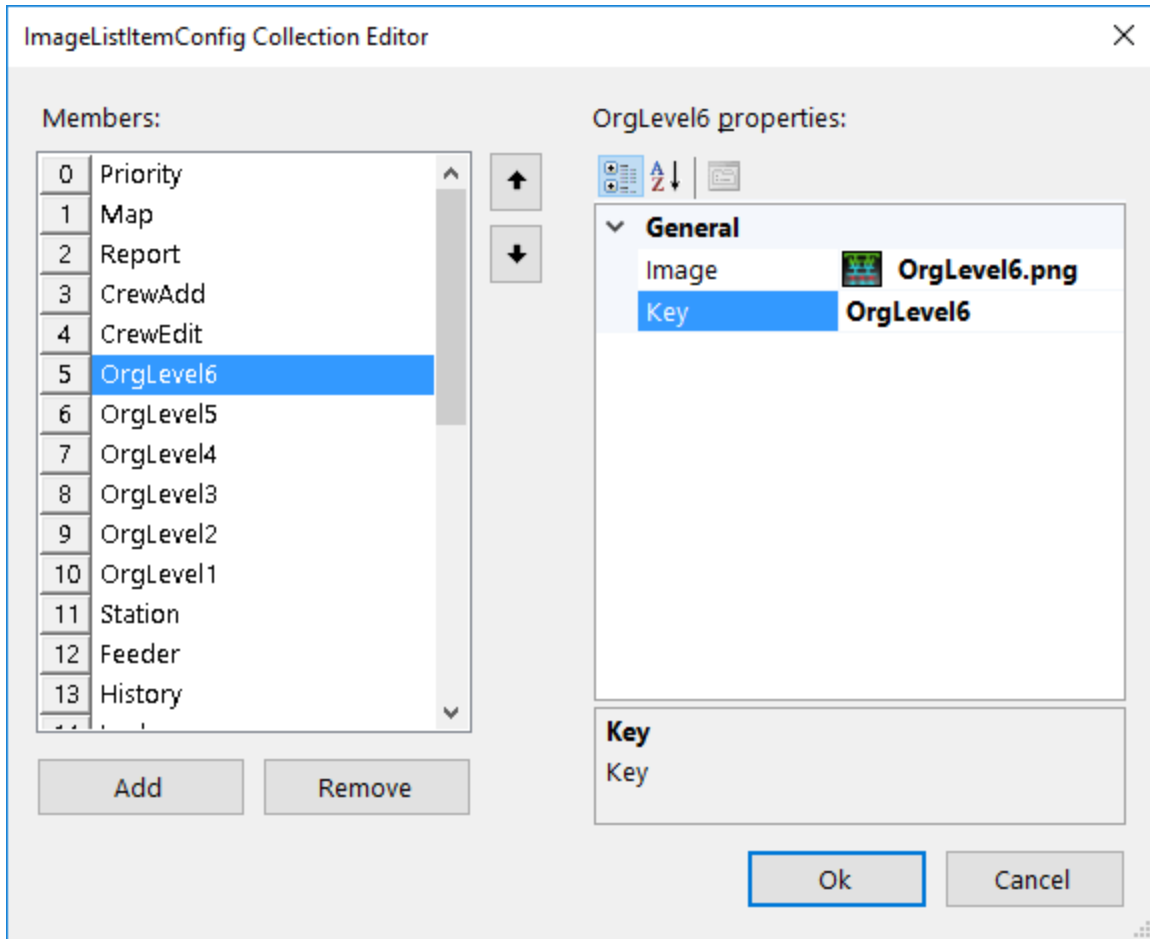


13.2.2. Configuring a Six-Level Management Areas Hierarchy

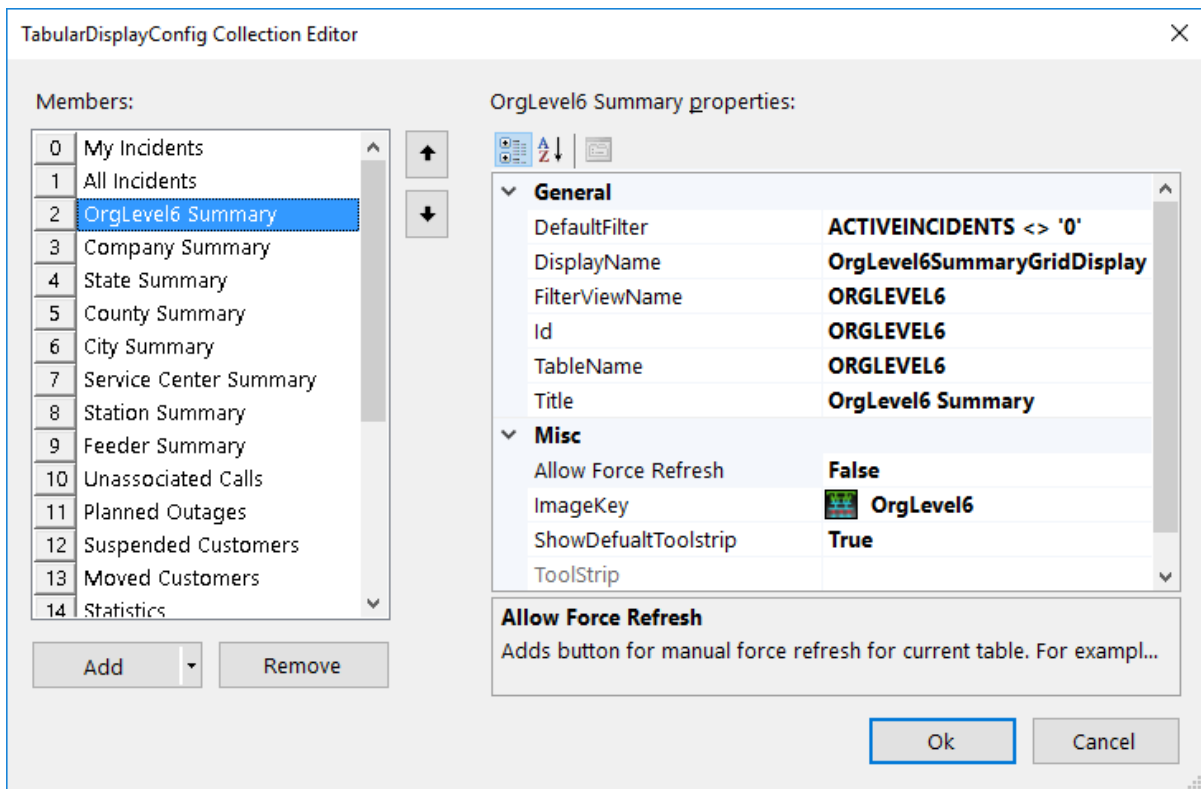
For a custom six-level management areas hierarchy, update the labels in the default five-level management areas hierarchy configuration and add configurations for the sixth management areas hierarchy level.

Modify the default `OmsDisplayPluginConfig.xml` as follows:

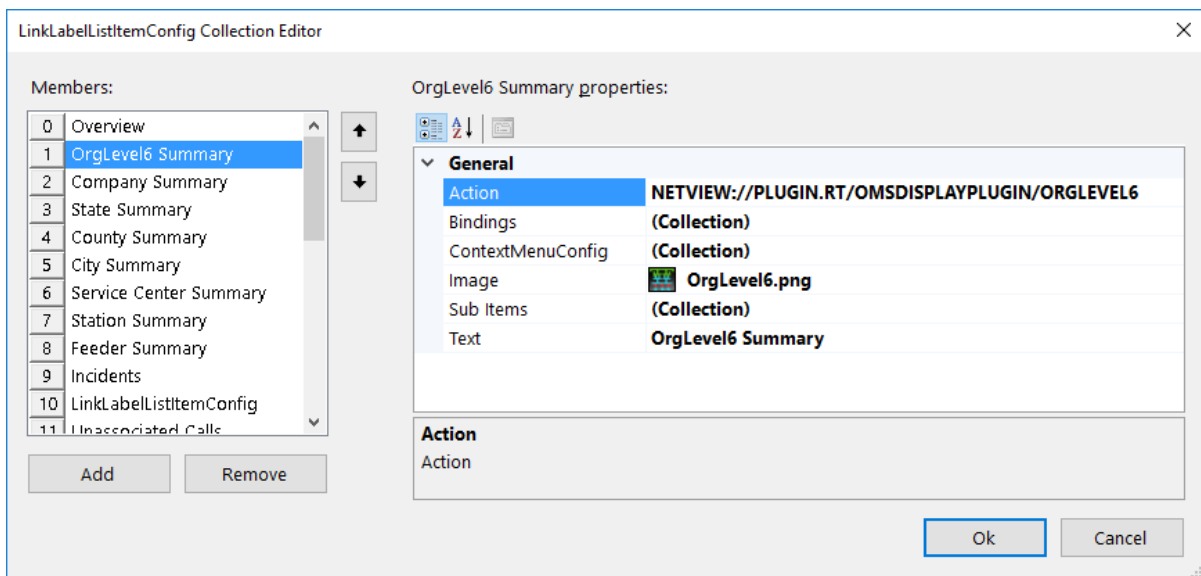
1. Open the OMS Client Configuration Editor dialog box.
2. On the Plugins tab, select `OmsDisplayPlugin`.
3. Update the labels as described in [Configuring a Five-Level Management Areas Hierarchy](#).
4. Define an image for the sixth hierarchy level, using the `ImageList` item in the Appearance group.

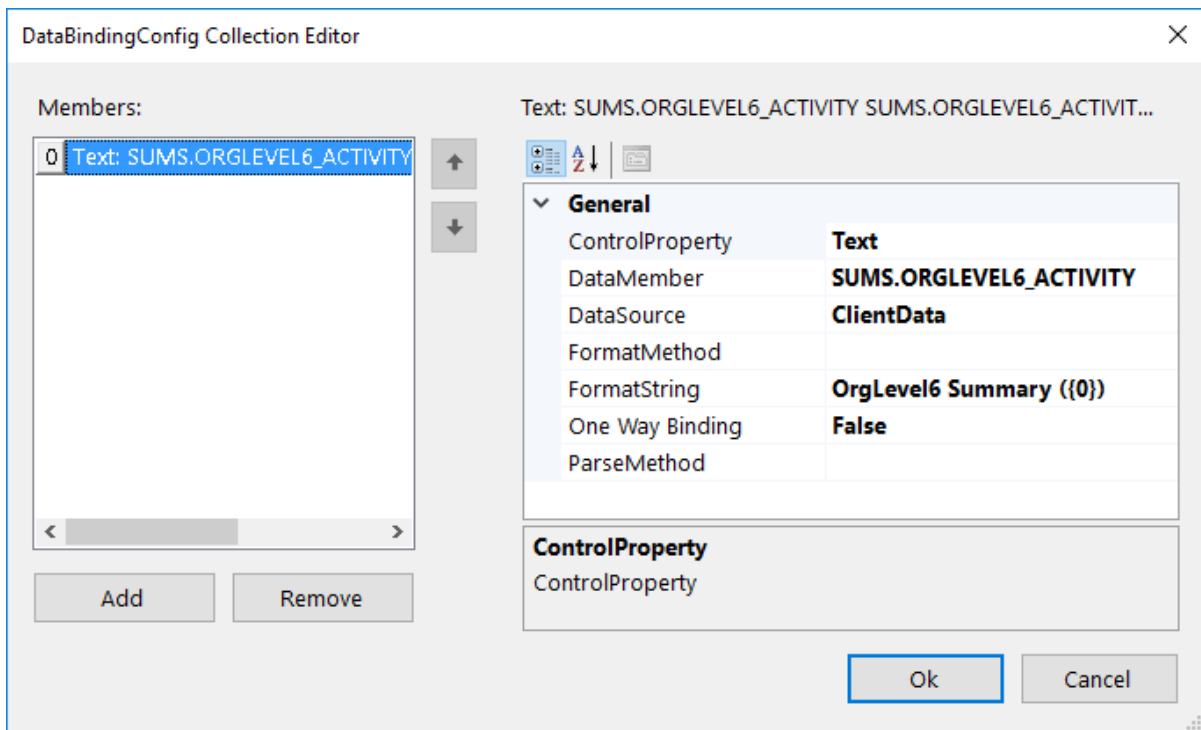


5. Add a tab for the sixth hierarchy level, using the TabDisplays item in the General group.
 - Copy the values of the Company Summary item.
 - Update the values to "ORGLEVEL6".

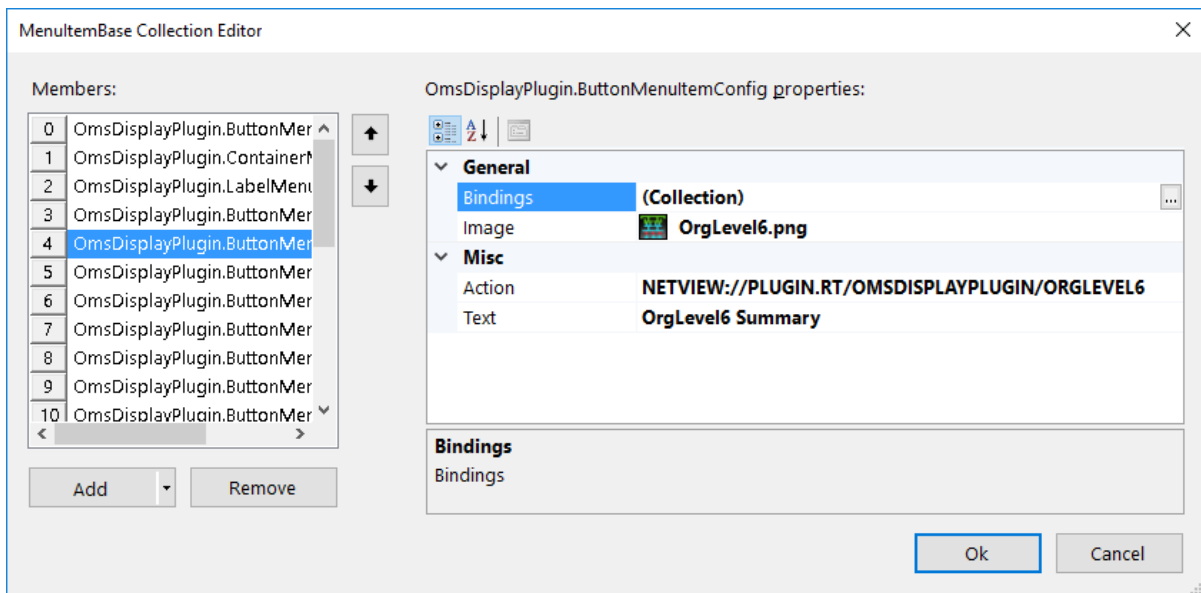


6. Add a display link for the sixth hierarchy level tab, using the Display Links item in the Appearance group.
 - Copy the values of the Company Summary item, including the bindings configurations.
 - Update the values to "ORGLEVEL6".

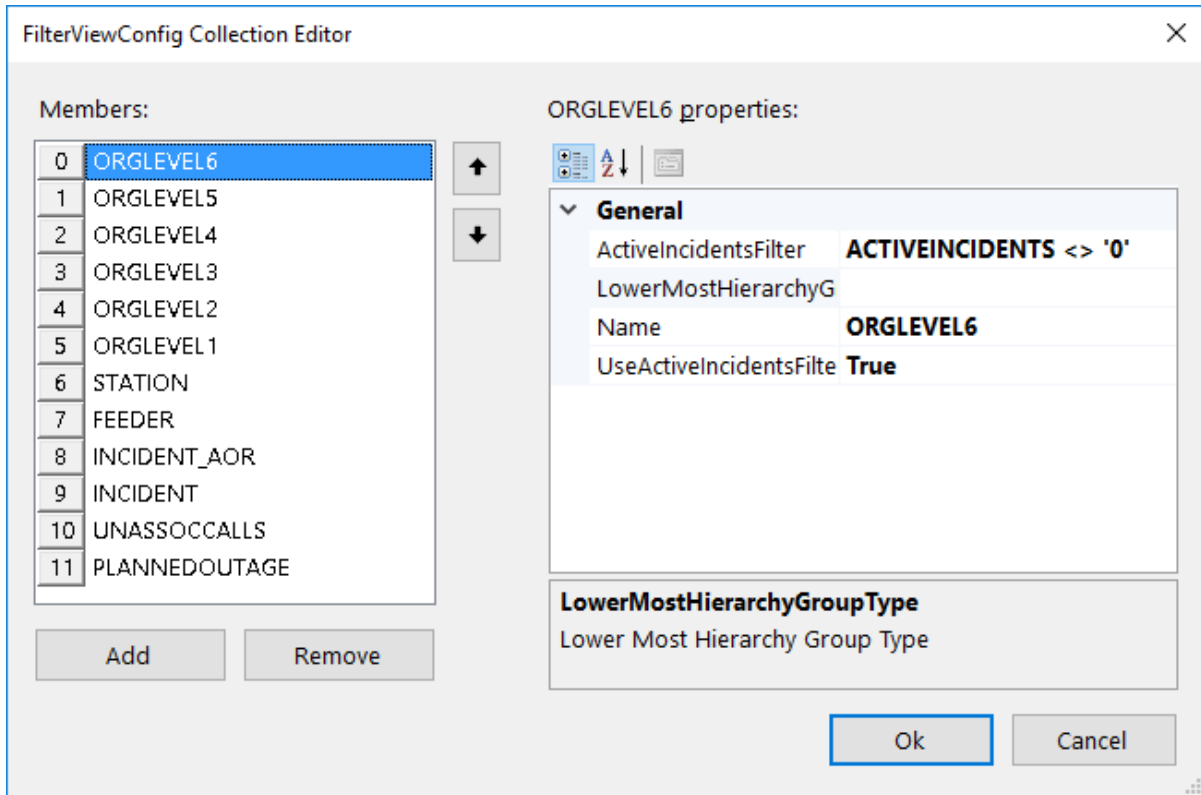




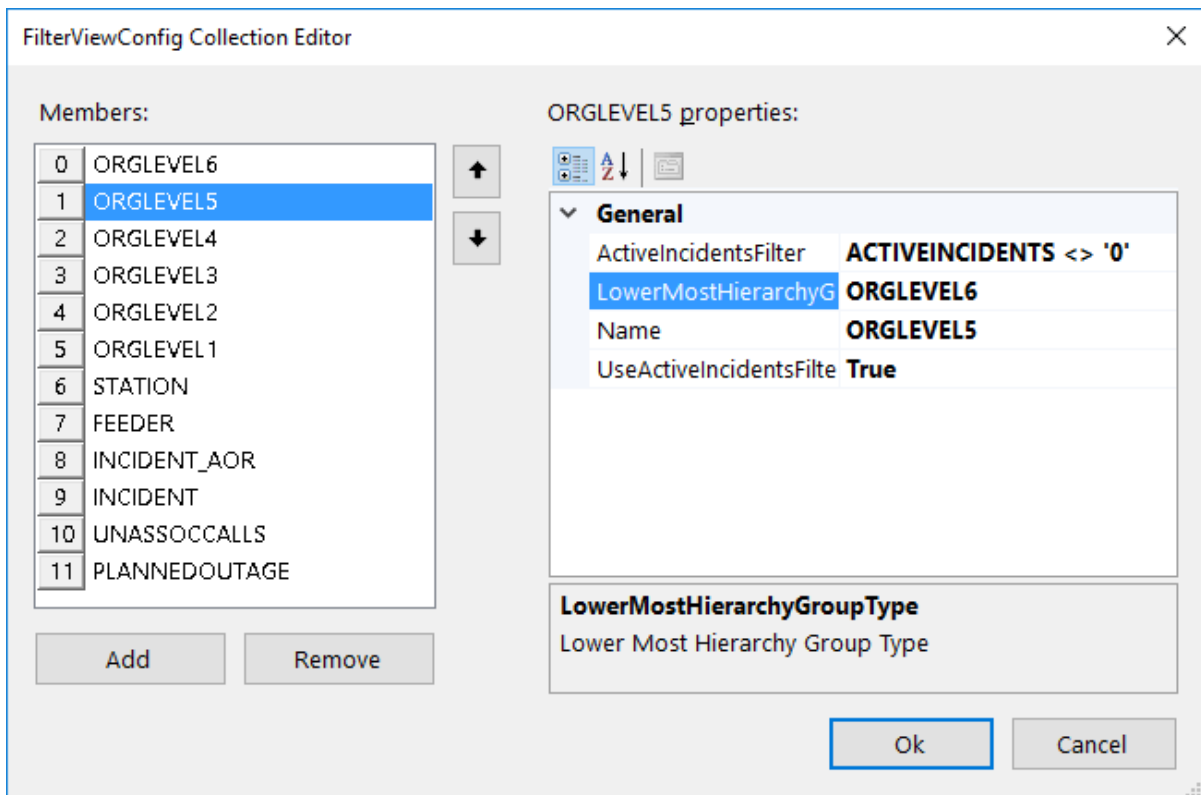
7. Add an OMS display plugin toolbar button for the sixth hierarchy level, using the Toolbar item in the General group.



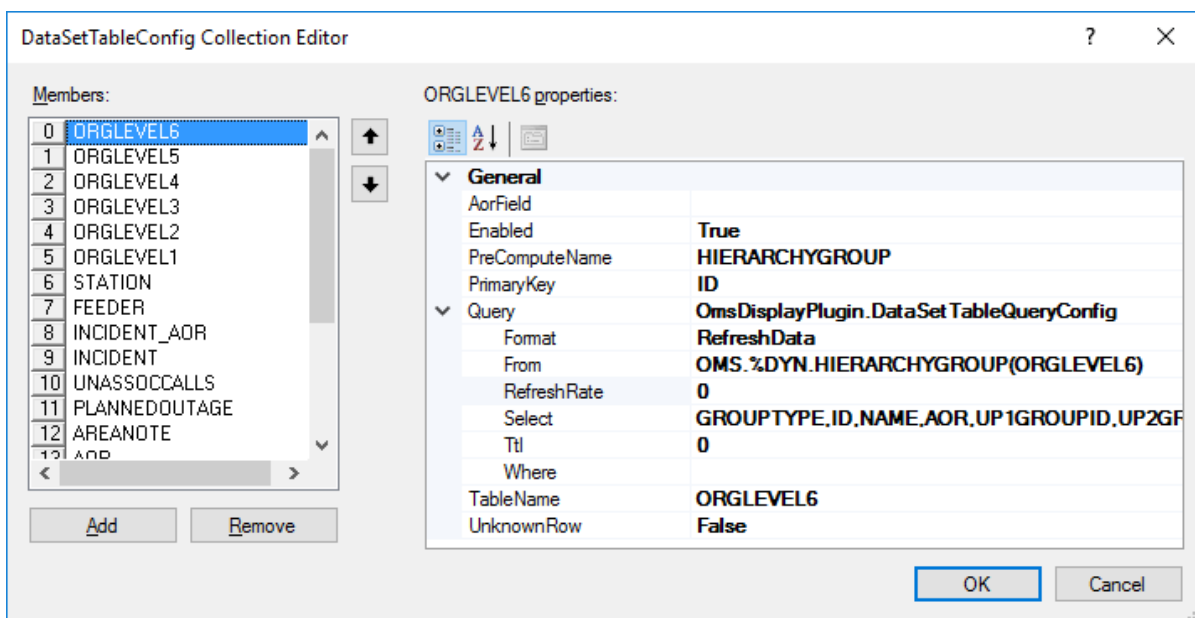
8. Add a filter for the sixth hierarchy level, using the FilterViewConfigs item in the General group.
 - Copy the values of the ORGLEVEL5 item.
 - Set the table name to "ORGLEVEL6".



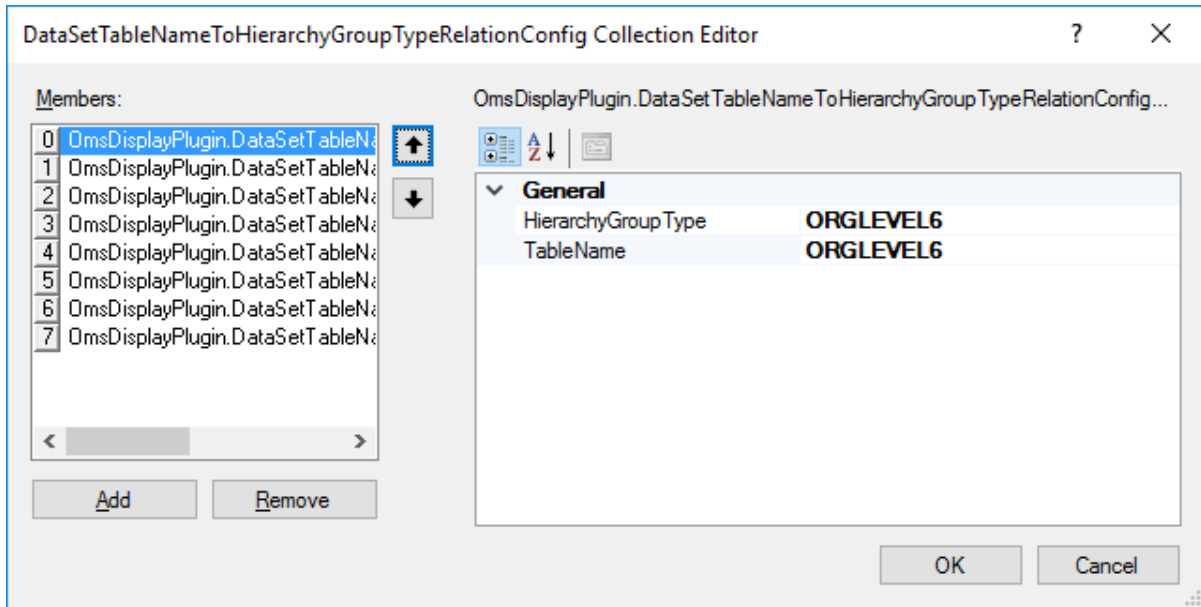
- Set LowerMostHierarchyGroupType = ORGLEVEL6 in a filter for the fifth hierarchy level, using the FilterViewConfigs item in the General group.



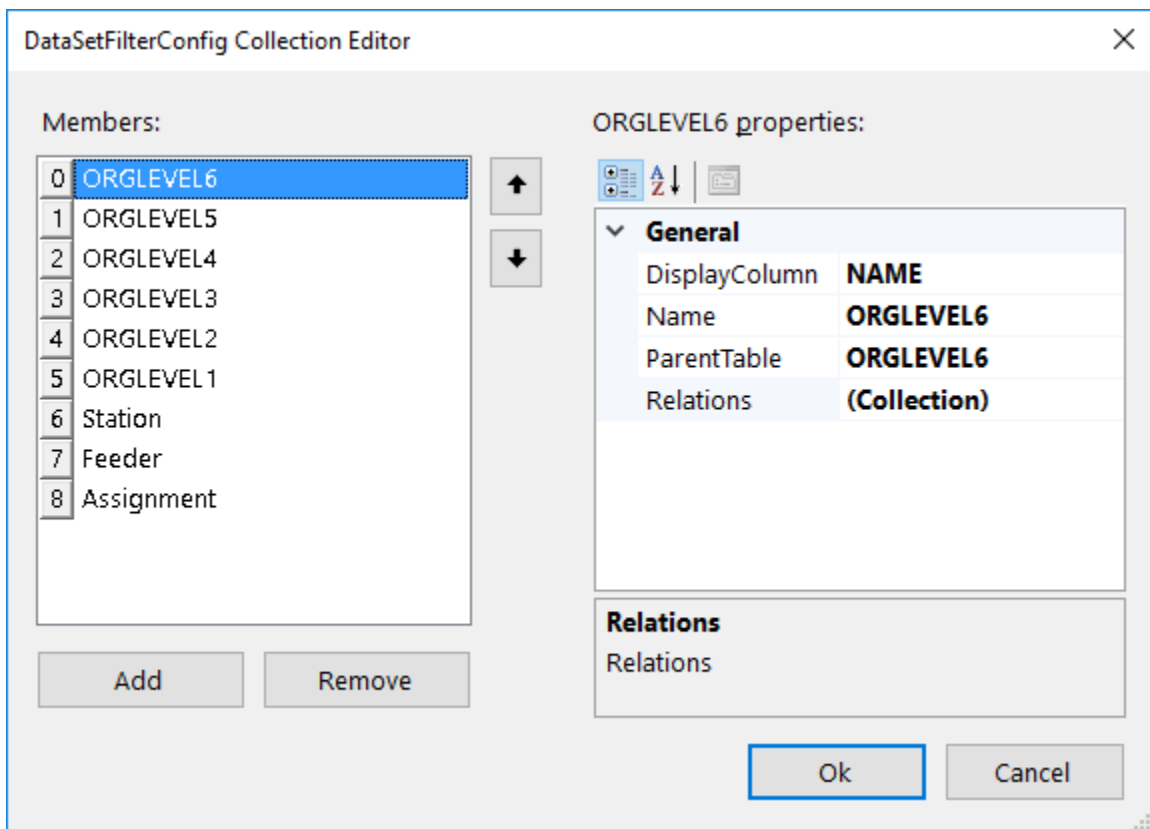
10. Add a table for the sixth hierarchy level, using the Tables item in the Client Data's Data Set Config group.
 - Copy the values of the ORGLEVEL5 item.
 - Set the From query to OMS. %DYN.HIERARCHYGROUP(ORGLEVEL6)".
 - Add the following values to the Select query: UP7GROUPID, UP7GROUPNAME.
 - Set the table name to "ORGLEVEL6".



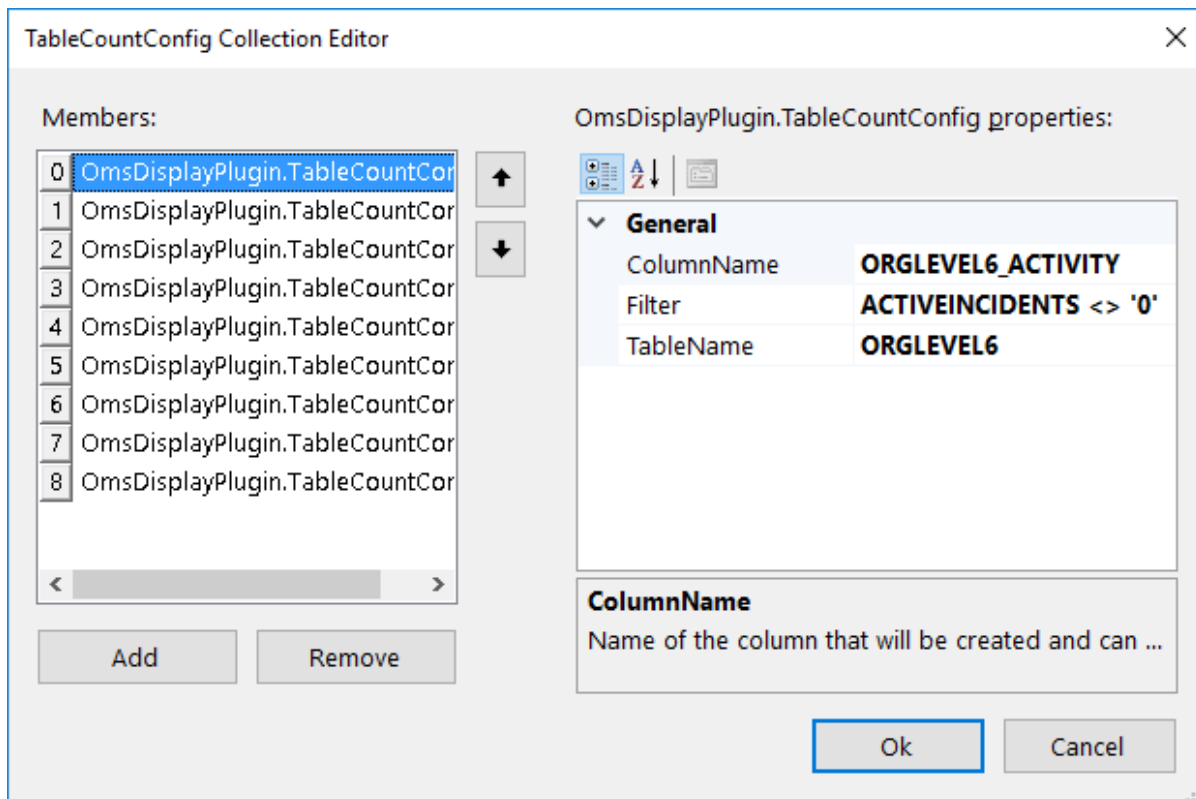
11. Add the ORGLEVEL6ID and ORGLEVEL6NAME values to the Query > Select value for the following tables, using the Tables item in the Client Data's Data Set Config group.
 - INCIDENT_AOR
 - INCIDENT
 - UNASSOCCALLS
 - 911CALLS
12. Add a "table name to hierarchy group" relation for the sixth hierarchy level, using the TableNameToHierarchyGroupTypeRelations item in the Client Data's Data Set Config group.
 - Set the HierarchyGroupType and TableName values to ORGLEVEL6.



13. Add a data set filter for the sixth hierarchy level, using the Filters item in the Client Data's Data Set Config group.
 - Copy the values of the ORGLLEVEL5 item.
 - Update the values to "ORGLLEVEL6".

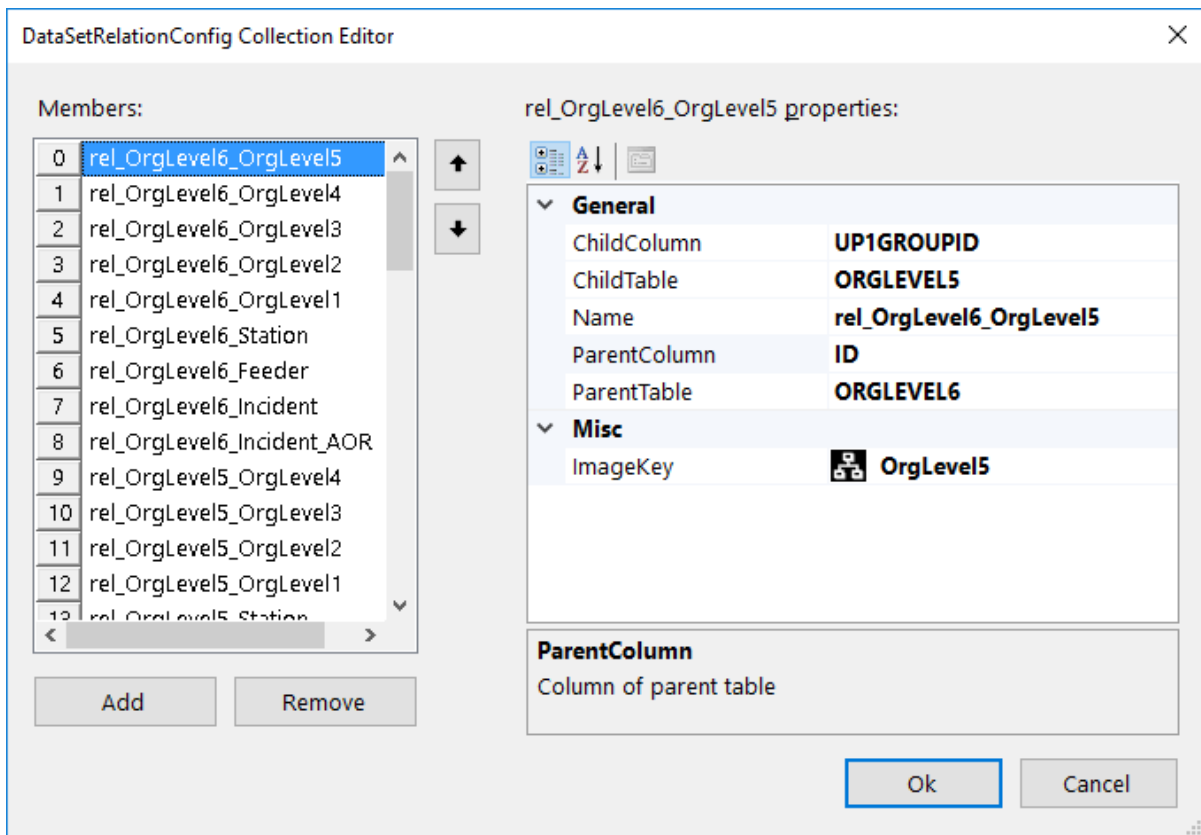


14. Add a table count configuration for the sixth hierarchy level, using the Table Counts item in the Client Data's Data Set Config group.
 - Copy the values of the ORGLEVEL5 item.
 - Update the values to "ORGLEVEL6".



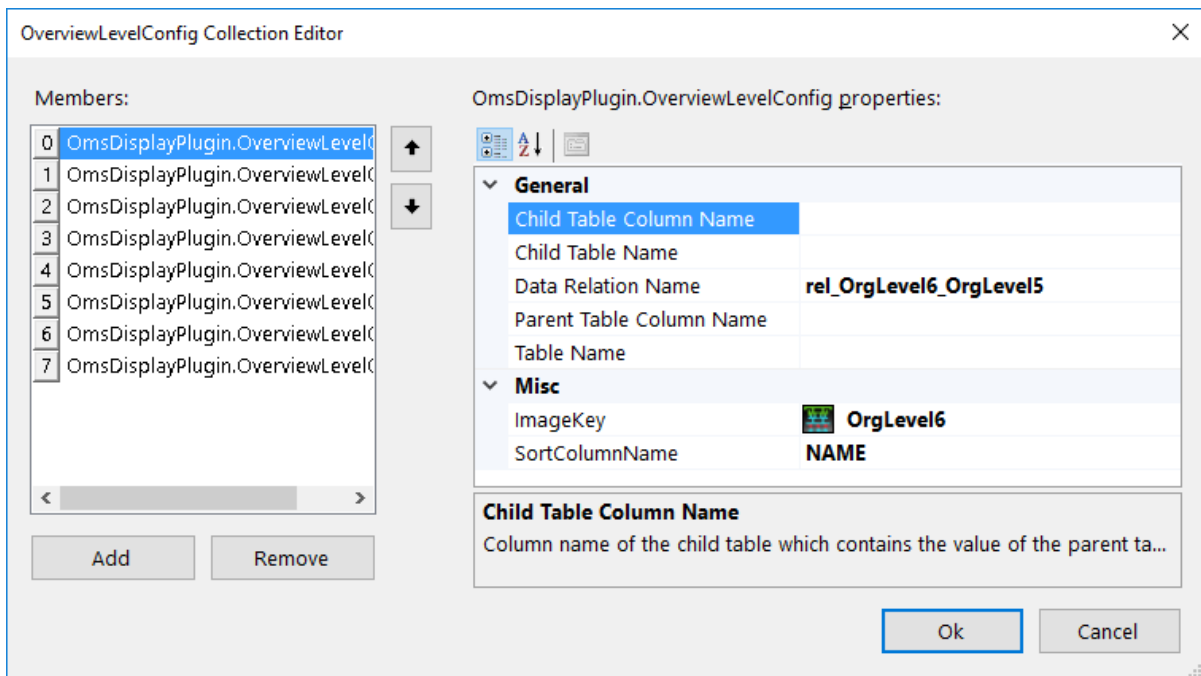
15. Add data set relations for the sixth hierarchy level, using the Relations item in the Client Data's Data Set Config group.
 - Add the following relations:
 - rel_OrgLevel6_OrgLevel5
 - rel_OrgLevel6_OrgLevel4
 - rel_OrgLevel6_OrgLevel3
 - rel_OrgLevel6_OrgLevel2
 - rel_OrgLevel6_OrgLevel1
 - rel_OrgLevel6_Station
 - rel_OrgLevel6_Feeder
 - rel_OrgLevel6_Incident
 - rel_OrgLevel6_IncidentAOR
 - Set the ChildColumn values as follows:
 -

- UP1GROUPID for rel_OrgLevel6_OrgLevel5
 - UP2GROUPID for rel_OrgLevel6_OrgLevel4
 - UP7GROUPID for rel_OrgLevel6_Feeder
 - ORGLEVEL6ID for rel_OrgLevel6_Incident and rel_OrgLevel6_IncidentAOR
- Set the ChildTable values as follows:
 - ORGLEVEL5 for rel_OrgLevel6_OrgLevel5
 - ...
 - ORGLEVEL1 for rel_OrgLevel6_OrgLevel1
 - STATION for rel_OrgLevel6_Station
 - FEEDER for rel_OrgLevel6_Feeder
 - INCIDENT for rel_OrgLevel6_Incident
 - INCIDENT_AOR for rel_OrgLevel6_IncidentAOR
- Set the ParentColumn values to ID.
- Set the ImageKey values as follows:
 - OrgLevel5 for rel_OrgLevel6_OrgLevel5
 - ...
 - OrgLevel1 for rel_OrgLevel6_OrgLevel1
 - Station for rel_OrgLevel6_Station
 - Feeder for rel_OrgLevel6_Feeder
 - Incident for rel_OrgLevel6_Incident
 - Incident for rel_OrgLevel6_IncidentAOR

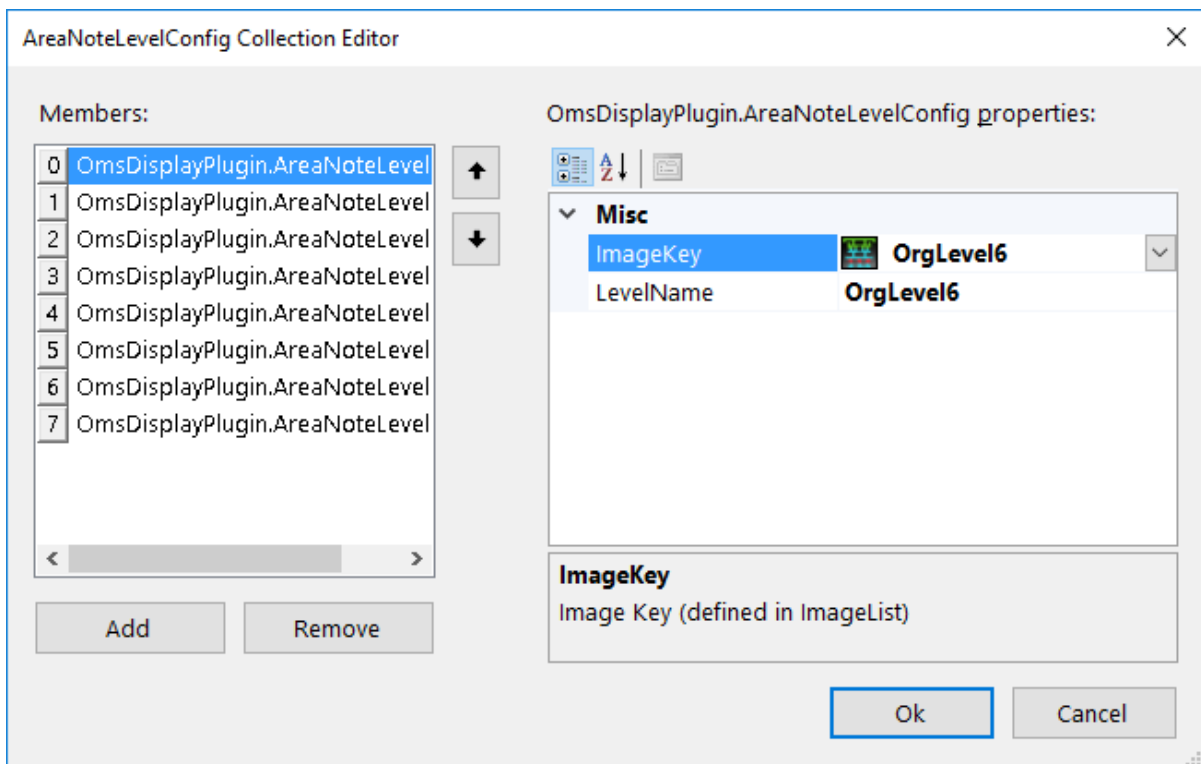


16. Add a level definition for the sixth hierarchy level, using the Level Definitions item in the Client Data's Overview Configuration group.

- Set the Data Relation Name value to rel_OrgLevel6_OrgLevel5
- Set the ImageKey value to OrgLevel6
- Set the SortColumnName value to NAME

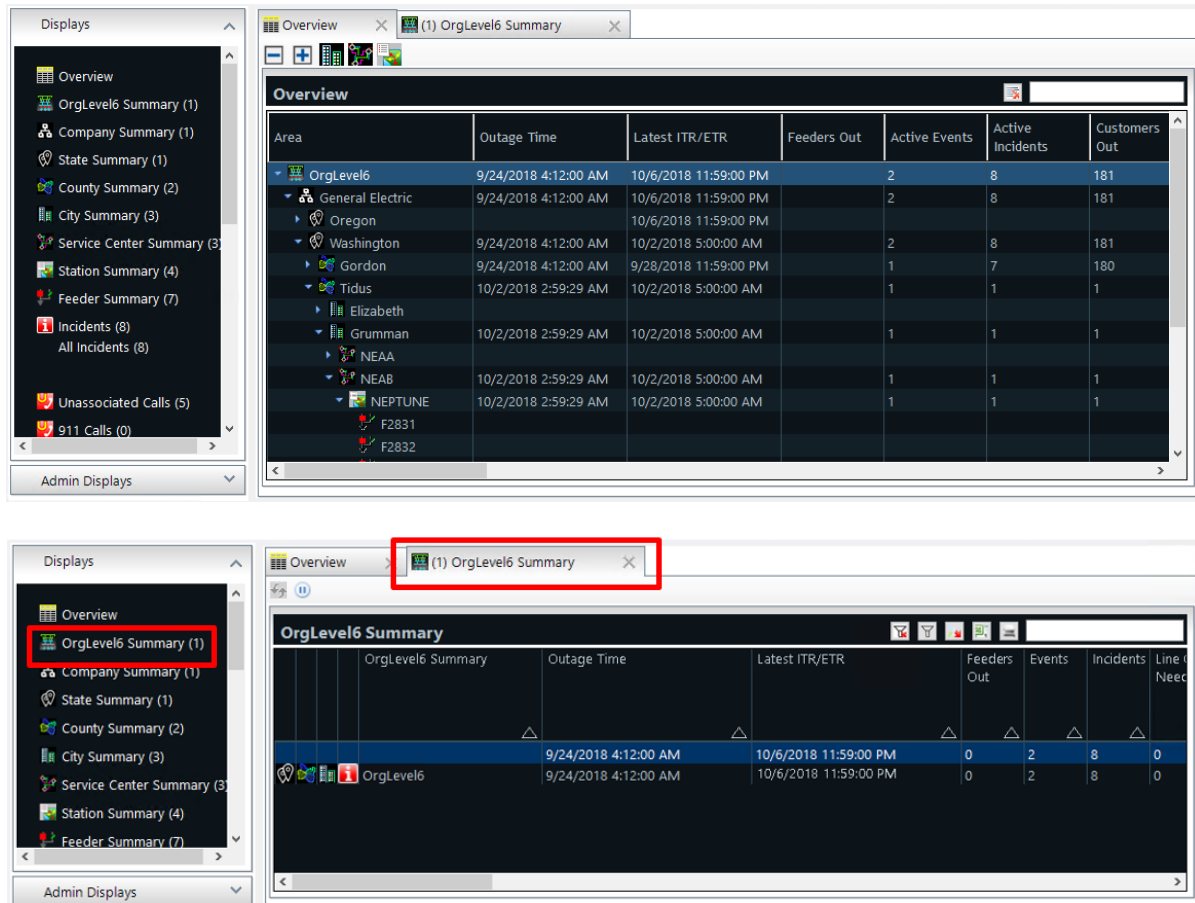


17. Add a level definition for the sixth hierarchy level, using the Level Definitions item in the Client Data's Area Settings Configuration group.
 - Set the ImageKey and LevelName values to OrgLevel6



18. Save your configurations.

The examples below show the Overview and OrgLevel6 Summary tabs on the OMS display plugin.

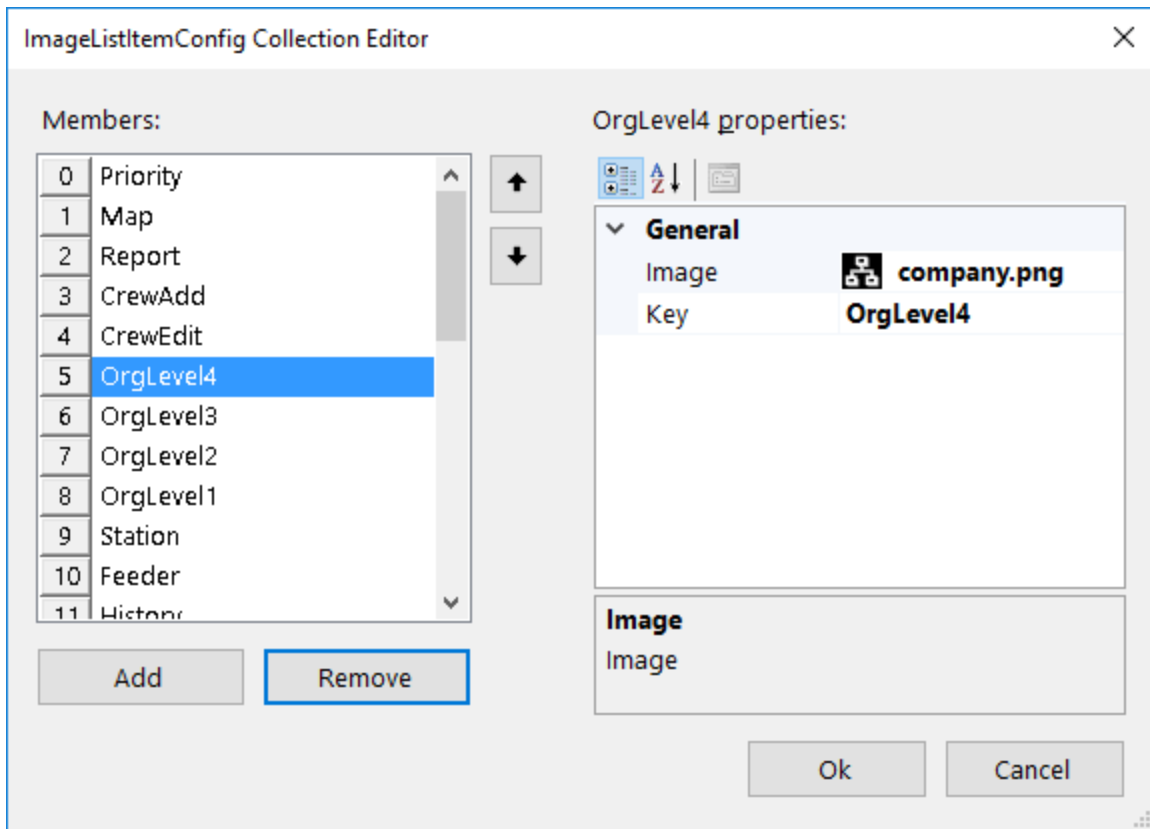


13.2.3. Configuring a Four-Level Management Areas Hierarchy

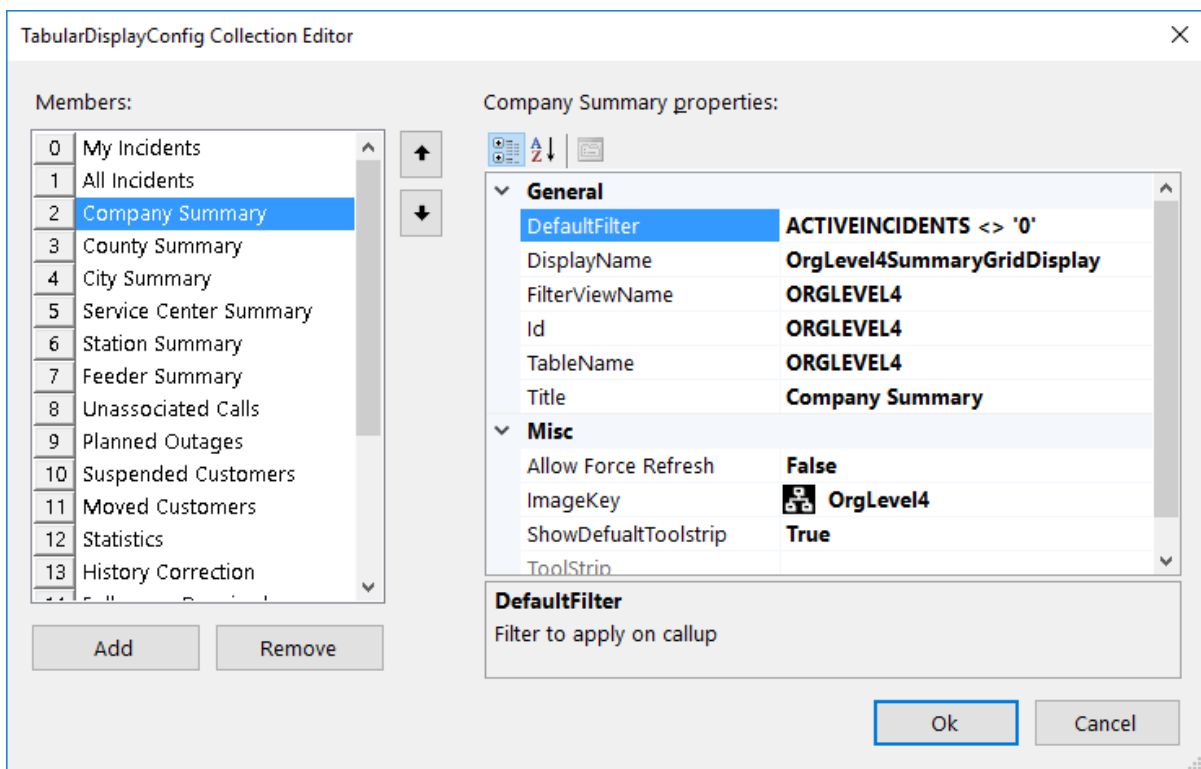
For a custom four-level management areas hierarchy, update the labels in the default five-level management areas hierarchy configuration, remove the fifth management areas hierarchy level, and update the configurations for the fourth management areas hierarchy level.

Modify the default `OmsDisplayPluginConfig.xml` as follows:

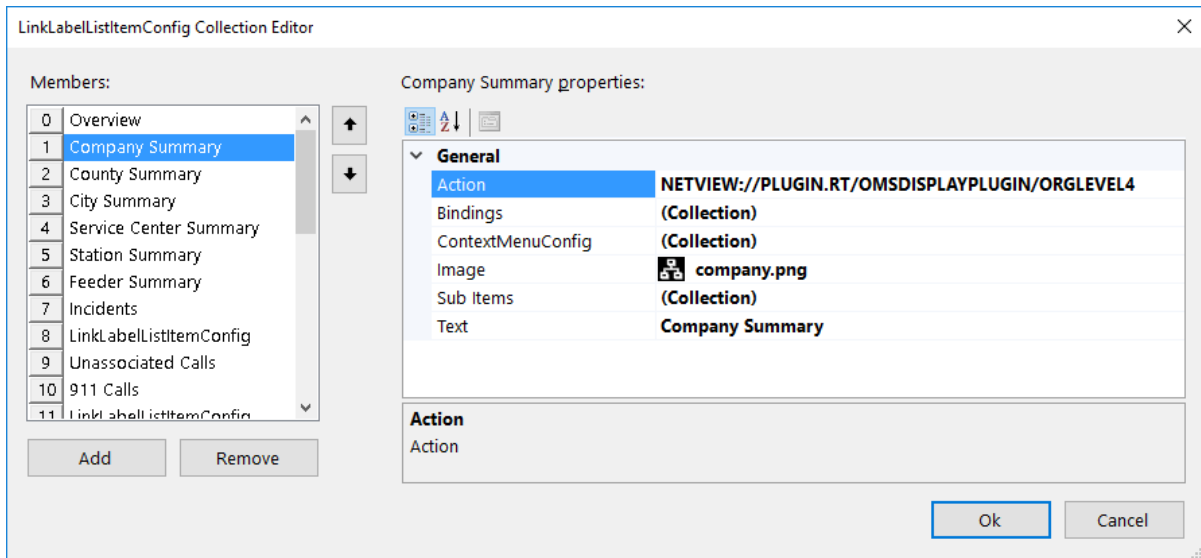
1. Open the OMS Client Configuration Editor dialog box.
2. On the Plugins tab, select `OmsDisplayPlugin`.
3. Update the labels as described in [Configuring a Five-Level Management Areas Hierarchy](#).
4. Remove the image definition for the fifth hierarchy level, using the `ImageList` item in the Appearance group.
5. Update the image for the fourth hierarchy level.



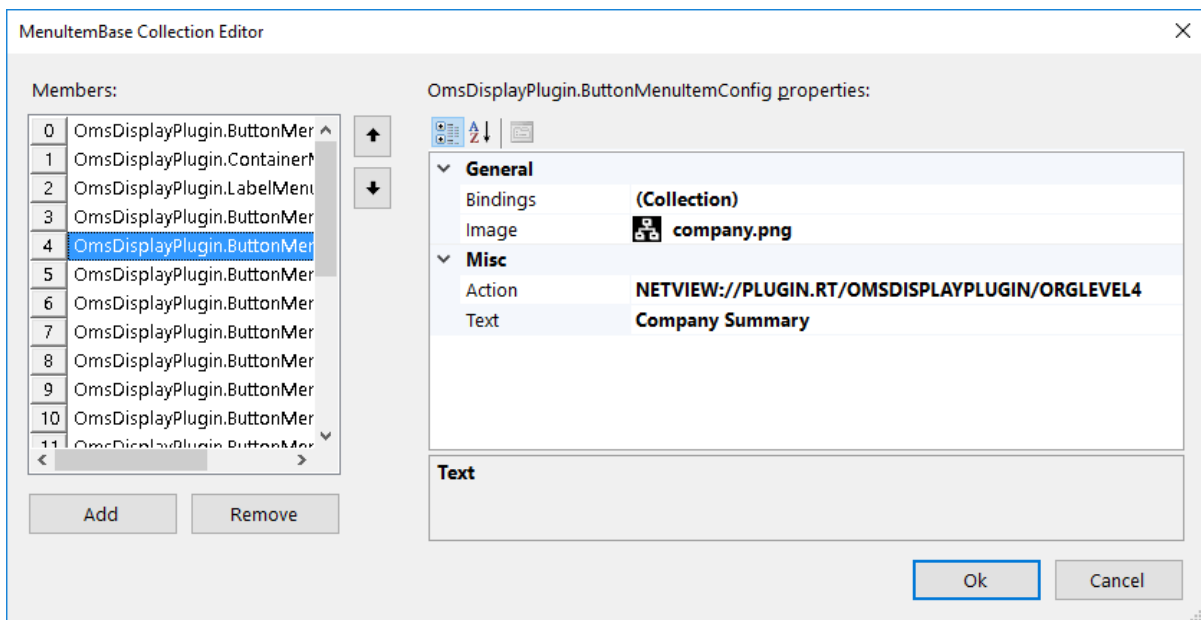
6. Remove the tab for the fifth hierarchy level, using the TabDisplays item in the General group.



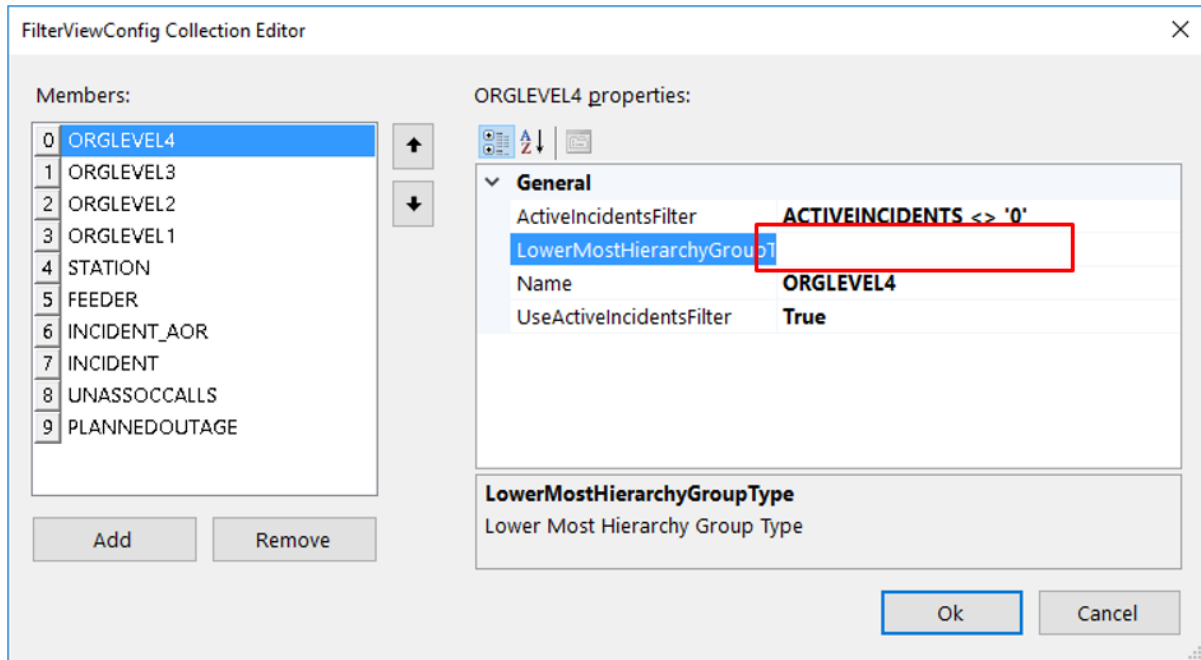
7. Remove the display link for the fifth hierarchy level tab, using the Display Links item in the Appearance group.



8. Remove the OMS display plugin toolbar button for the fifth hierarchy level, using the Toolbar item in the General group.



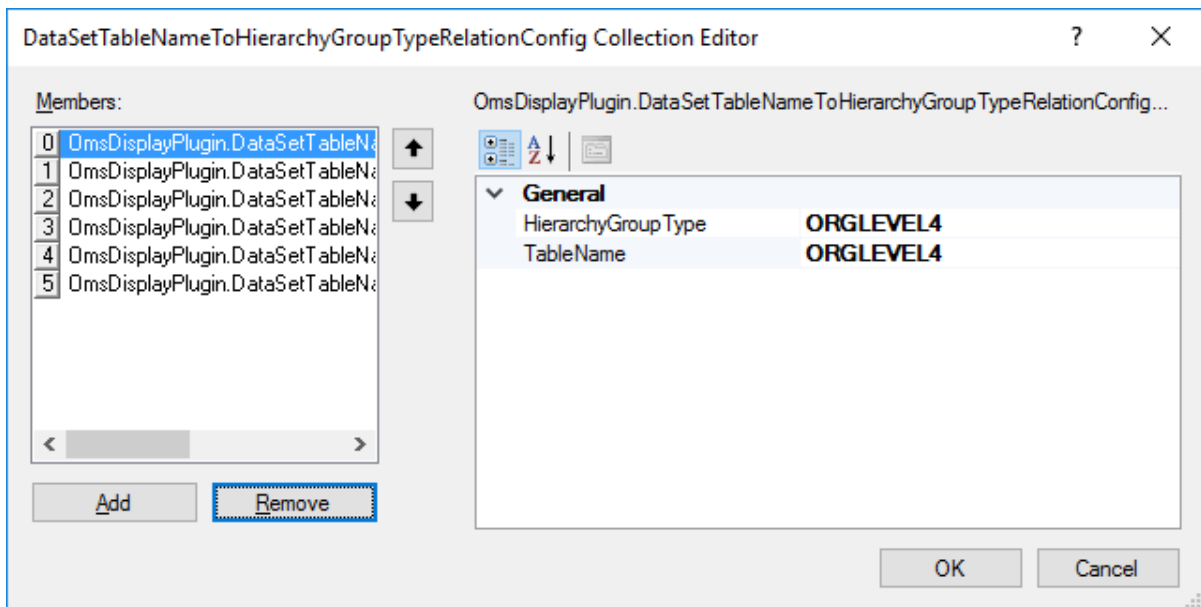
9. Remove the filter for the fifth hierarchy level, using the FilterViewConfigs item in the General group.
10. Clear the LowerMostHierarchyGroupType field in the filter for the fourth hierarchy level, using the FilterViewConfigs item in the General group.



11. Remove ORGLEVEL5ID and ORGLEVEL5NAME values from the Query > Select value for the following tables, using the Tables item in the Client Data's Data Set Config group.

- INCIDENT_AOR
- INCIDENT
- UNASSOCCALLS
- 911CALLS

12. Remove the "table name to hierarchy group" relation for the fifth hierarchy level, using the TableNameToHierarchyGroupTypeRelations item in the Client Data's Data Set Config group.



13. Remove the table for the fifth hierarchy level, using the Tables item in the Client Data's Data Set Config group.

DataSetTableConfig Collection Editor

Members:

0	ORGLEVEL4
1	ORGLEVEL3
2	ORGLEVEL2
3	ORGLEVEL1
4	STATION
5	FEEDER
6	INCIDENT_AOR
7	INCIDENT
8	UNASSOCCALLS
9	PLANNEDOUTAGE
10	ΔREFNOTE

ORGLEVEL4 properties:

General

AorField	
Enabled	True
PreComputeName	HIERARCHYGROUP
PrimaryKey	ID
Query	OmsDisplayPlugin.DataSet TableQueryC
TableName	ORGLEVEL4
UnknownRow	False

Buttons: Add, Remove, OK, Cancel

14. Remove the data set filter for the fifth hierarchy level, using the Filters item in the Client Data's Data Set Config group.

DataSetFilterConfig Collection Editor

Members:

0	ORGLEVEL4
1	ORGLEVEL3
2	ORGLEVEL2
3	ORGLEVEL1
4	Station
5	Feeder
6	Assignment

ORGLEVEL4 properties:

General

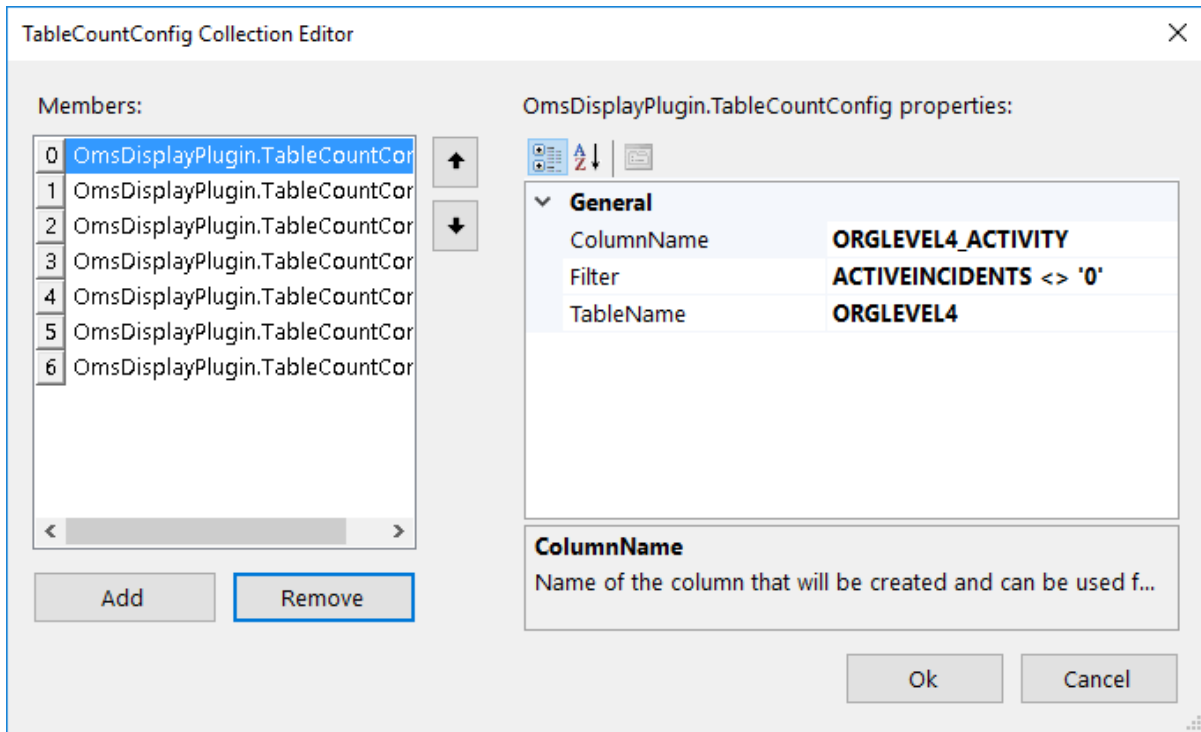
DisplayColumn	NAME
Name	ORGLEVEL4
ParentTable	ORGLEVEL4
Relations	(Collection)

Relations

Relations

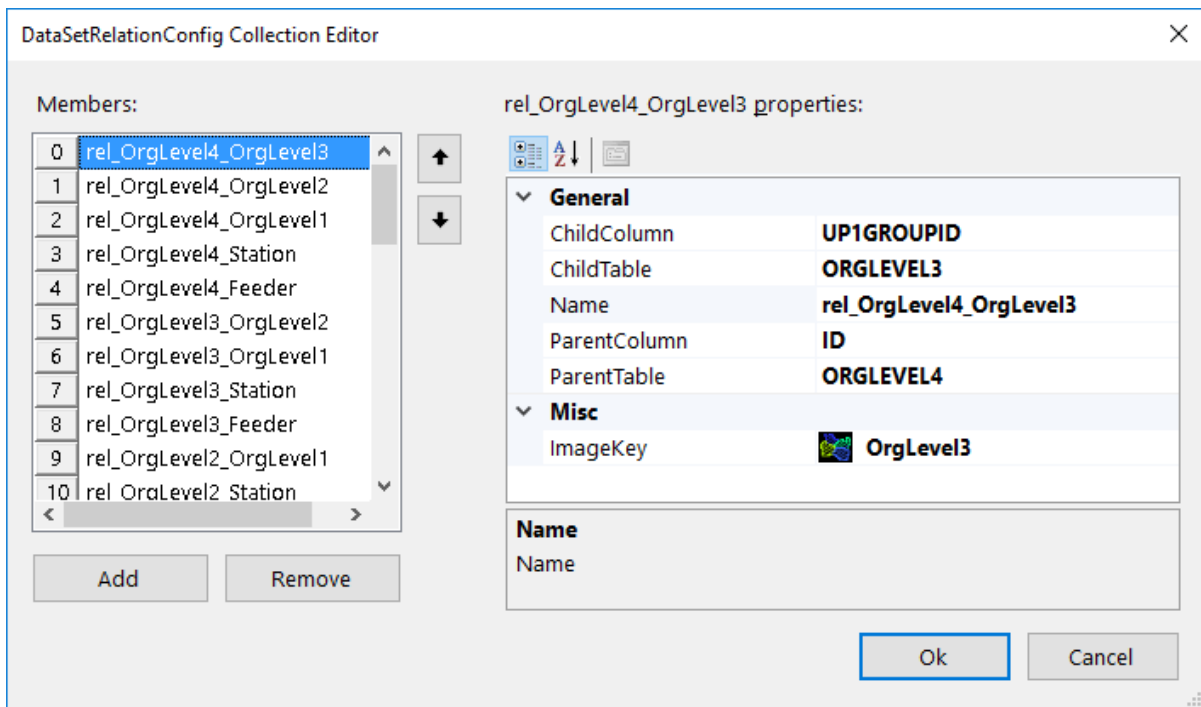
Buttons: Add, Remove, Ok, Cancel

15. Remove the table count configuration for the fifth hierarchy level, using the Table Counts item in the Client Data's Data Set Config group.

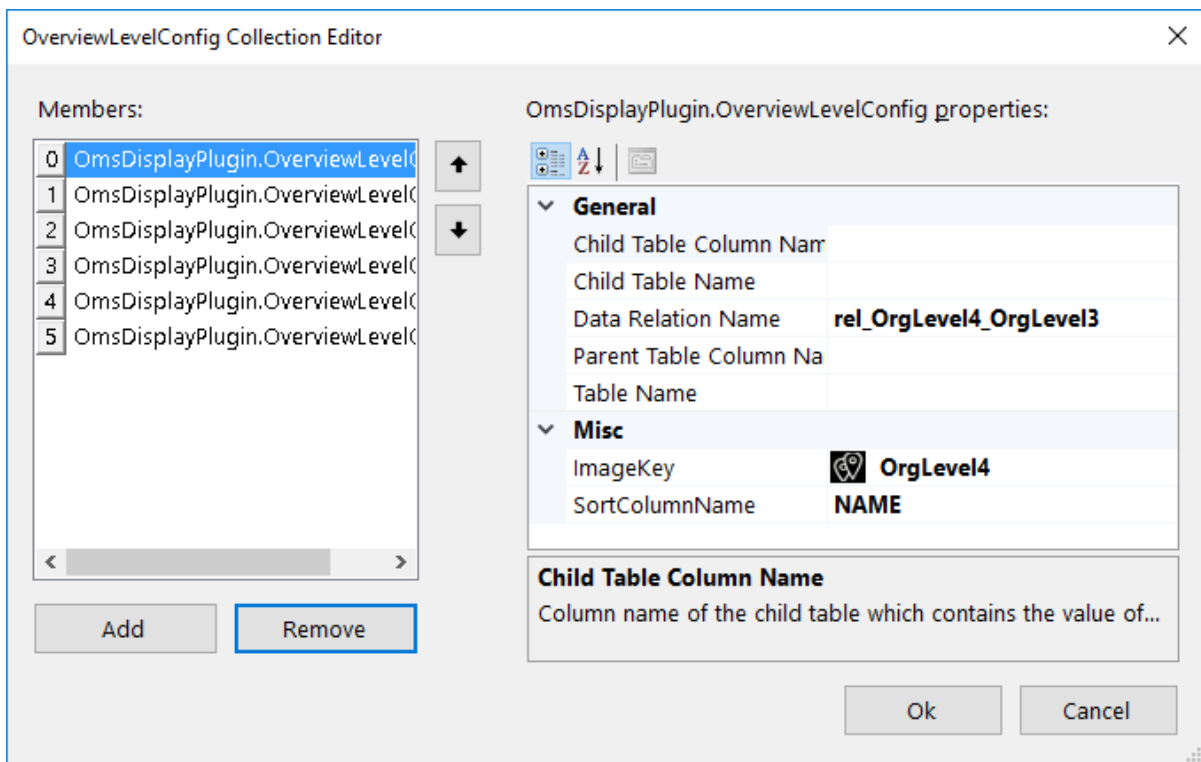


16. Remove the data set relations for the fifth hierarchy level, using the Relations item in the Client Data's Data Set Config group.

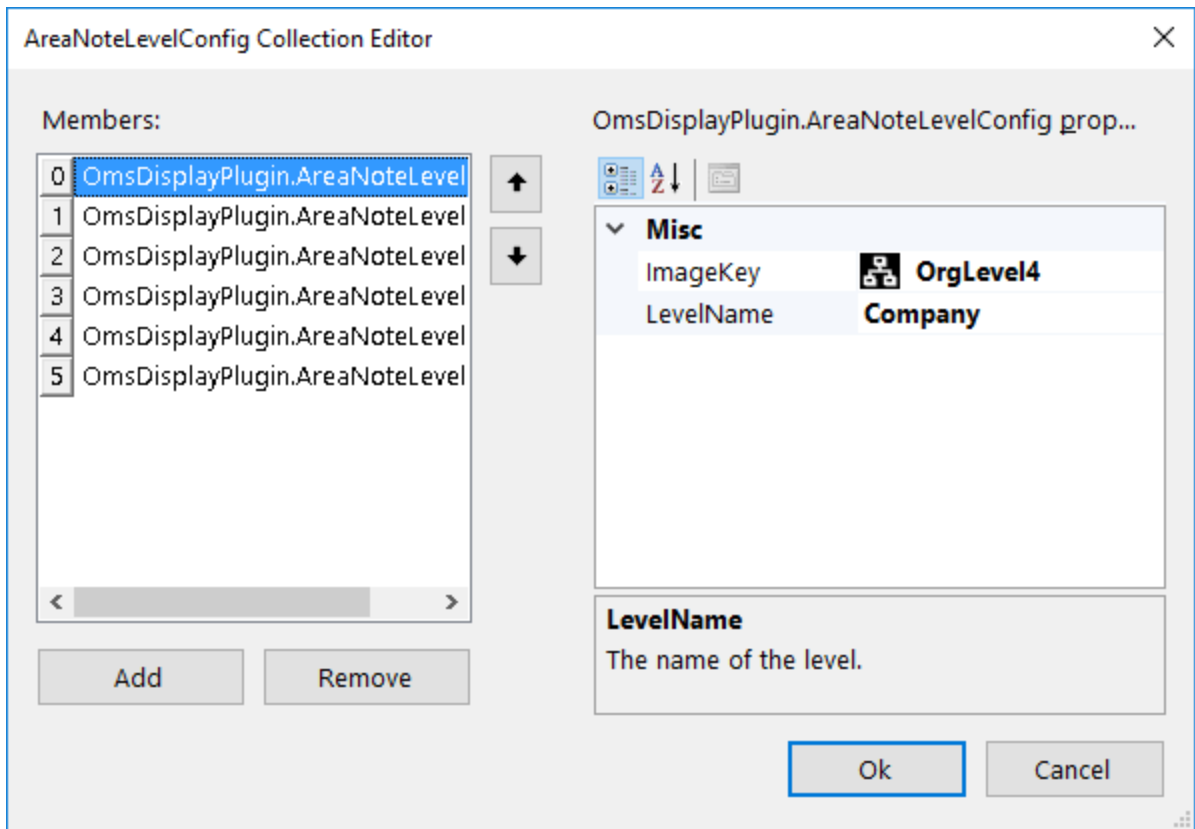
- Remove the following relations:
 - rel_OrgLevel5_OrgLevel4
 - rel_OrgLevel5_OrgLevel3
 - rel_OrgLevel5_OrgLevel2
 - rel_OrgLevel5_OrgLevel1
 - rel_OrgLevel5_Station
 - rel_OrgLevel5_Feeder
 - rel_OrgLevel5_Incident
 - rel_OrgLevel5_IncidentAOR



17. Remove the level definition for the fifth hierarchy level, using the Level Definitions item in the Client Data's Overview Configuration group.

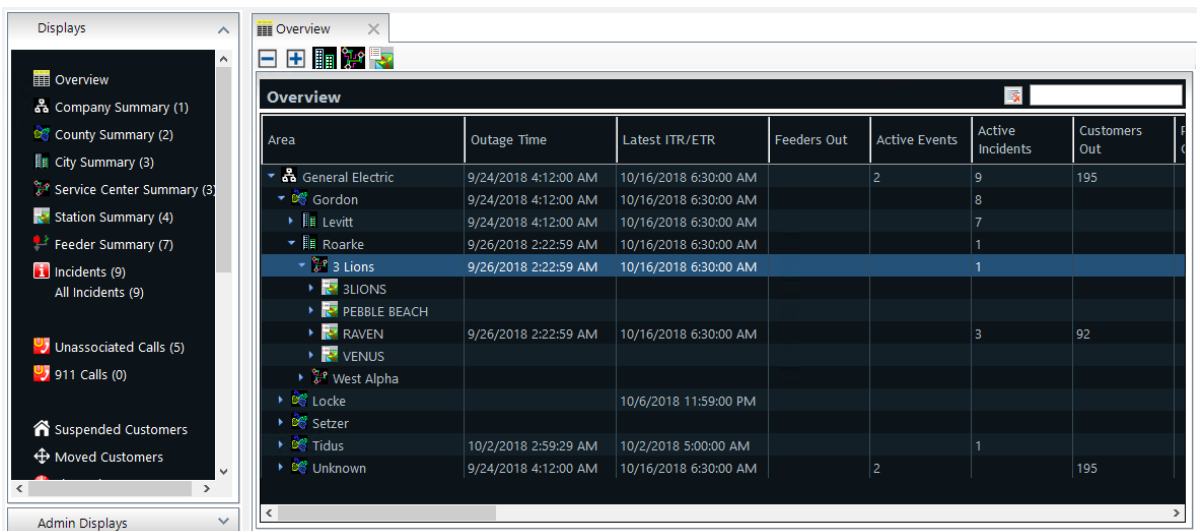


18. Remove the level definition for the fifth hierarchy level, using the Level Definitions item in the Client Data's Area Settings Configuration group.



19. Save your configurations.

The examples below show the Overview and Company Summary tabs on the OMS display plugin.



Displays

Overview

Company Summary (1)

County Summary (2)

City Summary (3)

Service Center Summary (3)

Station Summary (4)

Feeder Summary (7)

Incidents (9)

Admin Displays

Overview

(1) Company Summary

Company Summary

	State	Outage Time	Latest ITR/ETR	Feeders Out	Events
		9/24/2018 4:12:00 AM	10/16/2018 6:30:00 AM	0	2
 General Electric		9/24/2018 4:12:00 AM	10/16/2018 6:30:00 AM	0	2